F78PB Probability and Statistics B

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1 Moments in Statistics

1.1 Set up

The general set up we have in (frequentist) statistics is we are interested in the results of a repeatable experiment described by a probability model $(\Omega, \mathcal{F}, \mathbb{P})$. The outcome of the experiment will be modelled as a random variable X.

In Probability Theory (such as in F78PA Probability and Statistics A) we typically now the distribution of X and aim to calculate properties such as probabilities, expectations, variance, etc. In Statistics we do not know the distribution of X but aim to learn the distribution or estimate aspects of it such the location, spread, and correlation.

The setting we consider is that we have n i.i.d. copies of X, denoted $\underline{X} = (X_1, \ldots, X_n)$ which we consider to be repeating the experiment. Once we have performed these experiments the outcomes are known and we observe $X_1 = x_1, \ldots, X_n = x_n$ which is our data $\underline{x} = (x_1, \ldots, x_n)$.

We will expand on this is Section 2.

First we will discuss two concepts:

- Estimate of the moments (e.g. $\mathbb{E}[X^k]$) of the random variable X given the data $\underline{x}$.

- Moment Generating Functions, Probability Generating Functions and Cumulant generating functions which are useful probabilistic tools.

1.2 Moments

The kth order moment about the value c of the random variable X is $\mathbb{E}[(X-c)^k]$ we can estimate this from our data using the kth order *sample moments* about c as follows:

$$\frac{1}{n}\sum_{i=1}^{n}(x_i - c)^k.$$

When $c = 0$ we have the *simple* moments (about zero) which approximate $\mathbb{E}[X^k]$:

$$\nu_k = \frac{1}{n}\sum_{i=1}^{n} x_i^k$$

When $c = \bar{x}$ we have the *central* moments (about the mean $\bar{x}$) which approximate $\mathbb{E}[(X-\mathbb{E}[X])^k]$:

$$m_k = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^k$$

Special cases:

- Sample mean: $\bar{x} = \nu_1$.
- Sample variance: $s^2 = \frac{n}{n-1}m_2$.

1.3 Pearson's symmetry coefficient

A continuous (discrete) random variable with probability density function (probability mass function) f_X is called symmetric if there exists x_0 such that

$$f_X(x_0 + y) = f_X(x_0 - y) \quad \text{for all } y \in \mathbb{R}.$$

Note that this implies that $X - x_0$ has the same distribution as $x_0 - X$. Since they have the same distribution, the expected values are equal:

$$\mathbb{E}[X - x_0] = \mathbb{E}[x_0 - X].$$

Rearranging gives $x_0 = \mathbb{E}[X]$. Similarly all the moments of $X - \mathbb{E}[X]$ equal those of $\mathbb{E}[X] - X$. In particular the odd central moments must be zero:

$$\mathbb{E}[(X - \mathbb{E}[X])^{2k+1}] = 0 \quad \text{for any } k \in \mathbb{N}.$$

We use this property to motivate a measure of asymmetry.

The first central moment ($k = 0$) is always zero, so we will consider the third central moment. To make this quantity dimensionless (independent of units), we divide by a suitable power of the standard deviation, in this case σ^3, this gives the following measure of how symmetric the distribution is

$$\frac{\mathbb{E}[(X - \mathbb{E}[X])^3}{\sigma^3}, \quad \sigma = \sqrt{\mathbb{V}\text{ar}(X)}.$$

Alternatively if we set $\tilde{X} = (X - \mathbb{E}[X])/\sigma$ to be the X after it has been standardised, then we can write

$$\frac{\mathbb{E}[(X - \mathbb{E}[X])^3}{\sigma^3} = \mathbb{E}\left[\tilde{X}^3\right], \quad \sigma = \sqrt{\mathbb{V}\text{ar}(X)}.$$

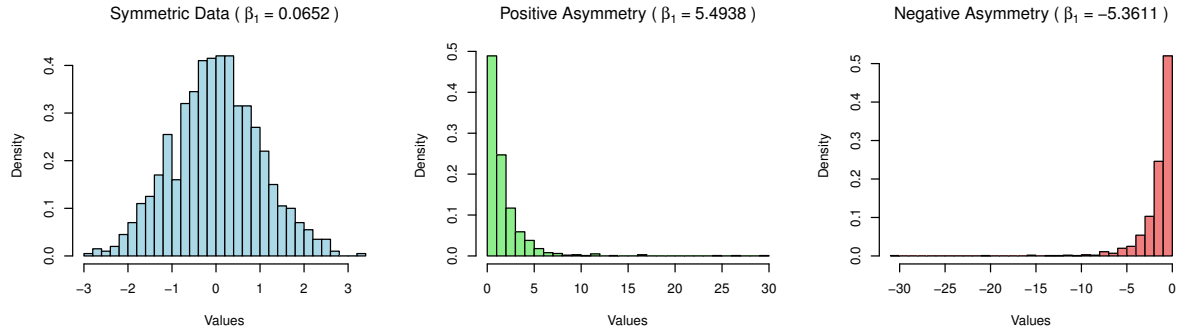
We can estimate this quantity using the Pearson's symmetry coefficient defined as:

$$\beta_1 = \frac{m_3}{s^3}.$$

- If data are symmetrically distributed: $\beta_1 \approx 0$
- If there is *positive* asymmetry (tail to the right): $\beta_1 > 0$
- For *negative* asymmetry (left tail): $\beta_1 < 0$

These are illustrated in Figure 1.

Figure 1: Pearson's symmetry coefficient for three examples



Note that the converse is not always true:
$\beta_1 = 0$ does not necessarily imply that data are symmetric as is demonstrated in the following example.

Example 1.1. Consider X to be a discrete random variable with image $\mathrm{Im}(X) = \{-2, -1, 1, 3\}$ and probability mass function

$$p_X(-2) = \frac{1}{30}, \quad p_X(-1) = \frac{11}{24}, \quad p_X(1) = \frac{1}{2}, \quad p_X(3) = \frac{1}{120}.$$

For this example the first and third moments are zero, and the fifth moment is positive:

$$\mathbb{E}[X] = -2p_X(-2) - p_X(-1) + p_X(1) + 3p_X(3) = \frac{-2}{30} - \frac{11}{24} + \frac{1}{2} + \frac{3}{120} = 0.$$
$$\mathbb{E}[X^3] = -2^3p_X(-2) - p_X(-1) + p_X(1) + 3^3p_X(3) = \frac{-2^3}{30} - \frac{11}{24} + \frac{1}{2} + \frac{3^3}{120} = 0.$$
$$\mathbb{E}[X^5] = -2^5p_X(-2) - p_X(-1) + p_X(1) + 3^5p_X(3) = \frac{-2^5}{30} - \frac{11}{24} + \frac{1}{2} + \frac{3^5}{120} = 1.$$

Since the expected value of X is zero the variance is given by the second moment

$$\mathbb{V}\mathrm{ar}(X) = \mathbb{E}[X^2] = 2^2p_X(-2) + p_X(-1) + p_X(1) + 3^2p_X(3) = \frac{4}{30} + \frac{11}{24} + \frac{1}{2} + \frac{9}{120} = \frac{7}{6}.$$

Therefore

$$\frac{\mathbb{E}[(X - \mathbb{E}[X])^3}{\sigma^3} = 0.$$

Now if we have a sample of n realisations from X then β_1 will approach 0 as n gets large. However the distribution of X is not symmetric.

Example 1.2. The number of car accidents per day that occurred during November in a particular area are given in Table 1.
Calculate the coefficient of symmetry β_1.

Accidents per day	0	1	2	3	4	5
Days (frequency)	2	4	5	9	7	3

Table 1: Car accidents per day

Solution: To calculate β we need μ_3 and s. The simplest way to approach this is to extend the table as follows:

Here we are given a set of possible outcomes $y_1, \ldots, y_m$ (in this case $m = 6$ and $(y_1, \ldots, y_m) = (0, 1, 2, 3, 4, 5)$) and the frequency f_i of how many times y_i was observed.

							Total	Notes
Accidents per day	0	1	2	3	4	5		y_i
Days (frequency)	2	4	5	9	7	3	30	f_i
Accidents per day $\times$ freq.	0	4	10	18	28	15	84	$y_i f_i$
(Accidents per day)$^2\times$ freq.	0	4	20	54	112	75	292	$y_i^2 f_i$
(Accidents per day)$^3\times$ freq.	0	4	40	162	448	375	1110	$y_i^3 f_i$

Table 2: Car accidents per day

From here we take the totals of each row to determine

$$\begin{aligned}
n &= \sum f_i = 30 \\
\bar{x} &= \frac{1}{\sum f_i} \sum y_i f_i = \frac{84}{30} = 2.8 \\
\nu_2 &= \frac{1}{n} \sum y_i^2 f_i = \frac{292}{30} \\
s^2 &= \frac{n}{n-1}(\nu_2 - \bar{x}^2) = \frac{30}{29}\left(\frac{292}{30} - 2.8^2\right) \approx 1.959 \\
\nu_3 &= \frac{1}{n} \sum y_i^3 f_i = \frac{1110}{30} = 37 \\
m_3 &= \nu_3 - 3\bar{x}\nu_2 + 3\bar{x}^2\nu_1 - \bar{x}^3 \approx -0.856.
\end{aligned}$$

Therefore

$$\beta_1 \approx \frac{-0.856}{1.959^{3/2}} = -0.312.$$

We can compare this with the bar plot given in Figure 2, where we see there is a small tail to the left so there is a negatively asymmetry.

1.4 Pearson's kurtosis coefficient

The kurtosis of a distribution gives a measure of how *peaked* the distribution is, relative to a normal distribution.

It is defined as

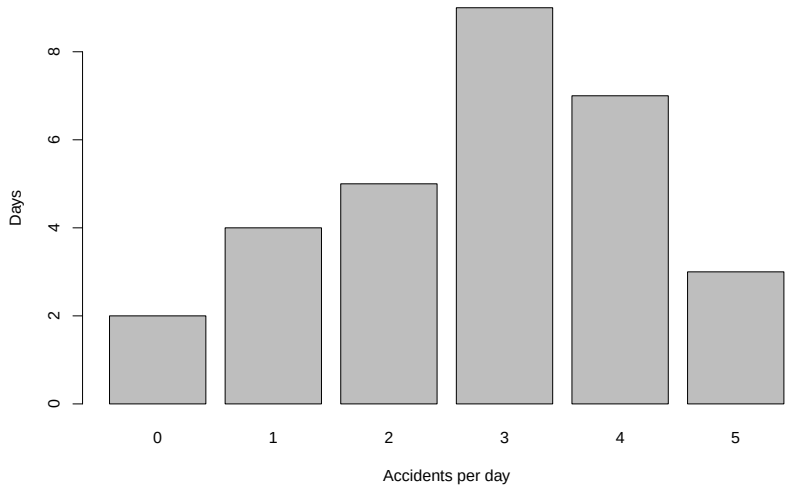
$$\gamma_2 = \frac{m_4}{s^4} - 3$$

This is an approximation of

$$\mathbb{E}\left[\left(\frac{X - \mathbb{E}[X]}{\sqrt{\mathbb{V}\mathrm{ar}(X)}}\right)^4\right] - 3.$$

Note the 3 here is the fourth moment of a standard normal random variable, i.e. $\mathbb{E}[Z^4] = 3$ for $Z \sim \mathrm{N}(0, 1)$.

Figure 2: Barplot of accidents per day



If $\gamma_2 = 0$ the data frequency distribution has the same kurtosis as a normal distribution; if $\gamma_2 > 0$ the distribution is more peaked than the normal; and if $\gamma_2 < 0$ it is less peaked. These are illustrated in Figure 3

Again, this measure must be used with caution, as the converse arguments do not necessarily hold.

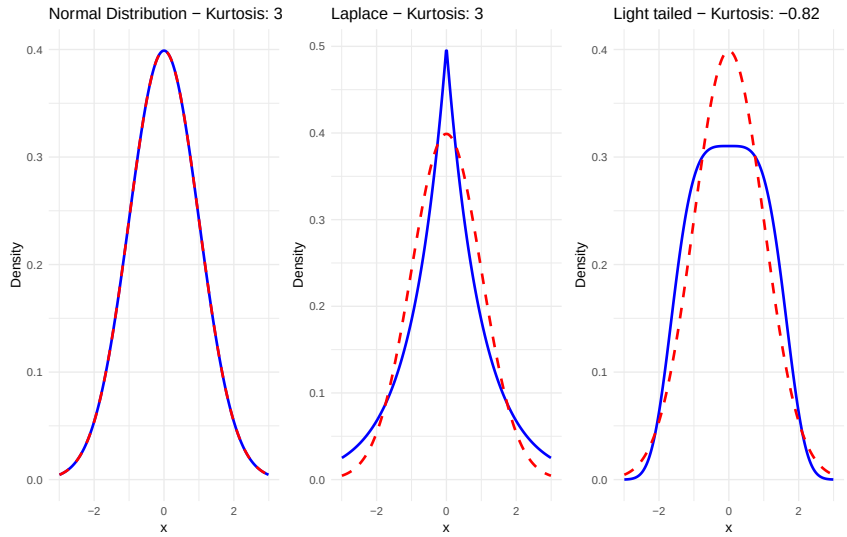


Figure 3: An illustration of 0 kurtosis, positive, and negative kurtosis. The red dashed line is the probability density function for a standard normal random variable.

1.5 Moment Generating Functions

We now develop the applications of generating functions in probability theory. We begin by first considering the type of question that generating functions can sometimes help us to answer.
A fundamental question in probability and statistical science:

Suppose $X_1, X_2, X_3, \ldots$ is a sequence of i.i.d. (independent and identically distributed) random variables.

Question: What is the distribution of $S_n = X_1 + X_2 + \cdots + X_n$?

or, alternatively,

Question: What is the distribution of $\bar{X} = \frac{S_n}{n} = \frac{X_1 + \cdots + X_n}{n}$

One approach: One approach to answering this question which is based on the ideas that we developed in F78PA. Suppose that for $k = 1, 2, ..., n$, $f_{X_k}(x_k)$ is the density for X_k. Then, since the variables $X_1, X_2, ..., X_n$ are independent, the joint density for $(X_1, X_2, \ldots, X_n)$ is given by

$$f_{X_1 \ldots X_n}(x_1, \ldots, x_n) = f_{X_1}(x_1) f_{X_2}(x_2) \ldots f_{X_n}(x_n).$$

So, we can find the distribution function $F_{S_n}(s)$ for the sum $S_n = X_1 + X_2 + \cdots + X_n$ by integrating the joint density function over an appropriate region:

$$F_{S_n}(s) = \mathbb{P}(S_n \leqslant s) = \mathbb{P}(X_1 + \cdots + X_n \leqslant s) = \int \cdots \int_{\{x_1 + \cdots + x_n \leqslant s\}} f_{X_1}(x_1) \ldots f_{X_n}(x_n) \, \mathrm{d}x_1 \ldots \mathrm{d}x_n.$$

Remark: The integral on the right-hand side of the equation above is an n-dimensional integral and it is rather messy and tedious to compute.

Alternative approach: In this section we will show that sometimes there is an easier way, based on generating function techniques, to determine the distribution of S_n.

Remark: Generating function arguments are examples of transform methods (e.g. Laplace transforms, Fourier transforms, etc) that are used in many branches of mathematics. It is important to remember that when we use generating functions, it is not the generating function *itself* that is interesting, but rather the way in which we use it to 'transform' our problem.

We start by defining the moment generating function of a random variable.

Definition 1.1. Suppose X is a random variable, then the moment generating function M_X for X is defined by

$$M_X(\theta) = \mathbb{E}(e^{\theta X}) \quad \forall \theta \in \mathbb{R} \text{ for which this expectation exists.}$$

Note: It follows from the definition above, that for any random variable X, we always have $M_X(0) = 1$.

Remark 1.1. Recall from F78PA Probability and Statistics A that we can calculate expectations of functions using probability mass functions (for discrete variables) or probability density functions (for continuous random variables). Therefore

$$M_X(\theta) = \sum_{x \in \mathrm{Im}(X)} e^{\theta x} p_X(x) \qquad \text{for } X \text{ discrete random variable}$$

$$M_X(\theta) = \int_{-\infty}^{\infty} e^{\theta x} f_X(x) \, dx \qquad \text{for } X \text{ continuous random variable.}$$

Example 1.3. Suppose that $X \sim \mathrm{Exp}(\lambda), \quad \lambda > 0$. Then we compute the moment generating

function for X as follows:

$$\begin{aligned} M_X(\theta) = \mathbb{E}(e^{\theta X}) &= \int_0^\infty e^{\theta x} f_X(x) \mathrm{d}x \\ &= \int_0^\infty e^{\theta x} (\lambda e^{-\lambda x}) \mathrm{d}x \\ &= \int_0^\infty \lambda e^{(\theta-\lambda)x} \mathrm{d}x \qquad \text{Note: This integral is finite only if } \theta < \lambda \\ &= \frac{\lambda}{\lambda - \theta} \underbrace{\int_0^\infty (\lambda - \theta) e^{(\theta-\lambda)x} \mathrm{d}x}_{=1} \\ &= \frac{\lambda}{\lambda - \theta} \end{aligned}$$

Remark: Since the integral above is finite only if $\theta - \lambda < 0$, i.e. $\theta < \lambda$, the moment generating function $M_X(\theta)$ is only defined for $\theta < \lambda$.

Example 1.4. Suppose that $X \sim \mathrm{N}(0,1)$ is a standard normal random variable. Then we have

$$\begin{aligned} M_X(\theta) = \mathbb{E}(e^{\theta X}) &= \int_{-\infty}^\infty \frac{e^{\theta x} e^{-\frac{x^2}{2}}}{\sqrt{2\pi}} \mathrm{d}x \\ &= \int_{-\infty}^\infty \frac{e^{-\frac{x^2}{2} + \theta x}}{\sqrt{2\pi}} \mathrm{d}x \\ &= \int_{-\infty}^\infty e^{\frac{\theta^2}{2}} \frac{e^{-\frac{1}{2}(x-\theta)^2}}{\sqrt{2\pi}} \mathrm{d}x \qquad \text{(completing the square in exponent)} \\ &= e^{\frac{\theta^2}{2}} \int_{-\infty}^\infty \underbrace{\frac{e^{-\frac{1}{2}(x-\theta)^2}}{\sqrt{2\pi}}}_{\text{density for } \mathrm{N}(\theta,1)} \mathrm{d}x \\ &= e^{\frac{\theta^2}{2}}. \end{aligned}$$

Remark: The moment generating function $M_X(\theta)$ for $X \sim \mathrm{N}(0,1)$ exists for *all* values of θ.

Example 1.5. Suppose that $X \sim \mathrm{Gamma}(\alpha, \lambda)$ is a Gamma random variable, then for $\theta < \lambda$, the moment generating function for X is given by

$$M_X(\theta) = \left(\frac{\lambda}{\lambda - \theta}\right)^\alpha$$

Proof: This is an exercise on Tutorial Sheet 1.

Check the file "Distributions" in the "Useful Formula Sheets" module on Canvas for more examples.

1.5.1 Properties of Moment Generating Functions

As the name suggests we can use the moment generating function to determine the moments of the distribution. We will only give an informal explanation to justify this. There are two approaches we will consider one based on differentiating the moment generating function and the other based on a series expansion.

In both cases we will make the standing assumption that the moment generating function exists in some neighbourhood of 0 (recall that the moment generating function is defined at 0 since $M_X(0) = 1$ but need not be defined for all θ). A neighbourhood of zero is an interval $(-\delta, \delta)$ for some $\delta > 0$.

Derivative Approach Consider the derivative of the moment generating function (assuming that we can exchange the derivative and the expectation)

$$\frac{d}{d\theta}M_X(\theta) = \frac{d}{d\theta}\mathbb{E}[e^{\theta X}] = \mathbb{E}\left[\frac{d}{d\theta}e^{\theta X}\right] = \mathbb{E}[Xe^{\theta X}]$$

Now setting $\theta = 0$ we have

$$\frac{d}{d\theta}M_X(\theta)|_{\theta=0} = M_X'(0) = \mathbb{E}[X]$$

Repeating this with k derivatives

$$\frac{d^k}{d\theta^k}M_X(\theta)|_{\theta=0} = M_X^{(k)}(0) = \mathbb{E}[X^k]$$

An implication of this is to find the k-th moment of X, calculate the k-th derivative of $M_X(\theta)$ then set $\theta = 0$.

Series Approach We can also use series expansions to calculate moments. To do this we need to recall the exponential series

$$e^x = \sum_{k=0}^{\infty}\frac{x^k}{k!}$$

and Taylor series for a smooth function g about the point x_0

$$g(x) = \sum_{k=0}^{\infty} g^{(k)}(x_0)\frac{(x-x_0)^k}{k!}.$$

Using the exponential series with $x = \theta X$ we can write the moment generating function as

$$M_X(\theta) = \mathbb{E}\left[e^{\theta X}\right] = \mathbb{E}\left[\sum_{k=0}^{\infty}\frac{\theta^k X^k}{k!}\right] = \sum_{k=0}^{\infty}\frac{\theta^k}{k!}\mathbb{E}\left[X^k\right].$$

If we compare this to the Taylor series of M_X about the point $\theta = 0$ (also known as the Maclaurin series of M_X) which is

$$M_X(\theta) = \sum_{k=0}^{\infty} M_X^{(k)}(0)\frac{\theta^k}{k!}.$$

Now if we equate the terms of these two expansions we see that

$$M_X^{(k)}(0) = \mathbb{E}\left[X^k\right].$$

In order to prove these results rigorously we would need to justify that we can swap the derivative (or infinite series) and expectation.

Uniqueness Another property of moment generating properties we will discuss is *uniqueness*. Suppose two random variables X and Y have moment generating functions M_X and M_Y with

$$M_X(\theta) = M_Y(\theta) \quad \text{for all } \theta \text{ in a neighbourhood of } 0.$$

Then we have that X and Y both have the same distribution (note that this does not imply that X and Y are equal). The proof of this beyond the scope of this course (it is a consequence of the Laplace Inversion Theorem).

These properties are summarised in the following theorem:

Theorem 1.1. *Suppose that X is a random variable and that the moment generating function $M_X(\theta) = \mathbb{E}(e^{\theta X})$ exists for all θ in a neighbourhood of 0 (i.e. $\exists \delta > 0$ such that the moment generating function $M_X(\theta)$ exists for all $-\delta < \theta < \delta$). Then:*

1. *All the moments of X exist, i.e. $\mathbb{E}(X^k)$ exists and is finite for all $k \geq 1$.*
2. *If $M_X(\theta)$ exists in a neighbourhood of 0, then for $k = 1, 2, \ldots$*

$$\mathbb{E}(X^k) = M_X^{(k)}(0),$$

where $M_X^{(k)}(0)$ = k^{th} derivative of $M_X(\theta)$ evaluated at $\theta = 0$.

3. *$M_X(\theta)$ has a power series expansion:*

$$M_X(\theta) = \sum_{k=0}^{\infty} \frac{\mathbb{E}(X^k)\theta^k}{k!} \qquad \text{for } -\delta < \theta < \delta.$$

4. **Uniqueness** *(NB This property is very important):*
If X and Y are random variable such that $M_Y(\theta) = M_X(\theta)$ for all $-\delta < \theta < \delta$, then the random variables X and Y have the same distribution.

Other important properties

1. Suppose that $Y = aX + b$, where X is a random variable with moment generating function and a, b are constants. Then $M_Y(\theta) = e^{b\theta} M_X(a\theta)$.

   Proof.

$$M_Y(\theta) = \mathbb{E}(e^{\theta Y}) = \mathbb{E}(e^{\theta(aX+b)}) = \mathbb{E}(e^{\theta b} e^{a\theta X}) = e^{b\theta}\mathbb{E}(e^{a\theta X}) = e^{b\theta} M_X(a\theta).$$

   □

2. Suppose that $X_1, X_2, \ldots, X_n$ are independent random variables with moment generating functions $M_{X_1}(\theta), M_{X_2}(\theta), \ldots, M_{X_n}(\theta)$ respectively, and let $Z = X_1 + \cdots + X_n$.
   Then $M_Z(\theta) = M_{X_1}(\theta) M_{X_2}(\theta) \cdots M_{X_n}(\theta)$.

   Proof.

$$\begin{aligned}
M_Z(\theta) &= \mathbb{E}(e^{\theta Z}) \\
&= \mathbb{E}(e^{\theta(X_1 + \cdots + X_n)}) \\
&= \mathbb{E}(e^{\theta X_1} e^{\theta X_2} \ldots e^{\theta X_n}) && \text{(by algebra)} \\
&= \mathbb{E}(e^{\theta X_1})\mathbb{E}(e^{\theta X_2}) \ldots \mathbb{E}(e^{\theta X_n}) && \text{(since } e^{\theta X_1}, \ldots, e^{\theta X_n} \text{ are independent!)} \\
&= M_{X_1}(\theta) M_{X_2}(\theta) \ldots M_{X_n}(\theta). && \text{(by definition of } M_X(\theta))
\end{aligned}$$

   □

Examples and applications

1. Suppose that X is an exponential random variable with parameter $\lambda > 0$ (i.e. $X \sim \text{Exp}(\lambda)$), then $M_X(\theta) = \frac{\lambda}{\lambda - \theta}$ for $\theta < \lambda$ (see above) and we can use the Corollary above to compute $\mathbb{E}(X^3)$:

$$\mathbb{E}(X^3) = M_X^{(3)}(0) = \frac{\mathrm{d}^3}{\mathrm{d}\theta^3}\left(\frac{\lambda}{\lambda - \theta}\right)\Big|_{\theta=0} = 3 \cdot 2 \cdot 1 \cdot \lambda \cdot (\lambda - \theta)^{-4}\big|_{\theta=0} = \frac{3!}{\lambda^3}$$

Alternatively, we can expand the moment generating function $M_X(\theta)$ as a power series and *equate coefficients* (using Theorem 1.1) to determine $\mathbb{E}(X^3)$:
We begin by noting that

$$M_X(\theta) = \frac{\lambda}{\lambda - \theta} = \frac{1}{1 - \frac{\theta}{\lambda}} = 1 + \frac{1}{\lambda}\theta + \frac{1}{\lambda^2}\theta^2 + \frac{1}{\lambda^3}\theta^3 + \ldots$$

Next, by using Theorem 1.1, we also have

$$M_X(\theta) = 1 + \mathbb{E}(X)\theta + \frac{\mathbb{E}(X^2)}{2!}\theta^2 + \frac{\mathbb{E}(X^3)}{3!}\theta^3 + \ldots$$

So, equating the θ^3 terms in the two series representations for $M_X(\theta)$ given above, we have

$$\frac{\mathbb{E}(X^3)\theta^3}{3!} = \frac{\theta^3}{\lambda^3} \quad \text{and so} \quad \mathbb{E}(X^3) = \frac{3!}{\lambda^3}.$$

2. Suppose that Y is a $\text{N}(4, 4)$ random variable with mean $\mu_Y = 4$ and variance $\sigma_Y^2 = 4$.
We know (see F78PA notes) that Y has the *same distribution* as $2Z + 4$, where Z is a $\text{N}(0, 1)$ random variable. Therefore, by the *uniqueness* of moment generating functions and using Important Property 1 from above, we have

$$M_Y(\theta) = M_{2Z+4}(\theta) \overset{\substack{\text{Important} \\ \text{Prop. 1}}}{=} e^{4\theta} M_Z(2\theta) = e^{4\theta} e^{\frac{4\theta^2}{2}} = e^{4\theta} e^{2\theta^2}$$

since $M_Z(\theta) = e^{\frac{\theta^2}{2}}$ (see above). **Note:** We can generalise this argument to obtain the following:

General result:
Suppose that Y is a $\text{N}\left(\mu, \sigma^2\right)$ random variable, then Y has the *same distribution* as $\sigma Z + \mu$ where Z is a $\text{N}(0, 1)$ random variable. So, by the *uniqueness* of moment generating functions and Important Property 1, we have

$$M_Y(\theta) = M_{\sigma Z+\mu}(\theta) = e^{\mu\theta} M_Z(\sigma\theta) = e^{\mu\theta} e^{\frac{\sigma^2\theta^2}{2}}$$

3. Suppose $X \sim \text{N}\left(\mu_X, \sigma_X^2\right)$ and $Y \sim \text{N}\left(\mu_Y, \sigma_Y^2\right)$ are independent normal random variables. Then the moment generating function of $X + Y$ is given by

$$M_{X+Y}(\theta) = M_X(\theta) M_Y(\theta) = e^{\mu_X + \frac{1}{2}\sigma_X^2\theta} e^{\mu_Y + \frac{1}{2}\sigma_Y^2\theta} = e^{\mu_X + \mu_Y + \frac{1}{2}(\sigma_X^2 + \sigma_Y^2)\theta}.$$

As this is equal to the moment generating function of a $\text{N}\left(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2\right)$ random variable we have by the uniqueness of moment generating functions

$$X + Y \sim \text{N}\left(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2\right).$$

4. Suppose $X_1, X_2, \ldots, X_n$ are independent, identically distributed $\text{Exp}(\lambda)$ random variables with $M_{X_k}(\theta) = \frac{\lambda}{\lambda-\theta}$ for $k = 1, 2, ..., n$. Let $Z = X_1 + X_2 + \cdots + X_n$. We can use moment generating functions to determine the distribution of the sum Z as follows:

   First, we note that by the independence of the variables $X_1, X_2, ..., X_n$ and Important Property 2, we have

$$M_Z(\theta) = M_{X_1}(\theta) \ldots M_{X_n}(\theta) = \left(\frac{\lambda}{\lambda-\theta}\right)\left(\frac{\lambda}{\lambda-\theta}\right)\cdots\left(\frac{\lambda}{\lambda-\theta}\right) = \underbrace{\left(\frac{\lambda}{\lambda-\theta}\right)^n}_{\text{mgf of Gamma}(n,\lambda)}$$

   Next we note that the right-hand side of the equation above is the moment generating function for a Gamma (n, λ) distribution. So, since $M_Z(\theta)$ the moment generating function for a Gamma (n, λ) distribution, it follows from the *uniqueness* of moment generating functions that the sum $Z + X_1 + X_2 + \cdots X_n$ must have a Gamma (n, λ) distribution.

1.6 Probability generating functions

If X is a random variable which takes non-negative integer values then it is sometimes useful to work with the *probability generating function* for X (rather than the moment generating function for X).

Definition 1.2. Suppose that X is a random variable which takes non-negative integer values. Then the **probability generating function** G_X for X is defined as

$$G_X(s) = \mathbb{E}(s^X) = \sum_{n=0}^{\infty} \mathbb{P}(X = n)s^n.$$

Remark 1.2. Probability generating functions arise in the study of random walks and branching processes.

Examples

1. Suppose that X is a Binomial, $\text{Bin}(n, p)$, random variable. Then we can use the Binomial Theorem to compute the probability generating function G_X for X:

$$\begin{aligned} G_X(s) = \sum_{k=0}^{n} \mathbb{P}(X = k)s^k &= \sum_{k=0}^{n} \binom{n}{k} p^k q^{(n-k)} s^k && \text{(letting } q = 1 - p\text{)} \\ &= (ps + q)^n && \text{(by the Binomial Theorem)} \end{aligned}$$

2. Suppose that $Y \sim \text{Geo}\,(p)$ is a geometric random variable with $\mathbb{P}(Y = n) = q^{n-1}p$ for $n = 1, 2, \ldots$, where $q = 1 - p$. Then using the summation formula for geometric series we can compute G_Y, the probability generating function for Y, as follows:

$$\begin{aligned} G_Y(s) = \sum_{n=1}^{\infty} \mathbb{P}(Y = n)s^n = \sum_{n=1}^{\infty} q^{n-1}ps^n &= ps + qps^2 + q^2ps^3 + \ldots \\ &= ps(1 + qs + (qs)^2 + \ldots) \\ &= \frac{ps}{1 - qs} && \text{(by summing the geometric series)} \end{aligned}$$

Important properties of probability generating functions

Suppose X is a discrete random variable with probability generating function $G_X(s) = \sum_{n=0}^{\infty} \mathbb{P}(X = n)s^n = \mathbb{E}(s^X)$. Then

1. $G_X(0) = \mathbb{P}(X = 0)$
2. $G_X(1) = 1$
3. $G_X^{(m)}(0) = m!\,\mathbb{P}(X = m)$
4. $G_X^{(m)}(1) = \mathbb{E}\left(X(X-1)\dots(X-m+1)\right)$, where $G_X^{(m)}(s)$ denotes the mth derivative of $G_X(s)$.

   Proof. Since the probability generating function $G_X(s)$ can be written as a power series, we can differentiate $G_X(s) = \sum_{n=0}^{\infty} \mathbb{P}(X = n)s^n$ m times by differentiating each term in the power series for $G_X(s)$ m times. This gives us a power series for $G_X^{(m)}(s)$:

   $$G_X^{(m)}(s) = \sum_{n=m}^{\infty} n(n-1)(n-2)\cdots(n-m+1)\mathbb{P}(X = n)s^{n-m}$$

   To obtain the result, evaluate the power series for at $G_X^{(m)}(s)$ at $s = 1$.

   □

   Special cases:

   $$G_X'(1) = \mathbb{E}(X)$$
   $$G_X''(1) = \mathbb{E}\left(X(X-1)\right) = \mathbb{E}(X^2) - \mathbb{E}(X)$$

   Note: It follows from these two facts that $\mathbb{V}\mathrm{ar}(X) = G_X''(1) + G_X'(1) - \left(G_X'(1)\right)^2$

5. Suppose that $X_1, X_2, \dots, X_n$ are independent random variables with probability generating functions $G_{X_1}(s), G_{X_2}(s), \dots, G_{X_n}(s)$.
   Let $Z = X_1 + \cdots + X_n$, then

   $$G_Z(s) = G_{X_1}(s) \cdot \cdots \cdot G_{X_n}(s)$$

   Proof. Exercise.
   Hint: The proof is very similar to the proof of the analogous result for moment generating functions.

   □

6. **Uniqueness**
   Suppose that the random variables X and Y have probability generating functions $G_X(s)$ and $G_Y(s)$ and that $G_X(s) = G_Y(s)$ for all s. Then X and Y have the *same distribution.*

Application: Compound Random Variables

Definition 1.3. Suppose that N is a counting random variable (i.e. $\text{Im}(N) \subseteq \{0, 1, 2, \dots\}$) and that $X_1, X_2, \dots$ is a sequence of independent, identically distributed random variables which are also *independent* of the variable N. Then we call the random sum

$$S_N = X_1 + X_2 + \cdots + X_N$$

a **compound random variable**.

Remarks

1. In a compound variable there are two sources of randomness. The terms in the sum are random variables *and* the *number* of terms in the sum, N, is also random.
2. Compound random variables arise in actuarial applications (e.g. risk theory).
   Example: Suppose that N equals the number of claims received by an insurance company in a year and suppose that X_i equals the size of the ith claim for $i = 1, 2, \dots$, then the compound variable $S_N = X_1 + X_2 + \cdots + X_N$ represents the total claim amount received in a year.

Proposition 1.1. *Suppose that $S_N = X_1 + X_2 + \cdots + X_N$ is a* **compound random variable**, *then the moment generating function $M_{S_N}(\theta)$ for S_N is given by*

$$M_{S_N}(\theta) = \mathbb{E}\big(e^{\theta S_N}\big) = G_N\left(M_X(\theta)\right)$$

where $G_N(s)$ is the probability generating function of N and $M_X(\theta)$ is the moment generating function for the variables $\{X_1, X_2, ...\}$.

Proof.

$$\begin{aligned}
M_{S_N}(\theta) &= \mathbb{E}\left(e^{\theta S_N}\right) \\
&= \sum_{n=0}^{\infty} \mathbb{E}\left(e^{\theta S_N} | N = n\right) \mathbb{P}(N = n) && \left(\begin{smallmatrix}\text{Partition Theorem}\\ \text{for Expectation -see Tut 8}\end{smallmatrix}\right) \\
&= \mathbb{P}(N = 0) + \sum_{n=1}^{\infty} \mathbb{E}\left(e^{\theta S_n} | N = n\right) \mathbb{P}(N = n) && \left(\begin{smallmatrix}\text{since } \mathbb{E}(e^{\theta S_n}|N=0)=\\ =\mathbb{E}(e^0|N=0)=1\end{smallmatrix}\right) \\
&= \mathbb{P}(N = 0) + \sum_{n=1}^{\infty} \mathbb{E}\left(e^{\theta(X_1+\cdots+X_n)} | N = n\right) \mathbb{P}(N = n) = && \left(\begin{smallmatrix}\text{since } S_N = X_1 + \cdots + X_n\\ \text{when } N = n\end{smallmatrix}\right) \\
&= \mathbb{P}(N = 0) + \sum_{n=1}^{\infty} \mathbb{E}\left(e^{\theta(X_1+\cdots+X_n)}\right) \mathbb{P}(N = n) && \left(\begin{smallmatrix}\text{since } X\text{'s and } N\\ \text{are independent}\end{smallmatrix}\right) \\
&= \mathbb{P}(N = 0) + \sum_{n=1}^{\infty} \mathbb{E}\left(e^{\theta X_1}\right) \cdots \mathbb{E}\left(e^{\theta X_n}\right) \mathbb{P}(N = n) && \left(\begin{smallmatrix}\text{since the } X\text{'s}\\ \text{are independent}\end{smallmatrix}\right) \\
&= \mathbb{P}(N = 0) + \sum_{n=1}^{\infty} \left(\underbrace{\mathbb{E}(e^{\theta X})}_{M_X(\theta)}\right)^n \mathbb{P}(N = n) && \left(\begin{smallmatrix}\text{since the } X\text{'s are}\\ \underline{\text{identically distributed}}\end{smallmatrix}\right) \\
&= \sum_{n=0}^{\infty} \mathbb{P}(N = n) \left(M_X(\theta)\right)^n && \left(\begin{smallmatrix}\text{since}\\ M_X(\theta)=\mathbb{E}(e^{\theta X})\end{smallmatrix}\right) \\
&= G_N\left(M_X(\theta)\right) && \left(\begin{smallmatrix}\text{since}\\ G_N(s)=\sum_{n=0}^{\infty}\mathbb{P}(N=n)s^n\end{smallmatrix}\right)
\end{aligned}$$

□

Example 1.6. Julie and Anne like to play chess. Each game that they play lasts an exponential amount of time (independent of all other games) and the expected length of a game is 1.5 hours. Also, each time that they play, independent of all other games, $\mathbb{P}(\text{Julie wins}) = \frac{1}{4}$.

Let N equal the number of games played until Julie wins her first game (so N is a Geo $\left(\frac{1}{4}\right)$ random variable). For $i = 1, 2, ...$, let X_i equal the length of the ith game played by Anne and Julie. Then the compound variable $S_N = X_1 + \cdots + X_N$ equals the length of the time that Julie and Anne play until Julie's first win.

Question: What is the distribution of S_N?

Strategy: To determine the distribution of S_N, we will determine the moment generating function $M_{S_N}(\theta)$ and see whether $M_{S_N}(\theta)$ corresponds to a known moment generating function. To use the Proposition above to determine M_{S_N}, we need to determine the probability generating function $G_N(s)$ and the moment generating function $M_X(\theta)$ for X, the length of a game.:

$$G_N(s) = \frac{\frac{1}{4} \cdot s}{1 - \frac{3}{4} \cdot s} \qquad \text{(from above, since } N \sim \text{Geo}\left(\frac{1}{4}\right)\text{)}$$

$$M_X(\theta) = \frac{\frac{2}{3}}{\frac{2}{3} - \theta} \qquad \text{(since } X \text{ is exponential with } \lambda = \frac{1}{\mathbb{E}(X)} = \frac{2}{3}\text{)}$$

So, by the Proposition above, we have:

$$M_{S_N}(\theta) = G_N\left(M_X(\theta)\right) = \frac{\frac{1}{4} \cdot \left(\frac{\frac{2}{3}}{\frac{2}{3} - \theta}\right)}{1 - \frac{3}{4} \cdot \left(\frac{\frac{2}{3}}{\frac{2}{3} - \theta}\right)} = \frac{\frac{1}{6}}{\frac{1}{6} - \theta} \quad \longleftarrow \quad \text{mgf for Exp}(\frac{1}{6})$$

Since the moment generating function for S_N is the same as the moment generating function for an Exp$(\frac{1}{6})$ distribution, it follows by the *uniqueness* of moment generating functions that S_N has an Exp$(\frac{1}{6})$ distribution with $\mathbb{E}(S_N) = 6$ hours.

1.7 Cumulant generating functions

A further generating function that can be useful is the *cumulant generating function*. This is defined in terms of the moment generating function and is obtained by taking its natural logarithm.

Definition 1.4. Suppose X is a random variable, then the *cumulant generating function* K_X for X is defined by

$$K_X(\theta) = \log(\mathbb{E}(e^{\theta X})) = \log(M_X(\theta)) \quad \forall \theta \in \mathbb{R} \text{ for which this expectation exists.}$$

Note: It follows from the definition above that, since for any random variable X have $M_X(0) = 1$, then $K_X(0) = 0$.

The *cumulants* $\kappa_i, i = 1, 2, \ldots$ of X are defined by expressing the power series expansion of $K_X(\theta)$ as

$$K_X(\theta) = \sum_{j=1}^{\infty} \frac{\kappa_j \theta^j}{j!} = \kappa_1 \theta + \frac{\kappa_2 \theta^2}{2!} + \frac{\kappa_3 \theta^3}{3!} + \ldots.$$

Therefore we can obtain the nth cumulant κ_n by differentiating n times and setting $\theta = 0$:

$$\kappa_n = K_X^{(n)}(0).$$

It can be shown that $\kappa_1 = \mathbb{E}(X)$, $\kappa_2 = \mathbb{V}\text{ar}(X)$. To do this we express

$$M_X(\theta) = 1 + \sum_{n=1}^{\infty} \frac{\mu_n \theta^n}{n!},$$

where $\mu_n = \mathbb{E}(X^n)$, $n = 1, 2, 3, ...$ Now, since $K_X(\theta) = \log M_X(\theta)$ we can use the Maclaurin series, $\log(1+y) = y - \frac{y^2}{2} + \frac{y^3}{3} - \frac{y^3}{4} \ldots ...$, to write

$$K_X(\theta) = \sum_{n=1}^{\infty} \frac{\mu_n \theta^n}{n!} - \frac{1}{2}\left(\sum_{n=1}^{\infty} \frac{\mu_n \theta^n}{n!}\right)^2 + \frac{1}{3}\left(\sum_{n=1}^{\infty} \frac{\mu_n \theta^n}{n!}\right)^3 \ldots.$$

Re-writing this expression as a power series in θ, we obtain

$$K_X(\theta) = \mu_1\theta + \frac{1}{2!}(\mu_2 - \mu_1^2)\theta^2 + \frac{1}{3!}(\mu_3 - 3\mu_1\mu_2 + 2\mu_1^3)\theta^3 \ldots$$

We can now see that the first three cumulants are given by

$$\kappa_1 = \mu_1 = \mathbb{E}(X),$$

$$\kappa_2 = \mu_2 - \mu_1^2 = \mathbb{V}\text{ar}(X),$$

$$\kappa_3 = \mu_3 - 3\mu_1\mu_2 + 2\mu_1^3.$$

Higher-order cumulants are expressed in terms of quite complicated expressions involving the moments of X and are not so frequently used. However, note that for $X \sim \text{N}\left(\mu, \sigma^2\right)$, it is easy to show that

$$K_X(\theta) = \mu\theta + \frac{\sigma^2\theta^2}{2}.$$

This means that for the normal distribution, $\kappa_n = 0$ for all $n \geq 3$. Thus for a given distribution the cumulants (beyond order 2) characterise how different the distribution is to the normal distribution. For example κ_3 captures the *skewness* of the distribution.

An important property of cumulants is that they are *additive* when we have sums of independent random variables. Suppose that $X_1, ..., X_n$ are independent random variables and let $Y = \sum_{i=1}^n X_i$. If $\kappa_{i,Y}$ denotes the ith cumulant of Y, then

$$\kappa_{i,Y} = \sum_{j=1}^{n} \kappa_{i,X_j}.$$

The proof of this is left as an exercise (but follows from the fact that $K_Y(\theta) = \sum_{j=1}^n K_{X_j}(\theta)$).

2 Construction and estimation of statistical models

A *mathematical model* is a simplified representation of a real phenomenon through a *mathematically expressed* system.

This can be a *statistical model* when any uncertainty in the system is expressed through *probabilities* and *statistical distributions*.

The process through which we *estimate* unknown quantities (*parameters*) in statistical models using available information (*data*) is called *statistical inference*.

Example 2.1. Suppose that a farmer has to choose between two fertilizers (A and B). Before placing a large order for either he obtains samples of each and apply A to 5 'test' plots and B to 5 plots of a certain crop. Then at the end of the season he measures the yield of the crop in *kgs* in each plot. The results are:

Fertiliser A: 12.5, 14.4, 15.1, 11.9, 13.1

Fertiliser B: 12.6, 13.2, 10.4, 10.9, 12.8

For fertilisers A and B the average yields are 13.4 and 11.98, respectively.

Does this tell us that A is better than B and he should place a large order for A?

Some points to consider:

- Data indicate that 'yield' is something that varies from plot to plot even if the same fertilizer is used.
- If we repeated the experiment we would surely obtain different measurements and, plausibly, could come to a different conclusion regarding which was better, i.e. it is possible that B may be better!

Remark 2.1.

- Statistical inference is the process whereby we will *quantify* how strong the evidence is that A is really better than B.
  We can then decide whether to order A, or whether, perhaps, we should do further experiments before making a decision.
- Not difficult to think of all kinds of scenarios that are analogous to Example 2.1: testing effectiveness of different drugs, comparing lifetimes of different brands of a component.
  - Every day we have to make potentially important decisions in the face of limited evidence;
  - statistics gives us a set of tools for trying to do this.

2.1 The concept of probability

Probability theory is used to construct the *models* from which observations are assumed to arise (see later) but also to express the degree of certainty in the conclusions.

I.e., in Example 2.1 can we associate a 'probability' with the conclusion "A is better than B"?

Example 2.2 (Example 2.1 (cont.))**.** Recall from earlier Statistics modules the notion of confidence interval (also considered later in this module). Suppose, in the context of Example 2.1, we knew that the measurements for fertilizer B follow a $\mathrm{N}(\mu_B, 1)$ distribution, where μ is unknown.

Then we could have calculated a 95% confidence interval as

$$\left(11.98 - 1.96\frac{1}{\sqrt{5}}, 11.98 + 1.96\frac{1}{\sqrt{5}}\right)$$

i.e. (11.1, 12.8).

This is an example of a probability being used to express the degree of certainty one might have in a conclusion regarding the range of values which μ_B could plausibly take.

2.2 Interpretations of probability

In the previous example the '95%' is indeed a probability. But of what event?

To a pure *frequentist*, the only meaningful definition of probability of an event is the frequency with which it occurs over a *long sequence of independent trials.*

Thus statements such as:

- the probability that a '6' is scored when a fair die is rolled is $\frac{1}{6}$
- the probability that a 'H' is uppermost when a fair coin is tossed is $\frac{1}{2}$

would be perfectly meaningful to a frequentist because there is a clearly defined, repeatable experiment in each case.

The probability has an interpretation as a frequency.

For example

- the 'probability' that I voted Labour in the last election
- the 'probability' that Celtic will win the Scottish League this year

would *not* be recognised by a frequentist as true probabilities.

These deal with propositions that must either be *true or false.*

There is no sequence of *identical* repeatable experiments for which these probabilities are frequencies.

For the last example, there is only one Scottish Premier League in the current season. Even though the SPL is played every year, the teams and their personnel change from year to year.

However,

- these probabilities - which express the degree of belief in a proposition regarding the world and are known as *subjective probabilities* - *are* meaningful;
- for example, a bookmaker, when setting the odds of Celtic winning the SPL this year, must consider his subjective probability that they will win.

Example 2.3 (Example 2.1 and 2.2 (cont.))**.** In Example 2.2 we calculated a 95% confidence interval for μ_B to be (11.1, 12.8).

The 95% must be interpretable as the frequency of occurrence of some event over a sequence of independent trials. Which event? Which experiment?

Here the repeatable *experiment* or trial is:

Sow 5 plots with the crop, fertilise them with fertiliser B and measure the yield of each plot. (We assume that on any experiment the observations $Y_1, Y_2, ..., Y_5$ are drawn from a $\mathrm{N}(\mu_B, 1)$ where μ_B is unknown but is identical for all trials).

What is the event whose frequency in a sequence of identical experiments is 0.95?

Is it the event that $\mu_B \in (11.1, 12.8)$?

No! Since μ_B never changes then this event either has frequency 0 or 1. The event that occurs with frequency 0.95 is *that the confidence interval that we calculate will contain* μ_B.

The classical argument then goes that since there is no reason to think that our particular experiment is 'atypical' then we can be 95% confident that confidence interval (11.1, 12.8) contains the value μ_B.

2.3 Main elements of frequentist inference

The experiment

This is a procedure that can be repeated independently many times where the outcome of the experiment is subject to uncertainty.

The outcome of the experiment can be considered to be a *random variable* X, i.e. a measurement on some real-valued quantity.

Typical examples of an experiment are:

1. Select an individual randomly (with replacement) from a large population and determine whether they carry a certain gene.
2. Select a component randomly from a large population and measure the lifetime.
3. Observe the night sky and measure the time elapsed before the first meteor appears.

The statistical model

The statistical model is simply the *distribution* that we assume for the observation X.

Exercise: Suggest suitable statistical models for the experiment examples above.

The statistical model is also referred to as the *population distribution* of the measurement X.

- Repeating the 'basic' experiment several times we obtain a sequence of measurements $X_1, X_2, ..., X_n$.
- When we have *independence* of the measurements we refer to them as a *random sample of size n.* This is equivalent to $X_1, X_2, ..., X_n$ being i.i.d. with probability function f_X.
- With independence, the multivariate probability density function or probability mass function of $\underline{X} = (X_1, X_2, ..., X_n)$ is

$$f_{\underline{X}}(x_1, x_2, ..., x_n) = f_X(x_1) \times f_X(x_2) \times \ldots \times f_X(x_n)$$

Model parameters

- Usually, although we might be sure of the family from which the distribution of X comes, we don't know the distribution's parameters.
- The main point of doing the experiment is (usually) to find out more about the parameters or some other fixed characteristic of the population distribution such as its *mean* or *variance*.
- We call this process *estimation*.
  In Example 2.1 we wanted to estimate $\mu_A - \mu_B$ to help us choose between the 2 fertilizers.

Exercise: Make sure you know the mean and variance of the Gamma, Exponential, Normal, Binomial, and Poisson distributions in terms of their parameters.

Remark 2.2. So far we have used notation such as $\mathbb{P}(X \leq x), f_X(x), p_X(x)$ for probabilities, probability density functions and probability mass functions. Since we will often in statistics be interested in also the parameters of the distribution we will include this in the notation as $\mathbb{P}(X \leq x; \theta), f_X(x; \theta), p_X(x; \theta)$ which denotes that the random variable X corresponds to the parameter θ.

Estimation

We carry out an experiment that yields a random sample $X_1, ..., X_n$ from some population distribution.

We wish to use the data to estimate a quantity θ which could be a population moment (*e.g.* mean or variance) or a parameter.

Since our estimate depends on the data, it must be some *function of the random sample* and we can write it as

$$\hat{\theta}(X_1, ..., X_n)$$

We refer to $\hat{\theta}$ as an *estimator* of θ.

Key point: The estimator $\hat{\theta}(\underline{X})$ is a random variable because it is a function of the random sample $\underline{X} = (X_1, \ldots, X_n)$. As such, its value will vary from one experiment (or sample) to another.

Note we distinguish between the random variable $\hat{\theta}(\underline{X})$, which depends on the random sample, $\underline{X}$, and varies across experiments, and the observed value or estimate $\hat{\theta}(\underline{x})$, which is computed from the actual data, $\underline{x}$, observed in a specific experiment, recall we use capital letters for random variables and lowercase letters for observed quantities.

Sometimes there will be a very natural choice for the function $\hat{\theta}$.

Example 2.4. Suppose we wish to estimate the population mean (usually called μ) from a random sample $X_1, ..., X_n$. Then a natural estimator is the *sample mean*

$$\hat{\mu}(X_1, ..., X_n) = \bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

If we need to estimate the population variance σ^2 from the random sample, then a natural estimator to use here would be the *sample variance*:

$$\widehat{\sigma^2}(X_1, ..., X_n) = S^2 = \frac{1}{n-1}\left\{\sum_{i=1}^{n} X_i^2 - \frac{(\sum_{i=1}^{n} X_i)^2}{n}\right\}$$

Remark 2.3. We can also write the sample variance as

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

Indeed, expanding this

$$\begin{aligned}\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i^2 - 2X_i\bar{X} + \bar{X}^2) &= \frac{1}{n-1}\left[\left(\sum_{i=1}^{n} X_i^2\right) - 2n\bar{X}^2 + n\bar{X}^2\right] \\ &= \frac{1}{n-1}\left[\left(\sum_{i=1}^{n} X_i^2\right) - n\bar{X}^2\right] = \frac{1}{n-1}\left\{\sum X_i^2 - \frac{(\sum X_i)^2}{n}\right\}.\end{aligned}$$

Why are $\bar{X}$ and S^2 sensible choices as estimators for μ and σ^2?

2.4 Properties of estimators

We conduct an experiment which yields a random sample $X_1, ..., X_n$ from a population. We wish to estimate some population quantity θ using an estimator $\hat{\theta}(\underline{X})$.

How can we decide whether this choice of estimator is sensible?

2.4.1 Bias of an estimator

We refer the distribution of $\hat{\theta}(\underline{X})$ as its *sampling distribution*.
It tells us how $\hat{\theta}(\underline{X})$ will vary over a large number of independent experiments.

Now if $\hat{\theta}(\underline{X})$ is a sensible estimator for θ then

- the values it typically takes should be close to θ in some sense
- if we conduct a number of identical experiments, each giving an estimate $\hat{\theta}(\underline{X})$, then the values should be scattered around the true value of θ.

This leads naturally the concept of *bias*. We define the bias of an estimator (for θ) $\hat{\theta}(\underline{X})$ to be:

$$\text{Bias}(\hat{\theta}(\underline{X})) = \mathbb{E}(\hat{\theta}(\underline{X}) - \theta).$$

If $\text{Bias}(\hat{\theta}(\underline{X})) = 0$ then the expectation of $\hat{\theta}(\underline{X})$ is θ and we say that $\hat{\theta}$ is *unbiased*.

Example 2.5 (Example 2.4 (cont.))**.** Let us return to the estimation of the population mean μ from random sample $X_1, ..., X_n$ in Example 2.4. What is the bias of the estimator $\hat{\theta}(\underline{X}) = \bar{X}$? Also, show that the sample variance, S^2 is an unbiased estimator of the population variance σ^2.
Solution: First consider the estimator $\bar{X}$ as an estimator of the population mean $\mu = \mathbb{E}[X_1]$. By the linearity of the expectation

$$\mathbb{E}[\bar{X}] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left[X_i\right] = \frac{1}{n}\sum_{i=1}^{n}\mu = \mu$$

Hence, the sample mean $\bar{X}$ is unbiased.
Before we consider the sample variance, let us calculate

$$\mathbb{E}\left[\left(\sum_{i=1}^{n} X_i\right)^2\right] = \mathbb{E}\left[\sum_{i=1}^{n}\sum_{j=1}^{n} X_i X_j\right] = \mathbb{E}\left[\sum_{i=1}^{n}\sum_{j=1}^{n} X_i X_j\right] = \sum_{i=1}^{n}\sum_{j=1}^{n}\mathbb{E}\left[X_i X_j\right].$$

Now we distinguish between when $i = j$ and $i \neq j$

$$\mathbb{E}\left[\left(\sum_{i=1}^{n} X_i\right)^2\right] = \sum_{i=1}^{n}\mathbb{E}\left[X_i^2\right] + \sum_{i=1}^{n}\sum_{j\neq i}\mathbb{E}\left[X_i X_j\right].$$

Using that $X_1, \ldots, X_n$ are i.i.d.

$$\mathbb{E}\left[\left(\sum_{i=1}^{n} X_i\right)^2\right] = \sum_{i=1}^{n}\mathbb{E}\left[X_1^2\right] + \sum_{i=1}^{n}\sum_{j\neq i}\mu^2 = n\mathbb{E}[X^2] + n(n-1)\mu^2.$$

We can use this now to calculate

$$\begin{aligned}\mathbb{E}\left\{\sum_{i=1}^{n} X_i^2 - \frac{(\sum_{i=1}^{n} X_i)^2}{n}\right\} &= n\mathbb{E}[X^2] - \frac{1}{n}\mathbb{E}\left[\left(\sum_{i=1}^{n} X_i\right)^2\right] \\ &= n\mathbb{E}[X^2] - \mathbb{E}[X^2] - (n-1)\mu^2 = (n-1)(\mathbb{E}[X^2] - \mu^2) = (n-1)\sigma^2.\end{aligned}$$

Therefore, S^2 is unbiased since

$$\mathbb{E}[S^2] = \mathbb{E}\left[\frac{1}{n-1}\left\{\sum X_i^2 - \frac{(\sum X_i)^2}{n}\right\}\right] = \sigma^2.$$

Note the factor of $1/(n-1)$ is necessary to ensure that S^2 is unbiased.

Observe that S^2 is an unbiased estimator of the population variance whenever the population mean and variance exist - it doesn't matter what the precise form of the distribution F_X is.
Recall from the *central limit theorem* (CLT): The sample mean $\bar{X}$ has *approximately* the $\mathrm{N}\left(\mu, \frac{\sigma^2}{n}\right)$ distribution for large n.

- This is exact if $X_i \sim \mathrm{N}\left(\mu, \sigma^2\right)$.
- Otherwise the approximation gets better as $n \to \infty$.

Example 2.6. Suppose that the observations $X_1, ..., X_n$ are a random sample from a Gamma (α, β) distribution. (This could arise if our observations were lifetimes of components or survival times of individuals.) Identify the precise form of the distribution of $\bar{X}$ in this case.
Solution: From the "Distributions" formula sheet or from Example 1.5 we have for $X \sim$ Gamma (α, β)

$$M_X(\theta) = \left(\frac{\beta}{\beta - \theta}\right)^{\alpha}.$$

Then using independence

$$M_{\bar{X}}(\theta) = \mathbb{E}\left[e^{\frac{\theta}{n}\sum_{i=1}^{n} X_i}\right] = \prod_{i=1}^{n} \mathbb{E}\left[e^{\frac{\theta}{n} X_i}\right] = M_X\left(\frac{\theta}{n}\right)^n.$$

Therefore for $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim}$ Gamma (α, β)

$$M_{\bar{X}}(\theta) = \left(\frac{\beta}{\beta - \frac{\theta}{n}}\right)^{n\alpha} = \left(\frac{n\beta}{n\beta - \theta}\right)^{n\alpha}.$$

Observe that this is the moment generating function for a Gamma $(n\alpha, n\beta)$, so by the uniqueness property of moment generating functions we have that $\bar{X} \sim$ Gamma $(n\alpha, n\beta)$.

Not all unbiased estimators are good.

Example 2.7. Consider a sample $X_1, \ldots, X_n$ which are i.i.d. copies of some random variable X. Consider the following two estimators for the population mean $\mu = \mathbb{E}[X]$:

$$\hat{\mu}_1(\underline{X}) = \bar{X},$$
$$\hat{\mu}_2(\underline{X}) = X_1.$$

These are both unbiased, but are they equally good?
Suppose $X \sim \mathrm{N}(\mu, 1)$ and we wish to estimate μ. For simplicity we use $\mu = 0$ as the true value, and we simulate 10 samples:

1.94 1.05 -0.70 -0.07 -1.60 -0.20 -0.38 -0.02 -1.10 -0.27

In which case

$$\hat{\mu}_1 = g_1(\underline{x}) = -0.13,$$
$$\hat{\mu}_2 = g_2(\underline{x}) = 1.94.$$

Clearly here $\hat{\mu}_1$ is much closer to the true value $\mu = 0$ than $\hat{\mu}_2$ but maybe that is just an unlucky sample. Lets have a look at the distributions of $\hat{\mu}_1(\underline{X}) \sim \mathrm{N}\left(0, \frac{1}{n}\right)$ and $\hat{\mu}_2(\underline{X}) \sim \mathrm{N}(0, 1)$. The density for these distributions are given in Figure 4 and as we can see both are centred around the true mean 0 but the spread for $\hat{\mu}_2$ is much larger.

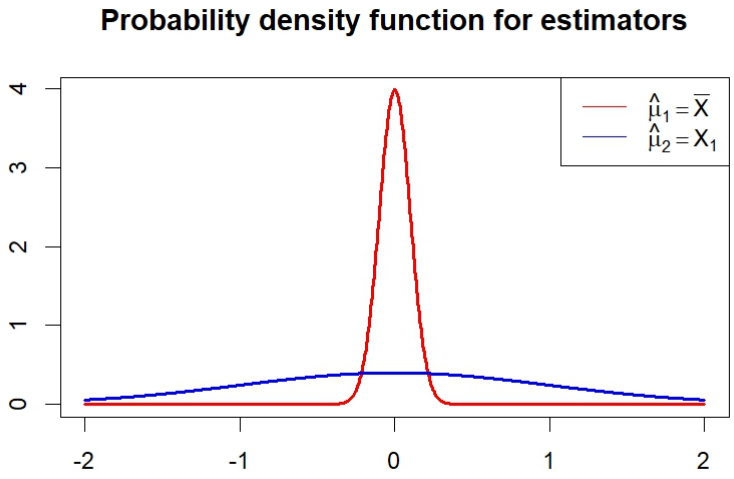


Figure 4: The probability density functions for $\hat{\mu}_1$ and $\hat{\mu}_2$ as given in Example 2.7.

An estimator that is unbiased could be of little value while one that is biased could actually be very useful.

Suppose we want to estimate the mean of a population, $\mu = 20$, and we use 2 estimators $\hat{\theta}_1(\underline{X}) \sim \mathrm{N}(20, 9)$ and $\hat{\theta}_2(\underline{X}) \sim \mathrm{N}(19, 1.2)$. The sampling distributions are shown in Figure 5.
Which estimator is more useful?
We need to consider other characteristics of estimators that are a better guide to how useful they are in practice.

2.4.2 Mean square error

For an estimate $\hat{\theta}(\underline{X})$ of θ we want the random error $\hat{\theta}(\underline{X}) - \theta$ to be small.
We define the *mean square error (MSE)* of $\hat{\theta}(\underline{X})$ as

$$\mathrm{MSE}(\hat{\theta}(\underline{X})) = \mathbb{E}[(\hat{\theta}(\underline{X}) - \theta)^2].$$

This can be written as

$$\mathrm{MSE}(\hat{\theta}(\underline{X})) = \mathrm{Bias}(\hat{\theta}(\underline{X}))^2 + \mathbb{V}\mathrm{ar}(\hat{\theta}(\underline{X}))$$

Proof. Let $\underline{X}$ be a random vector and $\hat{\theta}$ be a function. First expanding the $\mathrm{MSE}(\hat{\theta}(\underline{X}))$ term

$$\begin{aligned}
\mathrm{MSE}(\hat{\theta}(\underline{X})) &= \mathbb{E}\left[(\hat{\theta}(\underline{X}) - \theta)^2\right] \\
&= \mathbb{E}\left[\hat{\theta}(\underline{X})^2 - 2\theta\hat{\theta}(\underline{X}) + \theta^2\right] \\
&= \mathbb{E}[\hat{\theta}(\underline{X})^2] - 2\theta\mathbb{E}[\hat{\theta}(\underline{X})] + \theta^2 \\
&= \mathbb{E}[\hat{\theta}(\underline{X})^2] - \mathbb{E}[\hat{\theta}(\underline{X})]^2 + \mathbb{E}[\hat{\theta}(\underline{X})]^2 - 2\theta\mathbb{E}[\hat{\theta}(\underline{X})] + \theta^2 \\
&= \mathbb{V}\mathrm{ar}(\hat{\theta}(\underline{X})) + (\mathbb{E}[\hat{\theta}(\underline{X}) - \theta])^2 \\
&= \mathbb{V}\mathrm{ar}(\hat{\theta}(\underline{X})) + \mathrm{Bias}(\hat{\theta}(\underline{X})^2.
\end{aligned}$$

□

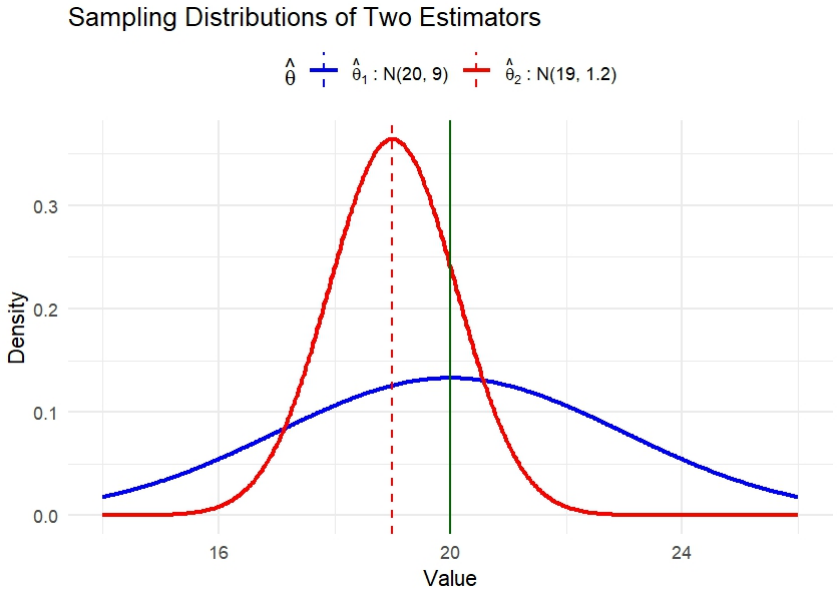


Figure 5: The figure shows the probability density for $\hat{\theta}_1$ in blue and $\hat{\theta}_2$ in red. The vertical dashed line gives the expected value of each estimator. We can see that the red line is not at 20 and hence the estimator $\hat{\theta}_2$ is biased.

Obviously, for unbiased estimators: $\text{MSE}(\hat{\theta}(\underline{X})) = \mathbb{V}\text{ar}(\hat{\theta}(\underline{X}))$

Example 2.8. Find the MSE of the sample mean $\bar{X}$ as an estimator of the population mean μ.

Solution:
We have seen that $\bar{X}$ is an unbiased estimator of μ.
It follows from the equation above that $\text{MSE}(\bar{X}) = \mathbb{V}\text{ar}(\bar{X}) = \frac{\sigma^2}{n}$.
(You should be able to prove this!!)

2.4.3 Efficiency

We can use the MSE as a measure to compare different estimators.
If $\hat{\theta}_1$ and $\hat{\theta}_2$ are estimators of θ, the (relative) efficiency of $\hat{\theta}_2$ compared to $\hat{\theta}_1$ is given by the ratio

$$\frac{\text{MSE}(\hat{\theta}_1)}{\text{MSE}(\hat{\theta}_2)}.$$

We say that $\hat{\theta}_1$ is more efficient than $\hat{\theta}_2$ if $\text{MSE}(\hat{\theta}_1) < \text{MSE}(\hat{\theta}_2)$.

Example 2.9. Consider a random sample (X_1, X_2) from a population with mean μ and variance σ^2. Now consider the following as estimators of μ:

$$\hat{\theta}_1(X_1, X_2) = \bar{X} \quad \text{and} \quad \hat{\theta}_2(X_1, X_2) = \frac{X_1}{3} + \frac{2X_2}{3}.$$

Determine which of the two estimators is more efficient.

Solution:
Clearly both of these are unbiased ($\mathbb{E}(\hat{\theta}_1(X_1, X_2)) = \mathbb{E}(\hat{\theta}_2(X_1, X_2)) = \mu$). It follows that

$$\text{MSE}(\hat{\theta}_1(X_1, X_2)) = \mathbb{V}\text{ar}(\hat{\theta}_1(X_1, X_2)) = \frac{\sigma^2}{2},$$
$$\text{MSE}(\hat{\theta}_2(X_1, X_2)) = \mathbb{V}\text{ar}(\hat{\theta}_2(X_1, X_2)) = \frac{\sigma^2}{9} + \frac{4\sigma^2}{9} = \frac{5\sigma^2}{9}.$$

Therefore

$$\text{MSE}(\hat{\theta}_2(X_1, X_2)) = \frac{5\sigma^2}{9} > \frac{\sigma^2}{2} = \text{MSE}(\hat{\theta}_1(X_1, X_2))$$

and, hence, $\hat{\theta}_1$ is more efficient that $\hat{\theta}_2$.

Example 2.10. In an experiment a bacteriologist must measure the temperature, θ of a growth chamber. She has two thermometers of differing quality. The first returns a random quantity $T_1 = \theta + E_1$ where E_1 is a random error with mean zero and variance σ_1^2. The second returns $T_2 = \theta + E_2$ where E_2 has zero mean and variance σ_2^2. Suppose further that E_1 and E_2 are independent. She decides to combine the two measurements via the estimator:

$$\hat{\theta}(\underline{T}) = aT_1 + (1-a)T_2$$

where $a \geq 0$. How should a be chosen to give the most efficient estimator of this form?

Solution: Since E_1 and E_2 have zero mean we have that $\mathbb{E}[T_1] = \mathbb{E}[T_2] = \theta$ and hence $\mathbb{E}[\hat{\theta}(\underline{T})] = \theta$ so the estimator is unbiased. Therefore it remains to find a value of a that minimises the variance. Since T_1 and T_2 are independent (as E_1 and E_2 are independent) we have

$$\mathbb{V}\text{ar}(\hat{\theta}(\underline{T})) = a^2\mathbb{V}\text{ar}(T_1) + (1-a)^2\mathbb{V}\text{ar}(T_2) = a^2\mathbb{V}\text{ar}(E_1) + (1-a)^2\mathbb{V}\text{ar}(E_2) = a^2\sigma_1^2 + (1-a)^2\sigma_2^2.$$

Consider the function $h(a) = a^2\sigma_1^2 + (1-a)^2\sigma_2^2$ we can minimise this by finding its stationary points: We first differentiate

$$h'(a) = 2a\sigma_1^2 - 2(1-a)\sigma_2^2.$$

Then solve $h'(a) = 0$ to find the critical point(s) are

$$a^* = \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}$$

Note that $0 \leq a^* \leq 1$ so this is a valid choice of a. Now $h''(a) = 2\sigma_1^2 + 2\sigma_2^2 \geq 0$ for all choices of a and hence also at a^* therefore by using the second derivative test a^* is a minimum. That is, setting $a = a^*$ gives the most efficient form of this estimator.

2.4.4 Standard Error

As already mentioned, good estimators should have small variance, i.e. large precision. The standard deviation of an estimator $\hat{\theta}$ is called its standard error:

$$\text{s.e.}(\hat{\theta}) = \sqrt{\mathbb{V}\text{ar}(\hat{\theta}(\underline{X})}.$$

It is especially useful when comparing estimators.

Example 2.11. On the day of HW's Students' Association elections, one of the SA representatives wants to predict the proportion of students that voted for a specific candidate A (that is, before the results are known!). She selects a sample of m students at random, who have just voted and asks them to answer (anonymously on a piece of paper) whether they voted for candidate A or not. She considers the outcome of this survey as a random sample $\underline{X} = (X_1, X_2, ..., X_m)$ of size m from a Ber(p) distribution where $X_i = 1$ if the ith student voted for A and is 0 otherwise. Is there an obvious estimator for p?

Yes, let

$$\hat{p}(\underline{X}) = \frac{1}{m}\sum X_i$$

(i.e. the number of students in the sample who voted for A, over the total number of voters in the sample).

Find the bias, s.e.($\hat{p}$) and MSE($\hat{p}$).

Solution:

Since $\sum X_i$ is the sum of m independent Bernoulli random variables we have that $\sum X_i \sim$ Bin(m, p) therefore

$$\mathbb{E}\left[\sum X_i\right] = mp, \quad \mathbb{V}\text{ar}\left(\sum X_i\right) = mp(1-p).$$

This gives

$$\mathbb{E}\left[\hat{p}(\underline{X})\right] = p, \quad \mathbb{V}\text{ar}\left(\hat{p}(\underline{X})\right) = \frac{p(1-p)}{m}.$$

Therefore, $\hat{p}$ is unbiased and has s.e.$(\hat{p}) = \frac{\sqrt{p(1-p)}}{\sqrt{m}}$, hence

$$\text{MSE}(\hat{p}) = \frac{p(1-p)}{m}.$$

3 Methods to construct Estimators

We now consider the problem of constructing an estimator for a parameter θ. We will consider two approaches:

1. Method of Moments estimation
2. Maximum likelihood estimation

3.1 Method of Moments estimation

Consider an experiment that yields an i.i.d. set of observations $X_1, ..., X_n$ from a distribution with density or mass function $f_X(x; \theta)$, where θ is unknown parameter(s).

Sometimes θ will be a *vector* of parameters, e.g. if $X \sim$ Gamma(α, β) then $\theta = (\alpha, \beta)$ (2-dimensional vector).

To find the Methods of Moments Estimator (MME) for θ we

i. Equate the sample moments with theoretical (population) moments.

ii. Then solve the equation(s).

Example 3.1 (Example 2.11 (cont.))**.** Recall in Example 2.11, the results of the *exit poll* give a random sample $X_1, ..., X_m$ from a Ber(p) distribution (where voting for candidate A is signified by $X_i = 1$. The obvious estimator is

$$\hat{p}(\underline{X}) = \frac{\sum X_i}{m}.$$

How could we have derived this systematically using the method of moments estimator?

Solution:

For the Ber(p) distribution, the first moment is:

$$\mathbb{E}(X) = \mu(p) = p$$

as a function of p, $\mu(p)$.

We equate this 'population' moment to the sample mean,

$$\bar{X} = \frac{\sum X_i}{m}$$

and solve for the parameter p to obtain the estimator

$$\mathbb{E}(X) = \bar{X} \Rightarrow \hat{p}(\underline{X}) = \bar{X}.$$

More generally the method of moments estimator for a 1-dimensional parameter θ is found by solving the equation

$$\mu(\theta) = \bar{X}.$$

Example 3.2. Suppose $X_1, ..., X_n$ is a random sample from Exp(β). What is the MME of β?

Solution: The first moment is:

$$\mathbb{E}[X] = \frac{1}{\beta}.$$

We equate this 'population' moment with the sample mean,

$$\bar{X} = \frac{\sum X_i}{m}$$

and solve for the parameter β to obtain the estimator

$$\mathbb{E}(X) = \bar{X} \Rightarrow \hat{\beta}(\underline{X}) = (\bar{X})^{-1}.$$

More generally we can perform method-of-moments estimation for distributions that have k unknown parameters: $\theta = (\theta_1, \theta_2, \ldots, \theta_k)$:

- Consider all moments up to order k and equate the rth sample moment, $\nu_r = \frac{1}{n}\sum x_i^r$, with the rth population moment, $\mathbb{E}[X^r]$, which is a function of the parameter θ, i.e. $\mathbb{E}[X^r] = \mu_r(\theta)$.
- This generates k simultaneous equations of the form

  $$\nu_r = \mu_r(\theta), r = 1, \ldots, k$$

  which we then solve for θ to get the estimators.

Note that equivalently we could work with central moments $\mathbb{E}[(X - \mu)^r]$ equated with their sample values

$$m_r = \frac{1}{n}\sum_i (x_i - \bar{x})^r.$$

Example 3.3. Now let $X_1, ..., X_n$ be a random sample from Gamma(α, β). What is the MME for $\theta = (\alpha, \beta)$?

Solution: We have two unknown parameters so we need two equations and hence we use the first two moments:

$$\mathbb{E}[X] = \frac{\alpha}{\beta}$$
$$\mathbb{E}[X^2] = \frac{\alpha(\alpha+1)}{\beta^2}$$

We equate these 'population' moments with the sample moments, then we need to solve two simultaneous equations

$$\nu_1 = \frac{\alpha}{\beta}$$
$$\nu_2 = \frac{\alpha(\alpha+1)}{\beta^2} = \frac{\alpha^2}{\beta^2} + \frac{\alpha}{\beta}.$$

In this case it may have been simpler to work with the 2nd central moment

$$\mathbb{V}\text{ar}(X) = \frac{\alpha}{\beta^2} \Rightarrow \mu_2 = \frac{\alpha}{\beta^2}.$$

In either case we have

$$\hat{\alpha} = \frac{\nu_1^2}{\mu_2} = \frac{\nu_1^2}{\nu_2 - \nu_1^2},$$
$$\hat{\beta} = \frac{\nu_1}{\mu_2} = \frac{\nu_1}{\nu_2 - \nu_1^2}.$$

Example 3.4. Suppose we have $X_1, \ldots, X_n$ which are i.i.d. from a $\text{N}\left(\mu, \sigma^2\right)$ where $\theta = (\mu, \sigma^2)$ is unknown. Find the MME of θ.

Solution:
In this case we have two unknowns so we use the first two moments:

$$\mathbb{E}[X] = \mu$$
$$\mathbb{E}[X^2] = \sigma^2 + \mu^2.$$

We equate these 'population' moments with the sample moments, then we need to solve two simultaneous equations

$$\nu_1 = \mu$$
$$\nu_2 = \sigma^2 + \mu^2.$$

Solving this gives the following estimators:

$$\hat{\mu}(\underline{X}) = \nu_1 = \bar{X},$$
$$\widehat{\sigma^2}(\underline{X}) = \nu_2 - \nu_1^2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2 - \left(\frac{1}{n}\sum_{i=1}^{n} X_i\right)^2 = \frac{n-1}{n}S^2.$$

Note that this estimator for σ^2 is biased.

3.1.1 Advantages and disadvantages of MME

- Main advantage is that it is generally simple to apply
- However, it has the disadvantage that in some circumstances it can give results that are clearly nonsensical.

Example 3.5. Suppose that you conduct an experiment in which you observe $X_1, ..., X_n$ where the X_i are i.i.d. from a $\text{U}(0, \theta)$, where $\theta > 0$. Find the method of moments estimator of θ.

Solution:
We know that $\mathbb{E}(X) = \frac{\theta}{2}$, so

$$\bar{X} = \frac{\theta}{2} \Rightarrow \hat{\theta}(\underline{X}) = 2\bar{X}.$$

Now suppose that for the particular experiment $n = 3$ and $(x_1, x_2, x_3) = (0.1, 0.2, 0.9)$.
Then in this case we would obtain the estimate $\hat{\theta} = 2\bar{x} = 0.8$.
However, this doesn't make sense: the data are not consistent with this value of θ. Suppose that the true value of θ was 0.8 then the image of X is $[0, 0.8]$ however we have observed that x_3 is not in this range.

3.2 Maximum likelihood estimation

Loosely speaking, maximum likelihood estimation is based on finding the value of the parameter(s) θ for which the available data *are most likely to have been observed.*
Consider the following example:

Example 3.6. A new driver is taking the driving test and passes on the 4th attempt. Based on this information (*data*) we want to estimate the probability θ of passing the test on any one attempt (*assuming that outcome of each attempt is independent, and probability does not change among attempts!*).

If we let $X_i = 1$ for success and 0 for failure, the available data are $\underline{x} = (0, 0, 0, 1)$.
If we know want to evaluate the probability that these data have been observed, this is given as

$$\begin{aligned} \mathbb{P}(\underline{x} \text{ is observed}) &= \mathbb{P}(\text{fails 1st, 2nd, 3rd tests, passes on 4th}) \\ &= \mathbb{P}(X_1 = 0, X_2 = 0, X_3 = 0, X_4 = 1) \\ &= (1-\theta)^3\theta \quad \text{(independence)}. \end{aligned}$$

Then, if we have to choose among, say, $\theta = 0.25, 0.5$ or 0.75 we notice that

$$\begin{aligned} \mathbb{P}(\underline{x} \text{ is observed}) &= 0.105, \qquad \text{for } \theta = 0.25 \\ &= 0.063, \qquad \text{for } \theta = 0.50 \\ &= 0.012, \qquad \text{for } \theta = 0.75 \end{aligned}$$

So, the given data set is *more likely to have been observed* when $\theta = 0.25$. By observing the curve $\theta(1-\theta)^3$, which is plotted in Figure 6, we see that the value of $\theta = 0.25$ leads to the highest probability of observing the data.

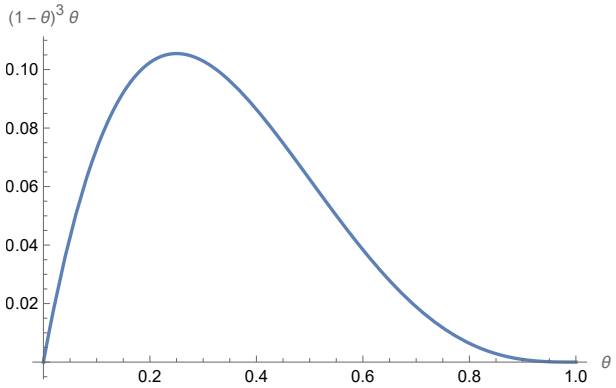


Figure 6: A plot of the curve $\theta(1-\theta)^3$.

We have used the idea of *maximum likelihood estimation.*

Informally, suppose we carry out an experiment and observe data $\underline{x}$.
Then the likelihood is a function of all possible parameter values θ and is 'defined' to be:

$$L(\theta; \underline{x}) = \mathbb{P}(\underline{X} = \underline{x}; \theta).$$

In other words is tells us how 'likely' the data are for given values of θ.
Note: here we are interested in how $L(\theta; \underline{x})$ varies with θ - the experimental observations $\underline{x}$ will typically be fixed.

3.2.1 Likelihood for i.i.d. observations from a discrete distribution

Let $Y_1, ..., Y_n$ be i.i.d. variables from a discrete distribution with probability mass function $p_Y(y; \theta)$.
Suppose the *realised* values from the experiment are $\underline{y} = y_1, \dots y_n$.
Then the likelihood function is given by

$$\begin{aligned} L(\theta; \underline{y}) &= \mathbb{P}(Y_1 = y_1, Y_2 = y_2, \dots, Y_n = y_n; \theta) \\ &= p_Y(y_1; \theta) \times p_Y(y_2; \theta) \times \cdots \times p_Y(y_n; \theta) \\ &= \prod_{i=1}^{n} p_Y(y_i; \theta) \end{aligned}$$

The above joint probability decomposes as a product since the observations are all independent of each other.
We then obtain the maximum likelihood estimator (MLE) by maximising $L(\theta; \underline{y})$ with respect to θ (often using calculus techniques).
Often it is more convenient to maximise the logarithm of the likelihood

$$\ell(\theta; \underline{y}) = \log(L(\theta; \underline{y}) = \sum_{i=1}^{n} \log(p_Y(y_i; \theta)).$$

Note that since log is a monotonic increasing function, the value of θ that maximises $\ell(\theta; \underline{x})$ also maximises $L(\theta; \underline{x})$ and *vice versa.*

Example 3.7. An astronomer counts the number of meteors in a section of the night sky for n consecutive nights obtaining data $\underline{x} = (x_1, \dots, x_n)$. She believes that these values are a random sample (i.i.d.) from a Poi(λ) distribution where λ is unknown. What is the maximum likelihood estimator of λ?

Solution:
Since the observations are assumed to be i.i.d. we have

$$L(\lambda; \underline{x}) = \prod_{i=1}^{n} p_X(x_i; \lambda) = \prod_{i=1}^{n} \frac{e^{-\lambda} \lambda^{x_i}}{x_i!} = \frac{e^{-n\lambda} \lambda^{\sum x_i}}{\prod_{i=1}^{n} x_i!}$$

Taking the logarithm we have the log likelihood for $\lambda > 0$

$$\ell(\lambda; \underline{x}) = -n\lambda + \log(\lambda) \left(\sum_i x_i \right) - \sum_i \log(x_i!).$$

As this function is smooth we can attempt to maximise ℓ using calculus techniques but we need to be careful of the constraint $\lambda = 0$. To maximise this function we observe that as $\lambda \to 0$

$\ell(\lambda;\underline{x}) \to -\infty$ and as $\lambda \to \infty$ $\ell(\lambda;\underline{x}) \to -\infty$. Therefore $\ell(\lambda;\underline{x})$ takes its maximum value within $(0,\infty)$. This point must be a critical point which we can find by differentiating, equating to zero and solving for λ. Differentiating gives us

$$\frac{d}{d\lambda}\ell(\lambda;\underline{x}) = -n + \frac{1}{\lambda}\left(\sum_i x_i\right).$$

Setting this equal to zero and solving for λ gives:

$$\hat{\lambda} = \frac{\sum x_i}{n} = \bar{x}.$$

Note that we need to confirm that this point is indeed a maximum (and not a minimum or inflection point) for l, in this case we have that $\hat{\lambda}$ is a maximum as it is the only critical point and as we have seen the function ℓ is smaller at the boundary $\lambda = 0$ and when $\lambda \to \infty$.

We see that the maximum likelihood estimator of λ is the same as we would obtain using method of moments. *This is not always the case!*

Example 3.8 (Example 2.11 and 3.1 (cont.))**.** In Example 2.11, suppose that the results of the *exit poll* gave that r students (out of m) voted for candidate A (and $m-r$ didn't). Find the maximum likelihood estimator for p.

Solution: Let $X_1,\ldots,X_m$ denote the results of the exit poll, i.e. $X_i = 1$ if the ith person voted for candidate A and $X_i = 0$ otherwise. Then $X_1,\ldots,X_m \overset{\text{i.i.d}}{\sim} \text{Ber}(p)$, the data we observe is the number of people who vote for candidate A, that is $Y = \sum X_i$, and we observe that $Y \sim \text{Bin}(m,p)$.

Then the likelihood of seeing r students vote for candidate A is

$$L(p;r) = \mathbb{P}(Y = r;r) = \binom{m}{r}p^r(1-p)^{m-r}$$

Taking the logarithm we have the log likelihood for $p \in [0,1]$

$$\ell(p;r) = \log(L(p;r)) = \log\left(\binom{m}{r}\right) + r\log(p) + (m-r)\log(1-p)$$

Now we wish to find the value of $p \in [0,1]$ which maximises $\ell(p;r)$, to find this we differentiate with respect to p

$$\frac{d}{dp}\ell(p;r) = \frac{r}{p} - \frac{m-r}{1-p} = \frac{r(1-p)-(m-r)p}{p(1-p)} = \frac{r-mp}{p(1-p)}.$$

Equating this equal to zero and solving for p we find that $\ell(\cdot;r)$ is maximised at

$$\hat{p} = \frac{r}{m}.$$

Note that this must be a maximum since $l(p;r) \to -\infty$ as $p \to 0$ or $p \to 1$ and there is only one critical point.

3.2.2 Likelihood for i.i.d. observations from a continuous distribution

Suppose now our experiment consists of i.i.d. observations $X_1,\ldots,X_n$ from a continuous distribution with density function $f_X(x;\theta)$.

How do we construct the likelihood in this case?

Problem: Since population distribution is *continuous*

$$\mathbb{P}(X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n|\theta) = 0 \quad \text{(for any } \theta\text{)}.$$

This assumes that our observations are measured to arbitrary high precision (an infinite number of decimal places).
However let's suppose that we can only measure a physical quantity to some finite precision (i.e. within $\pm\frac{\delta}{2}$) for some δ and that we observe $x_1, \ldots, x_n$.
Then we consider the event $X_i \in (x_i - \frac{\delta}{2}, x_i + \frac{\delta}{2})$,

$$\mathbb{P}\left\{X_i \in (x_i - \frac{\delta}{2}, x_i + \frac{\delta}{2})\right\} \approx f_X(x_i;\theta)\delta.$$

It follows that, since the likelihood is the probability of the data given θ (and using independence of X_i)

$$\begin{aligned} L(\theta;\underline{x}) &\approx \mathbb{P}\left\{X_1 \in (x_1 - \frac{\delta}{2}, x_1 + \frac{\delta}{2}), \ldots, X_n \in (x_n - \frac{\delta}{2}, x_n + \frac{\delta}{2})|\theta\right\} \\ &\approx \prod_{i=1}^{n} f_X(x_i;\theta)\delta^n, \end{aligned}$$

with the approximation becoming increasingly accurate as δ decreases (i.e. the precision increases).
Now, the positive factor δ^n does not depend on θ and has no bearing on the value of θ that maximises $L(\theta;\underline{x})$. Therefore, we can omit it from the the above expression and *define* the likelihood to be

$$L(\theta;\underline{x}) = \prod_{i=1}^{n} f_X(x_i;\theta).$$

The maximum likelihood estimate is the value of θ that maximises $L(\theta;\underline{x})$.

Example 3.9. Suppose $X_1, ..., X_n$ is a random sample from Exp(λ). Find the maximum likelihood estimator of λ.
Solution:
The probability density function is $f_X(x;\lambda) = \lambda e^{-\lambda x}$.
For given $\underline{x} = (x_1, ..., x_n)$, the likelihood is

$$L(\lambda;\underline{x}) = \prod_{i=1}^{n} f_X(x_i;\lambda) = \lambda^n e^{-\lambda \sum x_i}$$

Taking logs and maximising you should find that maximum likelihood estimate is

$$\hat{\lambda} = \frac{1}{\bar{x}}.$$

How does this estimate compare with the method-of-moments estimate?

Calculus will not always help to maximise the likelihood. Consider the following example.

Example 3.10. Let $X_1, \ldots, X_n$ be a random sample from a U$(0, \theta)$ distribution, where $\theta > 0$. What is the maximum likelihood estimator of θ?

Solution:
We know that

$$f(x;\theta) = \begin{cases} \frac{1}{\theta}, & \text{if } 0 < x < \theta \\ 0, & \text{otherwise} \end{cases}$$

Now the likelihood function is given by

$$L(\theta;\underline{x}) = \prod_{i=1}^{n} f(x_i;\theta) = \begin{cases} \left(\frac{1}{\theta}\right)^n, & \text{if } \theta \geq x_{(n)} = \max\{x_1,\ldots,x_n\} \\ 0, & \text{otherwise} \end{cases}$$

We cannot use calculus here - it is not clear which value of θ maximises $L(\theta)$. Instead we consider the *graph* of $L(\theta;x)$ as a function of θ which is given in Figure 7. Observe that $L(\theta;\underline{x})$

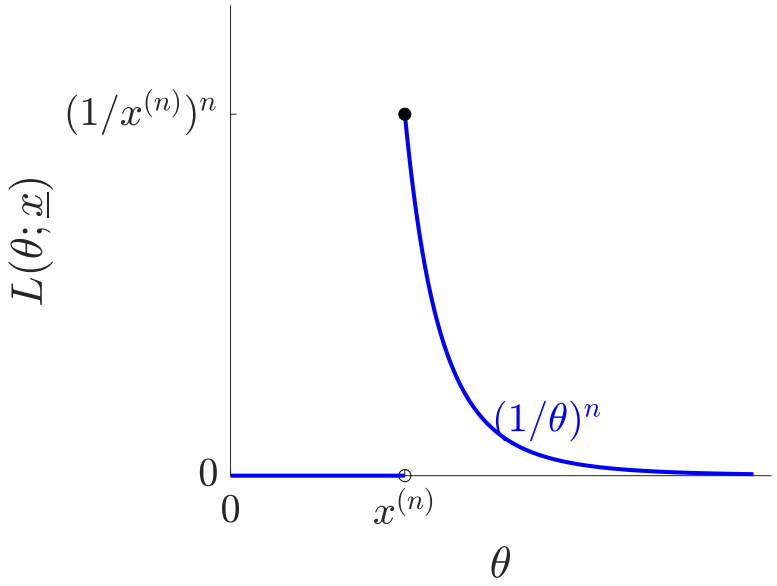


Figure 7: Likelihood function for a Uniform random variable. The function is zero for $\theta \leq x^{(n)}$, and follows $L(\theta;\underline{x}) = \theta^{-n}$ for $\theta > x^{(n)}$.

is maximised for smallest possible value of θ for which the likelihood is non-zero.
It follows that the maximum likelihood estimator of θ is

$$\hat{\theta}(\underline{X}) = X_{(n)} = \max\{X_1, X_2, ..., X_n\}.$$

(Note this is not the same as the method of moments estimator.)

Problem:
Find the Bias and MSE of the maximum likelihood estimator $\hat{\theta}$. Compare your results with the corresponding ones for the MME. Which estimator do you think is better?

3.2.3 Maximum likelihood estimators for several parameters

Suppose $\underline{X}$ is a random sample from a population probability density function/probability mass function $f_X(x;\underline{\theta})$ where $\underline{\theta} = (\theta_1,\ldots,\theta_k)$.
We can use maximum likelihood estimator to find a *vector* of estimators $\underline{\hat{\theta}} = (\hat{\theta}_1,\ldots,\hat{\theta}_k)$ of $\underline{\theta}$.

1. Write down the likelihood function $L(\underline{\theta};\underline{x}) = \prod_{i=1}^{n} f_X(x_i;\underline{\theta})$

2. Consider the log-likelihood function $\ell(\underline{\theta};\underline{x}) = \log L(\underline{\theta};\underline{x})$ and solve (*if possible*) the system of k equations
$$\frac{\partial}{\partial\theta_r}\ell(\underline{\theta};\underline{x}) = 0, \quad r = 1, 2, \ldots, k$$
simultaneously to obtain $\hat{\underline{\theta}} = (\hat{\theta}_1, \ldots, \hat{\theta}_k)$.

3. Check whether $L(\underline{\theta};\underline{x})$ is maximised at $\hat{\underline{\theta}}$ (solution in step (2)).
This step is not easy. For $k = 2$ the 2-dimensional 2nd derivative test can be used. If $k \geq 3$ other (e.g. Lagrangian) methods can be used.

Remark 3.1 (Second Derivative Test in Two Variables). Suppose $f : \mathbb{R}^2 \to \mathbb{R}$ is twice continuously differentiable and (x^*, y^*) is a *critical point* of f, meaning
$$\nabla f(x^*, y^*) = \mathbf{0}.$$
To classify the behaviour of f near (x^*, y^*), consider the second-order Taylor expansion:
$$f(x, y) \approx f(x^*, y^*) + \frac{1}{2}\mathbf{v}^T H \mathbf{v}, \quad \text{where } \mathbf{v} = \begin{pmatrix} x - x^* \\ y - y^* \end{pmatrix}.$$
Here H is the *Hessian matrix* and defined by:
$$H = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2}(x^*, y^*) & \frac{\partial^2 f}{\partial x \partial y}(x^*, y^*) \\ \frac{\partial^2 f}{\partial x \partial y}(x^*, y^*) & \frac{\partial^2 f}{\partial y^2}(x^*, y^*) \end{pmatrix}.$$
Since H is symmetric, its eigenvalues are real. The sign of the quadratic form
$$\mathbf{v}^T H \mathbf{v}$$
for all nonzero $\mathbf{v}$ determines whether H is *positive definite*, *negative definite*, or *indefinite*, which classifies the critical point.

Classification Using Determinant and Leading Principal Minor: Let
$$D = \det(H) = \frac{\partial^2 f}{\partial x^2}(x^*, y^*)\frac{\partial^2 f}{\partial y^2}(x^*, y^*) - \left(\frac{\partial^2 f}{\partial x \partial y}(x^*, y^*)\right)^2.$$

- If $D = 0$: Hessian is singular $\Rightarrow$ **Test inconclusive** (higher-order terms needed).
- If $D < 0$: Hessian is indefinite $\Rightarrow$ **Saddle point**.
- If $D > 0$: Hessian is definite $\Rightarrow$ sign of $\frac{\partial^2 f}{\partial x^2}$ decides:
  - $\frac{\partial^2 f}{\partial x^2}(x^*, y^*) > 0$: **Local minimum**.
  - $\frac{\partial^2 f}{\partial x^2}(x^*, y^*) < 0$: **Local maximum**.

In summary:

Condition	Classification
$D > 0$, $\frac{\partial^2 f}{\partial x^2} > 0$	Local minimum
$D > 0$, $\frac{\partial^2 f}{\partial x^2} < 0$	Local maximum
$D < 0$	Saddle point
$D = 0$	Inconclusive

Examples:

1. $f(x,y) = x^2 + y^2$: $H = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, D = 4 > 0, \ \frac{\partial^2 f}{\partial x^2} = 2 > 0 \Rightarrow$ **Local minimum** at $(0,0)$.

2. $f(x,y) = x^2 - y^2$: $H = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}, D = -4 < 0 \Rightarrow$ **Saddle point** at $(0,0)$.

Example 3.11. Suppose $X_1, ..., X_n$ is a random sample from $\mathrm{N}\left(\mu, \sigma^2\right)$. Find the maximum likelihood estimator for μ and σ^2.

Solution:
The likelihood function is given by

$$L(\mu, \sigma^2; \underline{x}) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2\right)$$

Taking the logarithm gives the log likelihood:

$$\ell(\mu, \sigma^2; \underline{x}) = \log L(\mu, \sigma^2; \underline{x}) = -n\log(\sqrt{2\pi}) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2.$$

Differentiate the log-likelihood function with respect to μ and σ^2 to get the following system of equations:

$$\frac{\partial}{\partial\mu}\ell(\mu, \sigma^2; \underline{x}) = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i - \mu) = 0 \quad (1)$$

$$\frac{\partial}{\partial\sigma^2}\ell(\mu, \sigma^2; \underline{x}) = \frac{-n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}(x_i - \mu)^2 = 0. \quad (2)$$

Solving (1) gives $\hat{\mu} = \bar{x}$.
Next, substitute $\bar{x}$ for μ into the second equation, and solve for σ^2. We get

$$\widehat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{n-1}{n}s^2$$

Now, check if $\ell(\mu, \sigma^2; \underline{x})$ is *maximised* at $\hat{\mu} = \bar{x}$, $\widehat{\sigma^2} = \frac{n-1}{n}s^2$.
Use the *2-dimensional* 2nd derivative test (see Remark 3.1):
Note that:

$$\frac{\partial^2}{\partial\mu^2}\ell(\hat{\mu}, \hat{\sigma}^2; \underline{x}) = \frac{-n}{\hat{\sigma}^2} < 0$$

$$\frac{\partial^2}{\partial(\sigma^2)^2}\ell(\hat{\mu}, \hat{\sigma}^2; \underline{x}) = \frac{n}{2\hat{\sigma}^4} - \frac{1}{\hat{\sigma}^6}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{n}{2\hat{\sigma}^4} - \frac{n\hat{\sigma}^2}{\hat{\sigma}^6} = \frac{-n}{2\hat{\sigma}^4} < 0$$

$$\frac{\partial^2}{\partial\mu\partial\sigma^2}\ell(\hat{\mu}, \hat{\sigma}^2; \underline{x}) = -\frac{1}{\hat{\sigma}^4}\sum_{i=1}^{n}(x_i - \bar{x}) = 0$$

Therefore,

$$\frac{\partial^2}{\partial\mu^2}\ell(\hat{\mu}, \hat{\sigma}^2; \underline{x}) \cdot \frac{\partial^2}{\partial(\sigma^2)^2}\ell(\hat{\mu}, \hat{\sigma}^2; \underline{x}) - \left(\frac{\partial^2}{\partial\mu\partial\sigma^2}\ell(\hat{\mu}, \hat{\sigma}^2; \underline{x})\right)^2 = \frac{n^2}{2\hat{\sigma}^6} > 0.$$

So, $L(\mu, \sigma^2; \underline{x})$ is *maximized* at $\hat{\mu} = \bar{x}, \hat{\sigma}^2 = \frac{n-1}{n}s^2$, and this is an global maximum (since there is only one critical point).

Note that the maximum likelihood estimator of σ^2 is *biased* (since $\mathbb{E}(S^2) = \sigma^2$).

3.2.4 Invariance of maximum likelihood estimators under 1-1 transformations

This is a very useful property of maximum likelihood estimators.
We can sometimes exploit it to simplify the problem of finding the maximum likelihood estimator of a parameter.

Suppose h is a 1-1 function and suppose two parameters θ and φ are related through h as $\varphi = h(\theta)$.
If $\hat{\theta}$ and $\hat{\varphi}$ are the maximum likelihood estimators of the two parameters respectively, we have

$$\hat{\varphi}(\underline{x}) = h(\hat{\theta}(\underline{x}))$$

Example 3.12. If $X \sim \text{Exp}(\beta)$, we know that $\mu = \mathbb{E}(X) = 1/\beta$. Use the invariance property to find the maximum likelihood estimator of μ using the maximum likelihood estimator of β.

Solution:
We know that the maximum likelihood estimator $\hat{\beta}P(\underline{X}) = 1/\bar{X}$.
Then, since the mean of the population is given as $\mu = h(\beta) = 1/\beta$, the invariance property of the maximum likelihood estimator gives that the maximum likelihood estimator $\hat{\mu}$ of μ is

$$\hat{\mu}(\underline{X}) = h(\hat{\beta}(\underline{X})) = \bar{X}.$$

3.2.5 Likelihood for censored observations

Observations that are only *partially known* are called *censored*.

Example 3.13. In a medical study, a random sample of 20 patients are followed up and the time T (in months) until they develop a specific condition is recorded. We assume that the observation starts for all patients at the same time (say $t = 0$), and the study ends after 2 months. We also assume that $T \sim \text{Exp}(\lambda)$. At the end of the 2-month period, 10 patients have developed the condition at times $t_1, t_2, ..., t_{10}$ with $\sum_{i=1}^{10} t_i = 7.3$. The remaining 10 patients are free of the condition.
Can we construct a likelihood and estimate λ in this case?
A possible realisation of the data is illustrated in Figure

Solution:
We assume that the times in the sample are *independent* of each other.
Therefore, 'the probability of the data given λ' can be naturally represented as a product of factors:

- For the patients developing the condition at $t_i < 2$ months, $1 \leq i \leq 10$, the probability is

$$f_T(t_i; \lambda) = \lambda e^{-\lambda t_i}$$

- Now, for the patients that did not develop the condition, the probability of the censored observation is the probability that a time exceeds 2 months:

$$\mathbb{P}(T > 2) = 1 - F_T(2; \lambda) = 1 - (1 - e^{-2\lambda}) = e^{-2\lambda}$$

Therefore we obtain the likelihood as

$$L(\lambda) = \prod_{i=1}^{10} \lambda e^{-\lambda t_i} \times \prod_{i=11}^{20} e^{-2\lambda} = \lambda^{10} e^{-\lambda(20 + \sum_{i=1}^{10} t_i)}$$

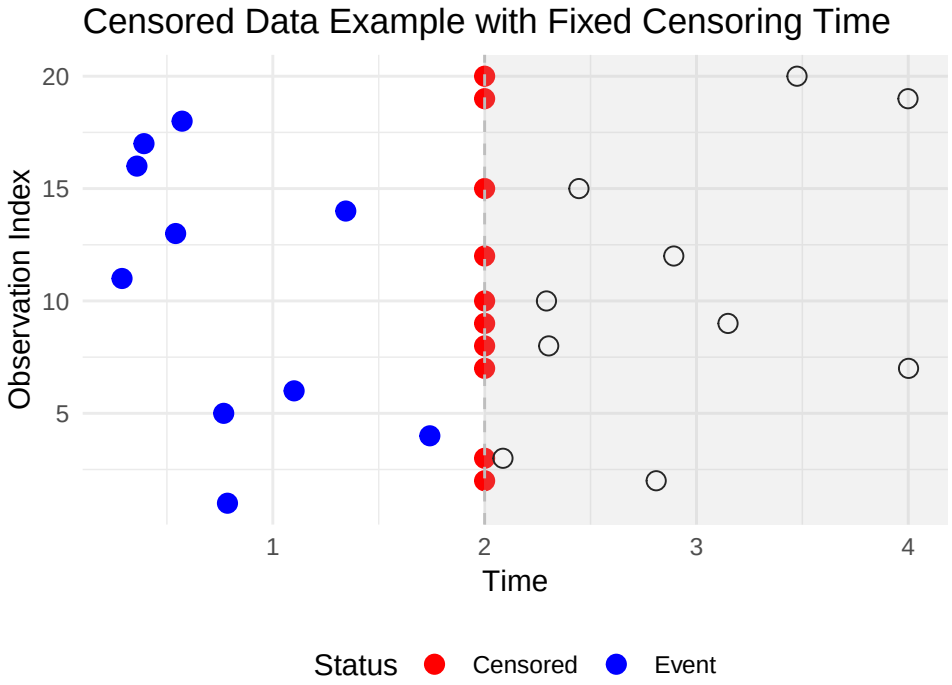


Figure 8: An illustration of a potential realisation of censored data. Here each event happens at a random time and if the event occurs before time 2 then it is observed, this is shown by a blue dot in the plot. If the event occurs after time 2 then it is not observed and the true value is given by a circle in the plot. For the events that that are not observed we only see the censored data that the event has not happened at time 2 represented by the red dot.

Taking logs we obtain the log-likelihood:

$$l(\lambda) = 10\log(\lambda) - \lambda(20 + \sum_{i=1}^{10} t_i).$$

Differentiating and equating to zero (*exercise*) we find that this is maximised by

$$\hat{\lambda} = \frac{10}{(20 + \sum_{i=1}^{10} t_i)}.$$

This answer is intuitively plausible. The denominator is the total number of months for which patients stayed free of the condition during the study.
Therefore the estimate of λ has the form (number of patients developing condition)/(total months of condition free) and seems a natural estimator of a 'failure rate'.

3.2.6 Numerical maximisation of the likelihood: Newton-Raphson method

When the likelihood (or log-likelihood) cannot be maximised analytically (calculus, or considering graphs etc.), we can use numerical methods.
Here we consider the *Newton-Raphson algorithm*:

- It provides a simple way to find a root of an equation of the form $f(x) = 0$ where f is differentiable

- Requires an initial guess of the root, x_0
- The idea is to use a linear approximation of f in a neighbourhood of the initial guess and use this to come up with an improved guess x_1.

This is illustrated in Figure 9.

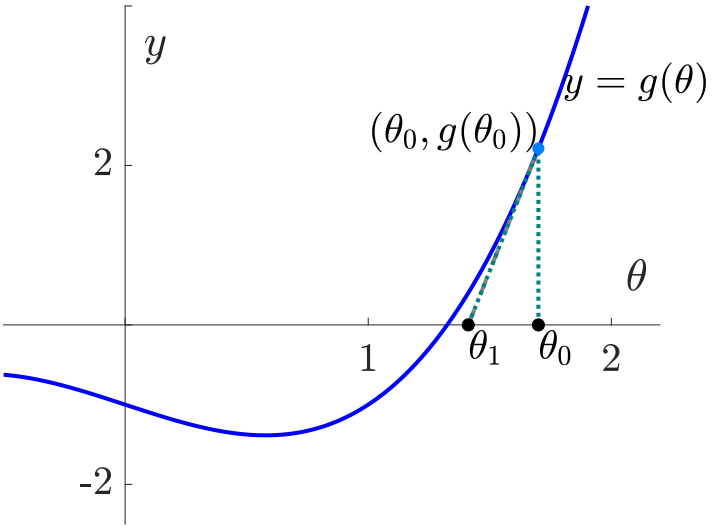


Figure 9: Illustration of one Newton–Raphson iteration for solving $f(x) = 0$. Starting from the initial guess x_0, the tangent at $(x_0, f(x_0))$ intersects the x-axis at x_1, providing the next approximation.

A linear approximation of f can be obtain using the Taylor series of f which gives

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0).$$

Finding the root of this linear equation gives our next approximation of the root of f as

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}.$$

Now we can apply the same procedure to x_1 to obtain an even better guess. In this way we can generate a sequence of estimates $x_1, x_2, \ldots, x_i, \ldots$ where

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}.$$

We repeat the process until the values of x_i converge (with sufficient accuracy). The algorithm will typically converge quickly to the root provided that the initial value x_0 is a reasonably good guess.

Example 3.14. (*highly artificial*)
Consider a population of sealed boxes each containing a random number of lead weights, X, where $X \sim \text{Poi}(\lambda)$. If a box contains 2 or more weights it will sink in water. Otherwise it will float. You select a random sample of 10 boxes, 5 of which float. What is the maximum likelihood estimator of λ?

Solution:
The maximum likelihood estimator satisfies then equation

$$2(1+\hat{\lambda})e^{-\hat{\lambda}} - 1 = 0 \quad \textbf{[Exercise]}$$

Need to solve $f(\lambda) = 2(1+\lambda)e^{-\lambda} - 1 = 0$. Now $f'(\lambda) = -2\lambda e^{-\lambda}$
and we use:

$$\lambda_{i+1} = \lambda_i + \frac{2(1+\lambda_i)e^{-\lambda_i} - 1}{2\lambda_i e^{-\lambda_i}}.$$

Also need an initial guess. Since 5 of the observed values of X are less than or equal to 1, and the other 5 are greater than or equal to 2 we could guess that the mean is around 1.5. So reasonable to take $\lambda_0 = 1.5$.
This gives the sequence of estimates: $\lambda_0 = 1.5, \lambda_1 = 1.673, \lambda_2 = 1.678$.

4 Confidence interval estimation

We will consider confidence interval (CI) estimation under the *classical inference* approach.
Let $X_1, X_2, \ldots, X_n$ denote a *random sample* from a distribution specified by a probability density function (or probability mass function if discrete) $f_X(x;\theta)$ where θ is an *unknown parameter*.
We have considered methods for obtaining a *point estimate* of θ from the random sample $\underline{X}$.
Now we consider *interval estimation* as a way of specifying *accuracy*.

Aim: Find a lower limit $\theta_L(\underline{X})$ and upper limit $\theta_U(\underline{X})$ such that

$$\mathbb{P}(\theta_L(\underline{X}) < \theta < \theta_U(\underline{X})) = 1 - \alpha$$

for $0 < \alpha < 1$.

The interval $(\theta_L(\underline{X}), \theta_U(\underline{X}))$ is called a $100(1-\alpha)\%$ confidence interval for θ.
Important remarks:

- θ is **not** random
- But $\theta_L(\underline{X})$ and $\theta_U(\underline{X})$ **are** random
- This means that $(\theta_L(\underline{X}), \theta_U(\underline{X}))$ is a random interval.
  Each time we repeat the 'experiment' we will obtain a different interval!

Then $(\theta_L(\underline{X}), \theta_U(\underline{X}))$ is a $100(1-\alpha)\%$ confidence interval if and only if $\mathbb{P}(\theta \in (\theta_L(\underline{X}), \theta_U(\underline{X}))) = (1-\alpha)$.
Let's look at a motivating example

Example 4.1. Construct a 95% confidence interval for population mean μ (assume σ^2 is known) when the sample size n is large. Note that we don't make any assumptions about the distribution of the population.

Solution:
We know that $\bar{X}$ is a good point estimator for μ. To obtain a confidence interval for μ we need information about the distribution of $\bar{X}$.

Remember, since n is large the central limit theorem gives

$$\bar{X} \dot{\sim} N\left(\mu, \frac{\sigma^2}{n}\right) \quad \Rightarrow \quad \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \dot{\sim} \mathrm{N}(0,1)$$

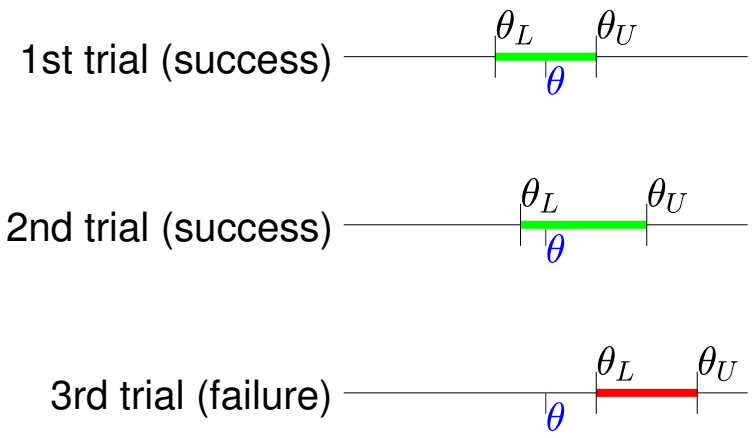


Figure 10: Illustration of three confidence interval realizations on the real line. Each line represents a different trial: the first two intervals contain the true parameter θ (success), while the third does not (failure). The endpoints of each interval are labelled as θ_L and θ_U.

$$\Rightarrow \mathbb{P}\left(-1.96 < \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} < 1.96\right) \approx 0.95 \quad \text{(Tables)}$$

Rearranging the inequality we obtain

$$\mathbb{P}\left(\bar{X} - 1.96\frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + 1.96\frac{\sigma}{\sqrt{n}}\right) \approx 0.95$$

So for large n and σ^2 known

$$\left(\bar{X} - 1.96\frac{\sigma}{\sqrt{n}}, \quad \bar{X} + 1.96\frac{\sigma}{\sqrt{n}}\right)$$

is *approximately* a 95% confidence interval for μ.

Notice that this is of the form $(\theta_L(\underline{X}), \theta_U(\underline{X}))$.

Remark 4.1. Further observations based on Example 4.1.

1. The interval $\left(\bar{X} - 1.96\frac{\sigma}{\sqrt{n}}, \quad \bar{X} + 1.96\frac{\sigma}{\sqrt{n}}\right)$ is **random**.

   Before we observe the data, the probability is 95% that the interval will contain the unknown μ. *After* observing the data, we have a value $\bar{x}$ and a specific interval

   $$\left(\bar{x} - 1.96\frac{\sigma}{\sqrt{n}}, \quad \bar{x} + 1.96\frac{\sigma}{\sqrt{n}}\right).$$

   This either contains μ or doesn't contain μ (and we are not certain about this!). Nevertheless, we can say that we are 95% confident that the interval contains μ. (If we repeat the process for a large number of times, on 95% of occasions the interval will contain μ.)

2. The confidence interval derived above is a 95% confidence interval for μ. In general we can obtain a $(1-\alpha)100\%$ confidence interval for μ using $z_{\alpha/2}$ instead of 1.96, i.e.

$$\left(\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \quad \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

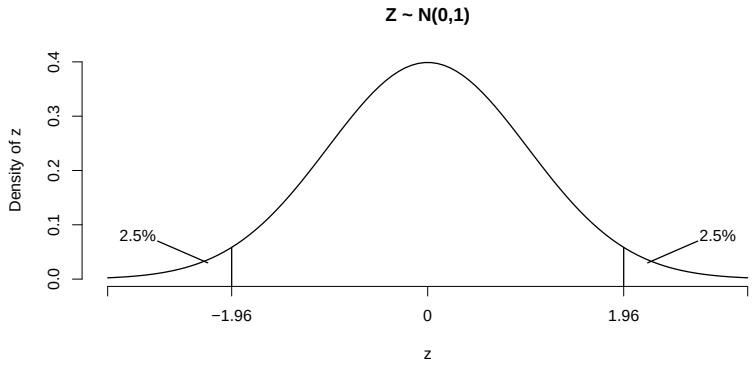
For example we get:

- a 90% confidence interval for μ with $z_{0.05} = 1.645$
- a 99% confidence interval for μ with $z_{0.005} = 2.576$
- a 50% confidence interval for μ with $z_{0.25} = 0.675$

3. In Example 4.1 we derived an *equal-tailed* confidence interval, i.e.

$$\mathbb{P}\left(\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} < -1.96\right) = \mathbb{P}\left(\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} > 1.96\right) \approx 0.025$$

We could have used the probability



$$\mathbb{P}\left(-1.75 < \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} < 2.33\right) \approx 0.95$$

to obtain an (*approximate*) 95% confidence interval for μ:

$$\left(\bar{X} - 2.33\frac{\sigma}{\sqrt{n}},\ \bar{X} + 1.75\frac{\sigma}{\sqrt{n}}\right).$$

Alternatively,

$$\mathbb{P}\left(-1.645 < \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} < \infty\right) \approx 0.95$$

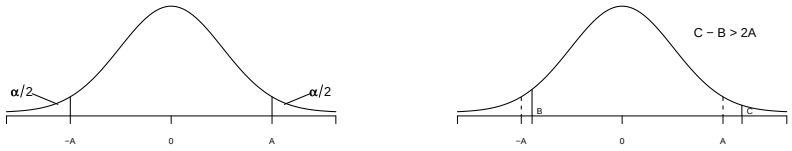
implies

$$\left(-\infty,\ \bar{X} + 1.645\frac{\sigma}{\sqrt{n}}\right)$$

is an (*approximate*) 95% confidence interval for μ. This is an example of a *one-sided confidence interval*.

4. Here it seems 'natural' to construct an equal-tailed confidence interval since the *sampling distribution* of $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$ is (approximately) symmetric.

Also, in this case, the equal-tailed confidence interval is the *narrower* 95% confidence interval (i.e. the most 'accurate').



4.1 Pivotal quantities

One standard approach to constructing a confidence interval is the pivotal method. This relies on using a *pivotal quantity*. A pivotal quantity is a function h with the following properties:

1. $h(\underline{X}, \theta)$ is a function of both the sample $\underline{X}$ and the parameter θ.
2. $h(\underline{X}, \theta)$ has a known (or *approximately* known) distribution that does not depend on θ.
   We use this knowledge to find constants a and b such that
   $$\mathbb{P}(a < h(\underline{X}, \theta) < b) = 1 - \alpha.$$
3. $h(\underline{X}, \theta)$ is monotonic (often linear) function of θ.
   So we can invert the inequality $a < h(\underline{X}, \theta) < b$ to produce an inequality with θ 'in the middle'.

Example 4.2 (Example 4.1 continued)**.** In Example 4.1 we can build the same confidence interval using the terminology of pivotal quantities. Here $\theta = \mu$ is our parameter of interest and the pivotal quantity is
$$h(\underline{X}, \mu) = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}.$$
This has the required properties:

1. $h(\underline{X}, \mu)$ is a function of $\underline{X}$ (via $\bar{X}$) and μ.
2. By the central limit theorem $h(\underline{X}, \mu)$ has an approximately normal distribution and this does not depend on μ.
3. In this case $h(\underline{X}, \theta)$ is a linear function of μ.

Suppose that $h(\underline{X}, \theta)$ is a pivotal quantity then we can use the (approximate) distribution of $h(\underline{X}, \theta)$ to find quantiles x^l_α, x^u_α such that
$$\mathbb{P}(x^l_\alpha \leq h(\underline{X}, \theta) \leq x^u_\alpha) = 1 - \alpha. \tag{3}$$
Note that since the distribution of $h(\underline{X}, \theta)$ does not depend on θ we have that x^l_α, x^u_α also do not depend on θ.
Now since $h(\underline{X}, \theta)$ is monotonic we can rearrange (3) to find $\theta_L(\underline{X}), \theta_U(\underline{X})$ such that
$$\mathbb{P}(\theta_L(\underline{X}) < \theta < \theta_U(\underline{X})) = 1 - \alpha$$
So far we have a method for constructing a confidence interval for μ based on
$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \dot{\sim} \mathrm{N}(0, 1) \quad \text{for } \textbf{large } n \text{ and } \textbf{known } \sigma^2$$
(no distributional assumptions).

How can we construct confidence intervals for μ (or other parameters) when:

- n is not large?
- σ is not known?
- population has a particular distribution?

Strategy: Try to find a *pivotal quantity* based on the maximum likelihood estimator of the parameter and then find its *sampling distribution.*

4.2 Sampling from a normal $\mathrm{N}(\mu, \sigma^2)$ population

Suppose $X_1, X_2, \ldots, X_n$ is a random sample from a $\mathrm{N}\left(\mu, \sigma^2\right)$ distribution with both μ and σ^2 *unknown.*
Remember:

- maximum likelihood estimator for μ is $\hat{\mu}(\underline{X}) = \bar{X}$
- maximum likelihood estimator for σ^2 is $\widehat{\sigma}^2(\underline{X}) = \frac{1}{n}\sum_i (X_i - \bar{X})^2 = \frac{n-1}{n}S^2$

Sampling distributions for $\bar{X}$ and S^2:

(1) $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$

(2) $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$

(3) $\bar{X}$ and S^2 are **independent** (*remember this*)

4.3 Important distributions

Chi-square with n degrees of freedom (χ^2_n)

1. χ^2_n is a particular type of Gamma distribution:

$$\chi^2_n = \mathrm{Gamma}\left(\frac{n}{2}, \frac{1}{2}\right).$$

   Note: n is a *parameter* in the density for χ^2_n.
   Calculations involving χ^2_n random variables are done using statistical tables.

2. χ^2_n has the **same distribution** as the sum of n i.i.d. χ^2_1 random variables (since the sum of i.i.d. Gamma random variables is another Gamma random variables).

3. **Key fact**: If $Z \sim \mathrm{N}(0,1)$, then $Z^2 \sim \chi^2_1$. It follows from this and from (2) that if $Z_1, Z_2, ..., Z_n$ are i.i.d. $\mathrm{N}(0,1)$ random variables, then

$$Z_1^2 + Z_2^2 + \cdots + Z_n^2 \sim \chi^2_n.$$

4. If the populations distribution is $\mathrm{N}\left(\mu, \sigma^2\right)$ then

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}.$$

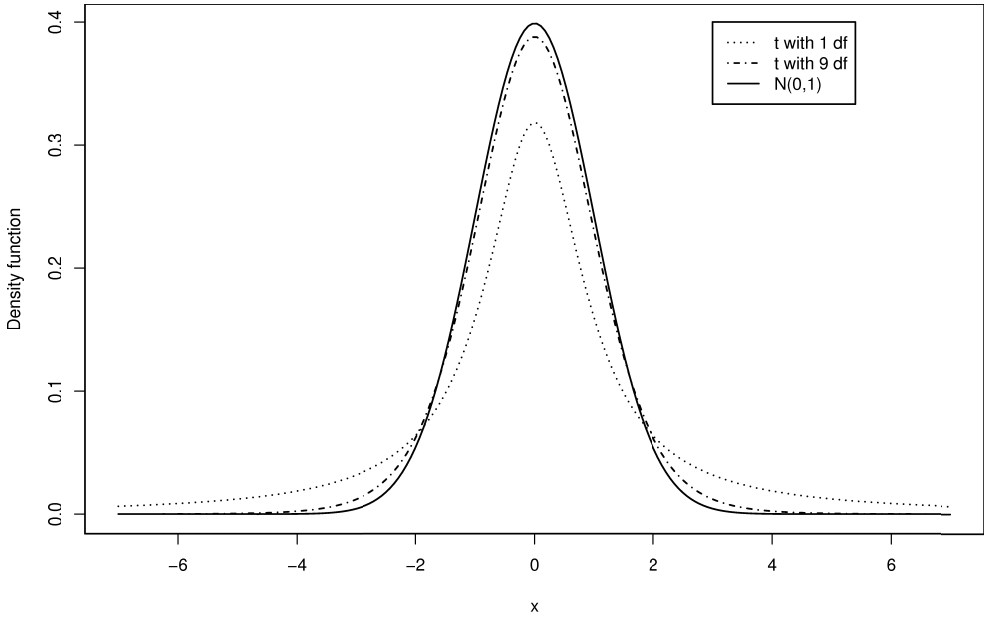


Figure 11: A plot of the probability density function for a t-distribution.

t-distribution with n degrees of freedom (t_n) The t distribution (or *Student's* t – named after the pseudonym 'Student' that W.S. Gosset used) is a continuous distribution.
The density for the t distribution with n degrees of freedom is given by

$$f(x;n) = \frac{\Gamma(\frac{n+1}{2})}{\sqrt{\pi n}\Gamma(\frac{n}{2})}\left(1+\frac{x^2}{n}\right)^{-\frac{n+1}{2}} \qquad \text{for} \quad -\infty < x < \infty.$$

A plot of the probability density function is given in Figure 11.
Note that:

1. The t distribution is symmetric around zero, with longer tails than the $\mathrm{N}(0,1)$.
2. As $n \to \infty$, the t distribution approaches the $\mathrm{N}(0,1)$ distribution.
3. The values of its cumulative distribution function are tabulated (e.g. see *NCST* p42–45).

The t distribution with n degrees of freedom can arise in the following way:

Theorem 4.1. *If $U \sim \mathrm{N}(0,1)$ and $V \sim \chi^2_n$ and U and V are **independent**, then the variable*

$$T = \frac{U}{\sqrt{V/n}} \tag{4}$$

has a t distribution with n degrees of freedom.

We will apply this characterisation to obtain the following key fact.

Corollary 4.1. *Suppose that* $\underline{X}$ *is a random sample from* $\mathrm{N}\left(\mu, \sigma^2\right)$*, then*

$$\frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t_{n-1}.$$

Proof of Corollary 4.1. We know that $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim \mathrm{N}\left(0,1\right)$ and $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$ and that these statistics are **independent**. So fact (2) above implies that

$$\frac{\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}}{\sqrt{\frac{S^2}{\sigma^2}}} = \frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t_{n-1}.$$

□

F distribution with (n_1, n_2) degrees of freedom (F_{n_1,n_2})

(1) An F random variable is constructed from the quotient of two χ^2 random variables.

In particular, suppose that $U \sim \chi^2_{n_1}$ and $V \sim \chi^2_{n_2}$ and that U and V are **independent**, then the distribution of the random variable

$$F = \frac{U/n_1}{V/n_2}$$

is called an F distribution with (n_1, n_2) degrees of freedom.

(2) An F random variable is always positive and the shape of the density is similar to the shape of a χ^2 density.

Calculations involving the F distribution are done using statistical tables.

4.4 Confidence intervals for means for μ - normal case

4.4.1 Population variance σ^2 <u>known</u>

Let $X_1, X_2, \ldots, X_n$ denote a random sample from $\mathrm{N}\left(\mu, \sigma^2\right)$. When the population variance σ^2 is *known*, the pivotal quantity $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim \mathrm{N}\left(0,1\right)$ (exactly).
A $(1-\alpha)100\%$ equal-tailed confidence interval for μ is given by

$$\left(\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \quad \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

where $z_{\frac{\alpha}{2}}$ is the appropriate percentage point of the $\mathrm{N}\left(0,1\right)$ distribution, i.e. it is such that $\mathbb{P}\left(Z > z_{\alpha/2}\right) = 1 - \Phi\left(z_{\alpha/2}\right) = \frac{\alpha}{2}$.

4.4.2 Population variance σ^2 <u>not known</u>

We can use the pivotal quantity $h(\underline{X}, \mu) = \frac{\bar{X}-\mu}{S/\sqrt{n}}$. To see that this is a pivotal quantity we observe:

1. The quantity is a function of both the data $\underline{X}$ and the parameter of interest μ.
2. By Corollary 4.1, $h(\underline{X}, \mu)$ is t-distributed with $n-1$ degrees of freedom.
3. Since $h(\underline{X}, \mu)$ is linear in μ it is also monotonic in μ.

Then, a $(1-\alpha)100\%$ equal-tailed confidence interval for μ is given by:

$$\left(\bar{X} - t_{n-1,\frac{\alpha}{2}}\frac{S}{\sqrt{n}}, \quad \bar{X} + t_{n-1,\frac{\alpha}{2}}\frac{S}{\sqrt{n}}\right)$$

where $t_{n-1,\frac{\alpha}{2}}$ is the point of the t_{n-1} distribution such that

$$\mathbb{P}(T > t_{n-1,\frac{\alpha}{2}}) = \frac{\alpha}{2}, \quad \text{with } T \sim t_{n-1}.$$

For example, 95% equal-tailed confidence interval for μ is given by

$$\left(\bar{X} - t_{n-1,0.025}\frac{S}{\sqrt{n}}, \quad \bar{X} + t_{n-1,0.025}\frac{S}{\sqrt{n}}\right) \qquad (\textit{use Tables})$$

Remark 4.2.
1. This result is *exact* for normal populations. It also gives an approximate confidence interval for non-normal populations.
2. For large n, the t_n distribution can be approximated by the $\mathrm{N}(0,1)$ distribution (e.g. $t_{n-1,0.025} \approx 1.96$).

Example 4.3. The population distribution of the length (measured in microns) of a certain kind of bacteria is considered to be normally distributed with *known* variance 0.25 and unknown mean μ. Eight bacteria are selected at random and their lengths measured. The resulting data are:

6.3 7.3 6.6 6.8 8.0 7.6 7.1 7.0

(a) Calculate a 90% confidence interval for the population mean μ.

(b) Suppose now that the population variance is unknown. Calculate a 90% confidence interval for μ in this case.

Solution

(a) Since the data is normally distributed with known variance we can use the pivotal quantity

$$\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim \mathrm{N}(0,1).$$

Note this is exact. Therefore we can find $z_{0.05} = 1.6449$ (from the statistical tables) such that

$$\mathbb{P}\left(-z_{0.05} \leq \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \leq z_{0.05}\right) = 0.9.$$

Rearranging this gives

$$\mathbb{P}\left(\bar{X} - z_{0.05}\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{0.05}\frac{\sigma}{\sqrt{n}}\right) = 0.9.$$

Therefore a 90% equal-tailed confidence interval is

$$\left(\bar{X} - z_{0.05}\frac{\sigma}{\sqrt{n}}, \bar{X} + z_{0.05}\frac{\sigma}{\sqrt{n}}\right).$$

We are given that $\sigma = 0.5$ and from the data we have

$$\begin{aligned} n &= 8 \\ \bar{x} &= 7.0875 \end{aligned}$$

Then we have

$$\begin{aligned} \bar{x} - z_{0.05}\frac{\sigma}{\sqrt{n}} &= 7.0875 - \frac{(1.6449)(0.5)}{\sqrt{8}} = 6.796728 \\ \bar{x} + z_{0.05}\frac{\sigma}{\sqrt{n}} &= 7.0875 + \frac{(1.6449)(0.5)}{\sqrt{8}} = 7.378272. \end{aligned}$$

Therefore a 90% equal-tailed confidence interval in this case is

$$(6.797, 7.378).$$

(b) If now the variance is unknown then we use the pivotal quantity

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$$

This leads to the 90% equal-tailed confidence interval

$$\left(\bar{X} - t_{n-1,0.05}\frac{S}{\sqrt{n}}, \quad \bar{X} + t_{n-1,0.05}\frac{S}{\sqrt{n}}\right)$$

From the data we have

$$\begin{aligned} n &= 8 \\ \bar{x} &= 7.0875 \\ s &= 0.5462535 \end{aligned}$$

and from the tables we have $t_{7,0.05} = 1.894579$. Then we have

$$\begin{aligned} \bar{x} - t_{n-1,0.05}\frac{s}{\sqrt{n}} &= 6.7216 \\ \bar{x} + t_{n-1,0.05}\frac{s}{\sqrt{n}} &= 7.4534. \end{aligned}$$

Therefore we have the 90% equal-tailed confidence interval

$$(6.722, 7.453).$$

4.5 Confidence intervals for σ^2

We use $\frac{(n-1)S^2}{\sigma^2}$ as a pivotal quantity for σ^2 since:

- it is a function of $\underline{X}$ and σ^2
- it has a *known* distribution (χ^2_{n-1}) which does not depend on σ^2
- it is monotonic in the parameter σ^2.

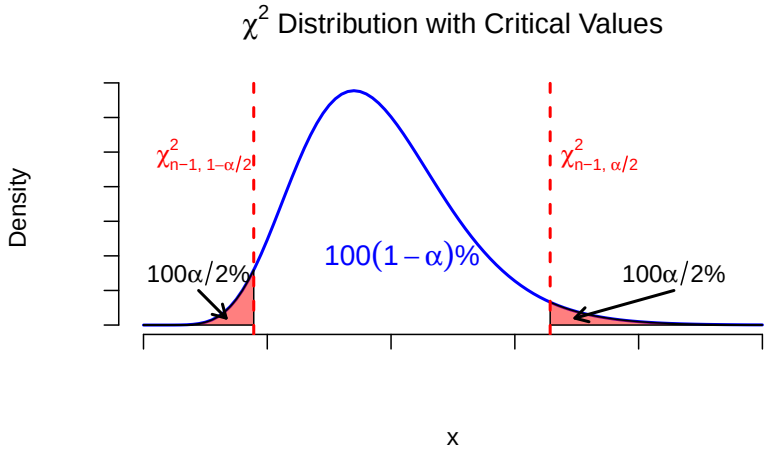


Figure 12: A plot of the probability density function for a χ^2 distribution with important quantiles marked on for the construction of $100(1-\alpha)\%$ Confidence intervals.

Use the quantile tables of the χ^2 distribution (e.g. NCST p40, 41) to find values $\chi^2_{n-1,1-\frac{\alpha}{2}}$ and $\chi^2_{n-1,\frac{\alpha}{2}}$ such that

$$\mathbb{P}(Y > \chi^2_{n-1,\frac{\alpha}{2}}) = \frac{\alpha}{2}, \quad \text{with } Y \sim \chi^2_{n-1}.$$

This gives us the quantiles as shown in Figure 12. Using these quantiles we have

$$\mathbb{P}\left(\chi^2_{n-1,1-\frac{\alpha}{2}} < \frac{(n-1)S^2}{\sigma^2} < \chi^2_{n-1,\frac{\alpha}{2}}\right) = 1-\alpha$$

Rearrange this inequality to obtain

$$\mathbb{P}\left(\frac{(n-1)S^2}{\chi^2_{n-1,\frac{\alpha}{2}}} < \sigma^2 < \frac{(n-1)S^2}{\chi^2_{n-1,1-\frac{\alpha}{2}}}\right) = 1-\alpha$$

So, for example, a 95% equal-tailed confidence interval for σ^2 is given by

$$\left(\frac{(n-1)S^2}{\chi^2_{n-1,0.025}}, \quad \frac{(n-1)S^2}{\chi^2_{n-1,0.975}}\right)$$

Note that this is **not** the narrowest 95% confidence interval, but it is *easy* to compute and *standard*.

Example 4.4. Find a 95% confidence interval for σ^2 for a sample of size $n = 12$ (from a normal population) that has sample variance 32.93.

Solution

We will use the confidence interval

$$\left(\frac{(n-1)S^2}{\chi^2_{n-1,0.025}}, \quad \frac{(n-1)S^2}{\chi^2_{n-1,0.975}}\right)$$

from above. We have $n = 12$ so we use the tables for the χ^2 distribution with $\nu = 11$ degrees of freedom and find:

$$\chi^2_{11,0.025} = 21.92$$
$$\chi^2_{11,0.975} = 3.816.$$

This leads to the confidence interval

$$\left(\frac{11s^2}{\chi^2_{11,0.025}}, \quad \frac{11s^2}{\chi^2_{11,0.975}}\right) = \left(\frac{(11)(32.93)}{21.92}, \quad \frac{(11)(32.93)}{3.816}\right) = (16.5, 94.924)$$

Example 4.5. Now find a 90% confidence interval for σ^2 for a sample of size $n = 25$ (from a normal population) where the sample variance 10.

Solution We use the confidence interval

$$\left(\frac{(n-1)S^2}{\chi^2_{n-1,\frac{\alpha}{2}}}, \quad \frac{(n-1)S^2}{\chi^2_{n-1,1-\frac{\alpha}{2}}}\right)$$

with $n = 25$, $\alpha = 0.1$ and $s^2 = 10$. From the tables we have

$$\chi^2_{n-1,\frac{\alpha}{2}} = \chi^2_{24,0.05} = 36.42,$$
$$\chi^2_{n-1,1-\frac{\alpha}{2}} = \chi^2_{24,0.95} = 13.85.$$

This gives

$$\left(\frac{24s^2}{\chi^2_{24,0.05}}, \quad \frac{24S^2}{\chi^2_{24,0.95}}\right) = \left(\frac{(24)(10)}{36.42}, \quad \frac{(24)(10)}{13.85}\right) = (6.59, \quad 17.33)\,.$$

Remark 4.3. Tables only give percentage points for $n \leq 100$.
What do we use for $n > 100$?

Recall:
if $Z_1, Z_2, \ldots, Z_n$ are i.i.d. $\mathrm{N}(0,1)$ random variables,
then $Z = Z_1^2 + Z_2^2 + \ldots + Z_n^2 \sim \chi^2_n$, with $\mathbb{E}(Z) = n$ and $\mathbb{V}\mathrm{ar}(Z) = 2n$.
Then CLT gives

$$\frac{Z - n}{\sqrt{2n}} \dot{\sim} \mathrm{N}(0,1)\,.$$

Application:

$$\frac{\frac{(n-1)S^2}{\sigma^2} - (n-1)}{\sqrt{2(n-1)}} \dot{\sim} \mathrm{N}(0,1)\,.$$

Use this as an approximate pivotal quantity.

4.6 Confidence interval estimation for two populations

4.6.1 Comparing two populations

We have two populations with means μ_1, μ_2 and variances σ_1^2, σ_2^2 respectively and we would like to compare these populations.

- What can we say about the difference $\mu_1 - \mu_2$?
- What can we say about the ratio σ_1^2/σ_2^2?

We will assume that we have two **independent** samples of size n_1 and n_2 from the first and second population respectively.

4.6.2 Comparison of means

How should we *estimate* the difference $\mu_1 - \mu_2$?
We know that

$$\bar{X}_1 = \frac{X_{1,1} + X_{1,2} + ... + X_{1,n_1}}{n_1}$$

is an unbiased estimator for μ_1, and likewise

$$\bar{X}_2 = \frac{X_{2,1} + X_{2,2} + ... + X_{2,n_2}}{n_2}$$

is an unbiased estimator for μ_2, so $\bar{X}_1 - \bar{X}_2$ is an unbiased estimator for $\mu_1 - \mu_2$.
Can we develop a confidence interval for the difference $\mu_1 - \mu_2$?

Case 1: (large sample sizes) Suppose that the sample sizes n_1 and n_2 are *large*. The Central Limit Theorem tells us that

$$\bar{X}_1 \dot{\sim} N\left(\mu_1, \frac{\sigma_1^2}{n_1}\right) \quad \text{and} \quad \bar{X}_2 \dot{\sim} \mathrm{N}\left(\mu_2, \frac{\sigma_2^2}{n_2}\right).$$

Since the samples are independent, the sample means $\bar{X}_1$ and $\bar{X}_2$ are **independent** too, and the sum of independent (approximately) normal random variables are also (approximately) normal. So we have

$$\bar{X}_1 - \bar{X}_2 \dot{\sim} N\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right).$$

(NB: $-\bar{X}_2 \dot{\sim} N(-\mu_2, \frac{\sigma_2^2}{n_2})$).
It follows that

$$\frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \dot{\sim} \mathrm{N}(0,1)$$

and we can use this to develop the following 95% confidence interval for $\mu_1 - \mu_2$:

$$\left(\bar{X}_1 - \bar{X}_2 - 1.96\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}, \bar{X}_1 - \bar{X}_2 + 1.96\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}\right) \quad \text{(Exercise)}$$

Remark 4.4. In all above results, if σ_1^2, σ_2^2 are *not known* we can replace them with S_1^2, S_2^2 for an approximate confidence interval since the Central Limit Theorem will still hold.

Case 2: (normal populations) Now suppose that we know the populations are $\mathrm{N}\left(\mu_1, \sigma_1^2\right)$ and $\mathrm{N}\left(\mu_2, \sigma_2^2\right)$ respectively. In this case we know the **exact** distributions of $\bar{X}_1$ and $\bar{X}_2$:

$$\bar{X}_1 \sim \mathrm{N}\left(\mu_1, \frac{\sigma_1^2}{n_1}\right) \qquad \text{and} \qquad \bar{X}_2 \sim \mathrm{N}\left(\mu_2, \frac{\sigma_2^2}{n_2}\right)$$

and

$$\frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim \mathrm{N}(0,1). \tag{5}$$

If σ_1^2 and σ_2^2 are **known** then we can use (5) as a pivotal quantity and we obtain the following 95% confidence interval for $\mu_1 - \mu_2$:

$$\left(\bar{X}_1 - \bar{X}_2 - 1.96\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}, \bar{X}_1 - \bar{X}_2 + 1.96\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}\right).$$

If σ_1^2 and σ_2^2 are **not known** then we need to use a *different* pivotal quantity.
We make the **additional assumption** that $\sigma_1^2 = \sigma_2^2 = \sigma^2$.
In this case we know that

$$\frac{(n_1-1)S_1^2}{\sigma^2} \sim \chi^2_{n_1-1} \quad \text{and} \quad \frac{(n_2-1)S_2^2}{\sigma^2} \sim \chi^2_{n_2-1}$$

and that $\frac{(n_1-1)S_1^2}{\sigma^2}$ and $\frac{(n_2-1)S_2^2}{\sigma^2}$ are **independent** (since the samples are independent).
It follows that

$$\frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{\sigma^2} \sim \chi^2_{n_1+n_2-2}. \tag{6}$$

(Note: This will **not** hold if the populations have *different* variances!).
It also follows from standard results that $\bar{X}_1 - \bar{X}_2$ and $\frac{(n_1-1)S_1^2+(n_2-1)S_2^2}{\sigma^2}$ are independent.
Using this fact and the distributional results (5) and (6) from above, we obtain the following pivotal quantity:

$$\frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\sigma^2(\frac{1}{n_1} + \frac{1}{n_2})}} \Big/ \sqrt{\frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{\sigma^2(n_1+n_2-2)}} = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{S_p^2(\frac{1}{n_1} + \frac{1}{n_2})}} \sim t_{n_1+n_2-2}$$

where $S_p^2 = \frac{(n_1-1)S_1^2+(n_2-1)S_2^2}{n_1+n_2-2}$. S_p^2 is called the *pooled variance*.
Using this pivotal quantity we obtain the 95% confidence interval for $\mu_1 - \mu_2$ as: (*Exercise*)

$$\left(\bar{X}_1 - \bar{X}_2 - t^* \sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}, \quad \bar{X}_1 - \bar{X}_2 + t^* \sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)} \right).$$

where $t^* = t_{n_1+n_2-2,0.025}$ is the appropriate percentage point of the t distribution.

Remark 4.5. The validity of this construction depends on the assumption that the populations are normally distributed and that the variances are equal:

- You should check these assumptions by looking at *plots* of the data *before* applying the result.
- If the plots appear to be 'bell-shaped' with similar spread, then the procedure outlined above will yield a reasonable confidence interval even if the populations are only approximately normal.
- However, if the variances (i.e. the spread of the data) appear to be significantly different, this approach should not be used.

Example 4.6. Potential buyers on an Internet company for personal auctioning can bid for an item over a period of time; they are also offered a 'buy-it-now' option at a fixed price. We want to investigate whether selling prices tend to be lower or higher under bidding or the 'buy-it-now' option. The data below are selling prices (in £) for a certain item over a period of one week:
'Buy-it-now': `459 449 449 464 482 472 434 439`
Bid: `444 473 489 479 433 464 452 479 456 470 459 474 464 469 484 446 459 424`

(a) Draw a suitable plot for comparing the two data sets.

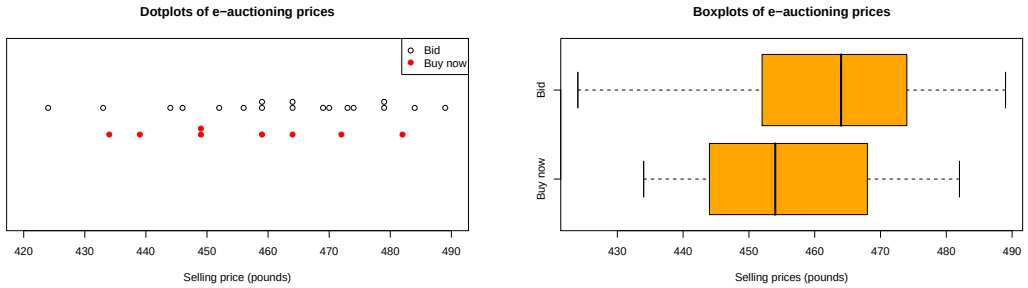
(b) Find a 90% confidence interval for the true difference in the mean price under these two options. Based on this interval, would you support that selling prices tend to be different under the two options?

(c) What are the assumptions that the above confidence interval is based on?

(d) Check these assumptions using the plot in (a).

Solution

(a) We can use dotplots on the same scale or boxplots



(b) We will use the confidence interval

$$\left(\bar{X}_1 - \bar{X}_2 - t^* \sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}, \quad \bar{X}_1 - \bar{X}_2 + t^* \sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}\right).$$

with $t^* = t_{n_1+n_2-2,\alpha/2}$ and $\alpha = 0.1$. Here 1 will denote the 'Buy now' data and 2 the 'bid' data.

From the data we have

$$\begin{aligned} n_1 &= 8, & \bar{x}_1 &= 456, & s_1^2 &= 268. \\ n_2 &= 18, & \bar{x}_2 &= 462.1, & s_2^2 &= 305.6. \end{aligned}$$

Therefore

$$s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2} = \frac{(7)(268)+(17)(305.6)}{24} = 294.7.$$

From the statistical tables $t_{24,0.05} = 1.711$. Therefore we have a 90% equal-tailed confidence interval

$$\left(\bar{x}_1 - \bar{x}_2 - t^* \sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}, \quad \bar{x}_1 - \bar{x}_2 + t^* \sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}\right)$$

$$= \left(456 - 462.1 - (1.714)\sqrt{294.7\left(\frac{1}{8} + \frac{1}{18}\right)}, \quad 456 - 462.1 + (1.714)\sqrt{294.7\left(\frac{1}{8} + \frac{1}{18}\right)}\right)$$

$$= (-18.6, 6.37).$$

(c) Here we have assumed that the populations are normally distributed and have the same population variance. We are also assuming that the two samples are independent.

(d) The graphs in part (a) do not present any evidence against the assumption of normal distributions or of equal variances.

4.6.3 Comparison of variances

Since σ_1^2 and σ_2^2 are measures of spread (or scale) it is natural to consider the *ratio* σ_1^2/σ_2^2 when comparing variances.
S_1^2 is a 'good' estimator for σ_1^2 and S_2^2 is a 'good' estimator for σ_2^2, so the ratio S_1^2/S_2^2 is a reasonable estimator for σ_1^2/σ_2^2.

To develop a *confidence interval* for the ratio σ_1^2/σ_2^2, we need to *assume that both populations are normally distributed* and use the F-distribution.
Recall that if $U \sim \chi^2_{n_1}$ and $V \sim \chi^2_{n_2}$ are independent random variables, then $F = \frac{U/n_1}{V/n_2}$ has an F_{n_1,n_2} distribution.
Apply this using:

$$\frac{(n_1-1)S_1^2}{\sigma_1^2} \sim \chi^2_{n_1-1} \quad \text{and} \quad \frac{(n_2-1)S_2^2}{\sigma_2^2} \sim \chi^2_{n_2-1}.$$

So

$$\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} = \frac{S_1^2}{S_2^2}\frac{\sigma_2^2}{\sigma_1^2} \sim F_{n_1-1,n_2-1}.$$

This provides a *pivotal quantity* for the confidence interval for the ratio σ_1^2/σ_2^2.
For an (equal-tailed) 95% confidence interval for the ratio σ_1^2/σ_2^2, we use

$$\mathbb{P}\left(F_{n_1-1,n_2-1,0.975} < \frac{S_1^2}{S_2^2}\frac{\sigma_2^2}{\sigma_1^2} < F_{n_1-1,n_2-1,0.025}\right) = 0.95.$$

where $F_{n_1-1,n_2-1,p}$ are percentage points for the appropriate F distribution (use Tables).
Note: Values for $F_{n_1-1,n_2-1,0.975}$ are not in the tables.
To find this value you must use the fact that (in general) for any n and m, $F_{n,m,0.975} = \frac{1}{F_{m,n,0.025}}$.

We can now show (*Exercise*) that the 95% (equal-tailed) confidence interval for σ_1^2/σ_2^2 is given by

$$\left(\frac{S_1^2/S_2^2}{F_{n_1-1,n_2-1,0.025}}, \frac{S_1^2/S_2^2}{F_{n_1-1,n_2-1,0.975}}\right).$$

Example 4.7 (Example 4.6 (cont.))**.** For the auctioning data in Example 4.6 check the assumption of equal population variances between the prices under the two selling options using a suitable 95% confidence interval.
Solution Recall

$$s_1^2 = 268, \quad s_2^2 = 305.6$$

From the statistical tables

$$F_{7,17,0.025} = 3.156.$$

To calculate the value $F_{7,17,0.975}$ we use the fact

$$F_{n,m,0.975} = \frac{1}{F_{m,n,0.025}}$$

However the value of $F_{17,7,0.025}$ is not in the tables so we need to use linear interpolation:

$$\begin{aligned} F_{17,7,0.025} &= \frac{24-17}{24-12}F_{12,7,0.025} + \frac{17-12}{24-12}F_{24,7,0.025} = \frac{7}{12}F_{12,7,0.025} + \frac{5}{12}F_{24,7,0.025} \\ &= \frac{7}{12}(4.666) + \frac{5}{12}(4.415) = 4.561417 \end{aligned}$$

As a reminder here we used the approximation

$$u(x) = \frac{b-x}{b-a}u(a) + \frac{x-a}{b-a}u(b) = u(a) + \frac{x-a}{b-a}(u(b) - u(a)).$$

Note this is just an approximation and we can compare with the value $F_{17,7,0.025} = 4.520627$ obtained in R. which gives

$$F_{7,17,0.975} = \frac{1}{F_{17,7,0.025}} = \frac{1}{4.561417} = 0.219.$$

Now we can construct the confidence interval

$$\left(\frac{s_1^2/s_2^2}{F_{7,17,0.025}}, \frac{s_1^2/s_2^2}{F_{7,17,0.975}}\right) = (0.278, 4.005).$$

Since 1 is included in this interval we have do not have sufficient evidence to reject the assumption of equal variance populations.

4.6.4 Paired data

The analyses given above depend in a crucial way on the assumption that the 2 samples are *independent*. This is not always the case!

Example 4.8. Suppose that 10 subjects have their blood pressure measured before and after taking a certain drug. For subject k, a *pair* of measurements $(X_{1,k}, X_{2,k})$ is recorded, representing subject k's blood pressure before and after the medicine. The variables $X_{1,k}$ and $X_{2,k}$ are *not* independent, and in fact, the pair $(X_{1,k}, X_{2,k})$ should be considered as a *single observation*.
We want an estimate (CI) of *difference* $\mu_1 - \mu_2$, where μ_1 is the mean blood measure before taking the drug and μ_2 is the mean blood pressure after taking the drug.

Model and assumptions

- We begin with a random sample of *paired data*, i.e. we have a sample of pairs

  $$(X_{1,1}, X_{2,1}), (X_{1,2}, X_{2,2}), ..., (X_{1,n}, X_{2,n})$$

  such that the *pairs* are i.i.d. with the same *joint distribution*.
- Let μ_1 and μ_2 denote the means of the marginal distributions for $X_{1,k}$ and $X_{2,k}$ respectively. Likewise, let σ_1^2 and σ_2^2 denote the variances for $X_{1,k}$ and $X_{2,k}$ respectively.
- Furthermore, we assume that $X_{1,k}$ and $X_{2,k}$ are **not** independent

Note:

$$\bar{X}_1 - \bar{X}_2 = \frac{1}{n}\sum_{k=1}^{n} X_{1,k} - \frac{1}{n}\sum_{k=1}^{n} X_{2,k} = \frac{1}{n}\sum_{k=1}^{n}(X_{1,k} - X_{2,k})$$

is an *unbiased* estimator of $\mu_1 - \mu_2$, but the variance of this estimator is (usually) smaller than it is in the case of independent samples.

In particular, if $\mathbb{C}\text{ov}(\bar{X}_1, \bar{X}_2) > 0$, then

$$\begin{aligned}\mathbb{V}\text{ar}(\bar{X}_1 - \bar{X}_2) &= \mathbb{V}\text{ar}(\bar{X}_1) + \mathbb{V}\text{ar}(\bar{X}_2) - 2\mathbb{C}\text{ov}(\bar{X}_1, \bar{X}_2) \\ &= \frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{n} - 2\mathbb{C}\text{ov}(\bar{X}_1, \bar{X}_2) < \frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{n}.\end{aligned}$$

Confidence interval for $\mu_1 - \mu_2$ (paired data) Let $D_k = X_{1,k} - X_{2,k}$, then $\underline{D} = (D_1, D_2, ..., D_n)$ is a random sample and $\bar{D} = \bar{X}_1 - \bar{X}_2$ is an unbiased estimator for $\mu_1 - \mu_2$.

(1) If n is **large**, then

$$\frac{\bar{D} - (\mu_1 - \mu_2)}{\sigma_D/\sqrt{n}} \dot{\sim} \mathrm{N}(0,1)$$

and $(\bar{D} - 1.96\frac{\sigma_D}{\sqrt{n}}, \bar{D} + 1.96\frac{\sigma_D}{\sqrt{n}})$ is an approximate 95% confidence interval for $\mu_1 - \mu_2$. If σ_D is *unknown*, it can be replaced by S_D.

(2) If n is small and the random sample $D_1, D_2, ..., D_n$ appear to be normally distributed, then

$$\frac{\bar{D} - (\mu_1 - \mu_2)}{S_D/\sqrt{n}} \sim t_{n-1}$$

and $(\bar{D} - t_{n-1,0.025}\frac{S_D}{\sqrt{n}}, \bar{D} + t_{n-1,0.025}\frac{S_D}{\sqrt{n}})$ is an (approximate) 95% confidence interval for $\mu_1 - \mu_2$.

Example 4.9 (Example 4.8 (cont.))**.** In Example 4.8 the blood pressure of the 10 subjects measured before and after taking the drug is given by:

Subject	1	2	3	4	5	6	7	8	9	10
Before	156	159	145	160	151	143	164	157	149	160
After	155	148	146	147	140	141	148	150	149	154

Find a 95% confidence interval for the true difference in the mean blood pressure before and after taking the drug, and use the interval to determine whether or not the drug is effective.
Solution: Since the data is paired we work with the differences, $d =$ `'Before' - 'After'`

```
1 11 -1 13 11  2 16  7  0  6
```

The sample mean and sample standard deviation of the differences are given by

$$\bar{d} = 6.6, \quad s_d = 5.985.$$

Since $n = 10$ is not large, we use a t-distribution to construct the confidence interval. We find the quantile

$$t_{9,0.025} = 2.262$$

from the statistical tables. Now we can construct a 95% confidence interval as follows:

$$\left(\bar{d} - t_{n-1,0.025}\frac{S_d}{\sqrt{n}}, \bar{d} + t_{n-1,0.025}\frac{S_d}{\sqrt{n}}\right) = \left(6.6 - \frac{(2.262)(5.985)}{\sqrt{10}}, 6.6 + \frac{(2.262)(5.985)}{\sqrt{10}}\right) = (2.32, 10.88).$$

Since $\mu_D = 0$ is not included in the confidence interval, the data suggests that there is evidence to show the drug is effective. In particular, since the confidence interval implies that μ_D is positive we have evidence that the drug lowers blood pressure.

4.6.5 Confidence intervals for Bin(m, p)

Assume that we want to estimate a *proportion* of a population having a specific characteristic. This can be expressed as a *probability* (e.g. probability next e-mail you receive contains a virus), *percentage* (e.g. percentage of people following a certain religion), or *rate* (e.g. mortality rate among pensioners of a certain age).

In many cases, we can express the above quantities as the probability p in a $\text{Bin}(m, p)$ distribution.

Suppose that X is a *single* observation from $\text{Bin}(m, p)$ (i.e., sample size is $n = 1$). The maximum likelihood estimator for p is given by

$$\hat{p}(X) = \frac{X}{m}.$$

We would like to construct a confidence interval for p which is based on the maximum likelihood estimator $\hat{p}$.

Approximate confidence interval for $\text{Bin}(m, p)$ To derive an approximate confidence interval for p, recall that if $X \sim \text{Bin}(m, p)$ and m is **large**, the CLT gives

$$X \dot{\sim} \text{N}\left(mp, mp(1-p)\right).$$

This also implies that

$$\hat{p}(X) = \frac{X}{m} \dot{\sim} N\left(p, \frac{p(1-p)}{m}\right)$$

which we can rearrange to

$$\frac{\hat{p}(X) - p}{\sqrt{\frac{p(1-p)}{m}}} \dot{\sim} \text{N}\left(0, 1\right).$$

Now, as this expression involves the **unknown** true proportion p in a non-linear manner, we can further approximate it by using the estimate $\hat{p}$ to obtain an appropriate *pivotal quantity*

$$h(\underline{X}, p) = \frac{\hat{p}(X) - p}{\sqrt{\frac{\hat{p}(X)(1-\hat{p}(X))}{m}}} \dot{\sim} \text{N}\left(0, 1\right).$$

It follows that we can construct a $(1-\alpha)100\%$ confidence interval for p, based on:

$$\begin{aligned} & \mathbb{P}\left(-z_{\alpha/2} < \frac{\hat{p}(X) - p}{\sqrt{\frac{\hat{p}(X)(1-\hat{p}(X))}{m}}} < z_{\alpha/2}\right) \approx 1 - \alpha \\ \Leftrightarrow \quad & \mathbb{P}\left(\hat{p}(X) - z_{\alpha/2}\sqrt{\frac{\hat{p}(X)(1-\hat{p}(X))}{m}} < p < \hat{p}(X) + z_{\alpha/2}\sqrt{\frac{\hat{p}(X)(1-\hat{p}(X))}{m}}\right) \approx 1 - \alpha \end{aligned}$$

Thus we obtain the $(1-\alpha)100\%$ confidence interval:

$$\left(\hat{p}(X) - z_{\alpha/2}\sqrt{\frac{\hat{p}(X)(1-\hat{p}(X))}{m}}, \quad \hat{p}(X) + z_{\alpha/2}\sqrt{\frac{\hat{p}(X)(1-\hat{p}(X))}{m}}\right)$$

confidence intervals for differences between proportions Suppose now that we want to estimate the difference between two proportions p_1 and p_2 based on two samples from the binomial populations $\text{Bin}(m_1, p_1)$ and $\text{Bin}(m_2, p_2)$.
For **large** m_1 and m_2 we can use the approximation

$$Z = \frac{(\hat{p}_1(X_1) - \hat{p}_2(X_2)) - (p_1 - p_2)}{\sqrt{\frac{p_1(1-p_1)}{m_1} + \frac{p_2(1-p_2)}{m_2}}} \dot{\sim} \text{N}\left(0, 1\right)$$

to obtain a $(1-\alpha)100\%$ confidence interval for the difference $p_1 - p_2$ of the form

$$(\hat{p}_1(X_1) - \hat{p}_2(X_2)) \pm z_{\alpha/2}\sqrt{\frac{\hat{p}_1(X_1)(1-\hat{p}_1(X_1))}{m_1} + \frac{\hat{p}_2(X_2)(1-\hat{p}_2(X_2))}{m_2}}.$$

Notice that, as before, the *unknown* proportions p_1 and p_2 have been substituted in the variance with their *estimates* $\hat{p}_1(X_1)$ and $\hat{p}_2(X_2)$, to give the above *approximate* confidence interval.

Example 4.10. This approximation can be used to report the *margin of error* in the results of opinion polls.

For example, suppose that we question $m = 1000$ people from a very large population and that $x = 300$ people respond 'yes' to some question. The problem is to estimate p, the proportion of people in the population who would answer 'yes' to the question.

We assume that X= number of people who respond 'yes' is $\text{Bin}(1000, p)$, so $\hat{p} = 0.3$.

We report that with 95% certainty, the true proportion p is between

$$0.3 - 1.96\sqrt{\frac{(0.3)(0.7)}{1000}} \quad \text{and} \quad 0.3 + 1.96\sqrt{\frac{(0.3)(0.7)}{1000}}.$$

In other words, with 95% certainty,

$$|p - \hat{p}| \leq 1.96\sqrt{\frac{(0.3)(0.7)}{1000}} \approx 0.03$$

so we say that the margin of error is 3%.

Exact confidence intervals for $\text{Bin}(m, p)$ An exact 95% (say) confidence interval for p can be obtained as follows:

We need to find values $p_L(X)$ and $p_U(X)$ such that

$$\mathbb{P}(p_L(X) < p < p_U(X)) = 0.95.$$

Suppose that $X = x$ has been observed from $\text{Bin}(m, p)$, then p_L is the value such that

$$\mathbb{P}(X \geq x | p = p_L) = \sum_{k=x}^{m} \binom{m}{k} (p_L)^k (1-p_L)^{m-k} = 0.025$$

and p_U is the value such that

$$\mathbb{P}(X \leq x | p = p_U) = \sum_{k=0}^{x} \binom{m}{k} (p_U)^k (1-p_U)^{m-k} = 0.025.$$

These equations must be solved numerically (by using tables) to give an exact 95% confidence interval for p of the form (p_L, p_U).

Example 4.11. Suppose that $m = 10$ and that $x = 4$. Let $F(k; m, p)$ denote the distribution function for the $\text{Bin}(m, p)$ distribution, i.e. for $0 \leq k \leq m$, $F(k; m, p) = \mathbb{P}(X \leq k)$ where $X \sim \text{Bin}(m, p)$.

To find p_L and p_U we must solve

$$P(X \geq 4 | p = p_L) = \sum_{k=4}^{10} \binom{10}{k} (p_L)^k (1-p_L)^{10-k} = 1 - F(3; 10, p_L) = 0.025$$

$$P(X \leq 4 | p = p_U) = \sum_{k=0}^{4} \binom{10}{k} (p_U)^k (1-p_U)^{10-k} = F(4; 10, p_U) = 0.025$$

Using the tables, we get

$$1 - F(3; 10, p_L) = 0.025 \quad \Rightarrow \quad F(3; 10, p_L) = 0.975 \quad \Rightarrow \quad p_L \approx 0.12$$

and

$$F(4; 10, p_U) = 0.025 \quad \Rightarrow \quad 1 - F(5; 10, 1 - p_U) = 0.025$$

$$\Rightarrow \quad F(5; 10, 1 - p_U) = 0.975 \quad \Rightarrow \quad 1 - p_U \approx 0.26 \quad \Rightarrow \quad p_U \approx 0.74.$$

So the 95% confidence interval for p is (0.12. 0.74).
Recall: For $p > 0.5$, $F(k; m, p) = 1 - F(m - k - 1; m, 1 - p)$.

5 Hypothesis Testing

Until now we have developed methods for estimating an unknown parameter θ either by point estimation or by confidence interval estimation. Now we consider a different question and start with a *hypothesis*.

Example 5.1. Imagine that a company states in its sales pitch that a particular model of its mobile phones lasts for 150 hours before it needs to be charged up. If you were thinking of buying one, you may like to obtain some proof that this assertion is true.

One way of doing this is to take a sample of phones and calculate the mean number of hours between charging. It would be impossible to do this for every phone produced since the population is so large, so the best that can be done is to calculate a sample mean.

A sample of 40 was taken and this produced a mean value of 147.4 hours (with a standard deviation of 12 hours). Does this mean that the manufacturer's claim has been disproved? Clearly 147.4 is less than 150 so it looks as if the manufacturer is over-estimating the time between charging.

However, it must be appreciated that this was just one sample; if another sample was taken it might give a very different result (for example, it could give a value of 152.3 hours, in which case the phones are doing better than the manufacturers claim!).

The method of *hypothesis testing* starts by making an assertion about the population, usually an assumption that the mean is equal to a stated result.

In this case it is hypothesised that the population mean μ for the mobile phones is 150 hours.

Method of Solution

Assert that the population mean is what the company claims it to be. Call this assertion H_0: the Null Hypothesis.

The following analysis will allow us to either reject this assertion or not reject it.

This can be written as

$$H_0 : \mu = 150 \quad \text{versus } H_1 : \mu \neq 150.$$

H_1 is called the Alternative Hypothesis.

Using theory developed before, assuming H_0 to be true implies that

$$\frac{\bar{X} - 150}{\frac{s}{\sqrt{n}}} \dot{\sim} \mathrm{N}(0, 1).$$

For our particular sample we can calculate a value for

$$\frac{\bar{x} - 150}{\frac{s}{\sqrt{n}}}$$

This calculates as

$$\frac{147.4 - 150}{\frac{12}{\sqrt{40}}} = -1.370.$$

Now recall that for a $\mathrm{N}(0,1)$ distribution, 95% of its values, say, are between -1.96 and 1.96.

So the observed value -1.370 (based on the sample) looks like a *typical value* from this distribution.

Therefore we do not have sufficient evidence to reject the null hypothesis that the mean time between charging for this type of phone is 150 hours.

Remark 5.1. At first glance it might appear that failing to reject H_0 is the same as accepting H_0. However these are fundamentally different: failing to reject means we do not have sufficient evidence against H_0, whereas accepting would imply we conclude H_0 is true.

Consider the hypothesis test:

$$H_0 : X \sim \mathrm{N}(0,1) \text{ versus } H_1 : X \text{ is not distributed according to } \mathrm{N}(0,1).$$

If H_0 is true then $\mathbb{P}(-3 \leq X \leq 3) \approx 0.997$ so we reject H_0 if $X \leq -3$ or $X \geq 3$.

Now suppose the true distribution of X is $\mathrm{Ber}(0.5)$. Then X never falls outside $[-3, 3]$ so we do not reject the null hypothesis. Does this imply that $X \sim \mathrm{N}(0,1)$? Clearly not— failure to reject does not confirm the null hypothesis; it only indicates insufficient evidence to reject it.

To make this idea more intuitive consider the following analogy: We can view a hypothesis test as being similar to a jury trial. The null hypothesis plays the role of the defendant and the jury are being asked to determine if there is sufficient evidence to conclude beyond a reasonable doubt that the defendant is guilty. If the jury delivers the verdict of not guilty this does not mean the defendant is innocent; it only means there was not enough evidence to convict.

5.1 Hypotheses and Test Statistics

A statistical hypothesis is an assertion about the distribution of one or more random variables. The basic hypothesis to be tested is called the *null hypothesis* H_0.

This can be regarded as expressing knowledge or belief about an unknown parameter being tested. (This is before you see the data). To be able to construct suitable criteria for testing, we also need an *alternative hypothesis* H_1, against which H_0 is tested.

If a statistical hypothesis completely specifies the considered distribution, it is called a *simple hypothesis*, e.g. $\mu = 150$. Otherwise it is called a *composite hypothesis* e.g. $\mu > 150$. A *statistical test* is a rule that divides the sample space into two subsets:

- the *acceptance region*, i.e. the region of values that are consistent with H_0;
- the *rejection (or critical)* region, i.e. the region of values inconsistent with H_0.

The decision regarding the test is based on the value of the test statistic, which is a suitable function of the data and the parameter(s) being tested (and its distribution is known). The range of values of a *test statistic* is also divided into two regions:

- An *acceptance region* where the values of the statistic are consistent with H_0;
- Its complement the *rejection region* where the values of the statistic are inconsistent with H_0.

5.1.1 Errors Associated with Statistical Tests

The probability of a Type I Error is the probability to **reject** H_0 when H_0 is **true**. It is denoted by α and is also referred to as the significance level of the test.

$$\mathbb{P}(\text{type I error}) = \alpha$$

The probability of a Type II Error is the probability of **failing to reject** H_0 when H_1 is **true**. This is denoted by β.

$$\mathbb{P}(\text{type II error}) = \beta$$

Also define the **power** of a test as:

$$\mathbb{P}(\textbf{reject } H_0 \text{ when it is } \textbf{false}) = 1 - \beta.$$

These are summarised in the following table:

	H_0 is true	H_1 is true
Fail to reject H_0	Correct decision	Type II error (with probability β)
Reject H_0	Type I error (with probability α)	Correct decision (with probability $1-\beta$)

Ideally we wish to minimise both α and β simultaneously - not achieved in practice (for fixed sample size n). More on errors and power of tests in later statistics courses.

Example 5.2. A random sample of size $n = 9$ is taken from a $\mathrm{N}(\mu, 900)$ population to test the hypothesis
H_0: $\mu = 100$ v H_1: $\mu \neq 100$
at the 5% level of significance ($\alpha = 0.05$).

(a) Find the critical (rejection) region of this test in terms of $\bar{X}$.

(b) Calculate the probability of a Type II error of this test when $\mu = 105$.

Solution:

(a) Under the null hypothesis our random sample $X_1, \ldots, X_9 \sim \mathrm{N}(100, 900)$ which means that

$$\bar{X} \sim \mathrm{N}\left(100, \frac{900}{n}\right) \Rightarrow \frac{\sqrt{n}}{\sqrt{900}}\left(\bar{X} - 100\right) \sim \mathrm{N}(0,1).$$

Since the alternative hypothesis is $\mu \neq 100$ we can reject the null hypothesis is the sample mean is either two small or two large. Therefore to find our rejection region we find $z_{\alpha/2}$ such that

$$\mathbb{P}(Z > z_{\alpha/2}) = \alpha \text{ for } Z \sim \mathrm{N}(0,1).$$

Then under the null hypothesis

$$\mathbb{P}(-z_{\alpha/2} \leq \frac{\sqrt{n}}{\sqrt{900}}\left(\bar{X} - 100\right) \leq z_{\alpha/2}; \mu = 100) = 1 - \alpha.$$

That is, we reject the null hypothesis if

$$\frac{\sqrt{n}}{\sqrt{900}}\left(\bar{X} - 100\right) \leq -z_{\alpha/2} \text{ or } \frac{\sqrt{n}}{\sqrt{900}}\left(\bar{X} - 100\right) > z_{\alpha/2}.$$

If we rearrange these we reject the null hypothesis if

$$\bar{X} \leq 100 - \frac{\sqrt{900}}{\sqrt{n}} z_{\alpha/2} \text{ or } \bar{X} > 100 + \frac{\sqrt{900}}{\sqrt{n}} z_{\alpha/2}.$$

In our case $n = 9$ and $\alpha = 0.05$ so $z_{\alpha/2} = 1.96$ and we reject if

$$\bar{X} \leq 100 - 19.6 = 80.4 \text{ or } \bar{X} > 100 + 19.6 = 119.6.$$

This means our critical rejection region for this test in terms of $\bar{X}$ is $(-\infty, 80.4) \cup (119.6, \infty)$.

(b) For a Type II error, we suppose now that the true value of $\mu = 105$ so $X_1, \ldots, X_9 \sim \mathrm{N}(105, 900)$, we want to determine the probability that we have failed to reject the null hypothesis. That is given $\mu = 105$ determine

$$\mathbb{P}(80.4 \leq \bar{X} \leq 119.6; \mu = 105)$$

If we standardise the sample mean (recalling the variance of $\bar{X} = 100$) we have

$$\mathbb{P}(80.4 \leq \bar{X} \leq 119.6; \mu = 105) = \mathbb{P}(-2.46 \leq \frac{\bar{X} - 105}{10} \leq 1.46; \mu = 105) = \mathbb{P}(-2.46 \leq Z \leq 1.46)$$

where $Z \sim \mathrm{N}(0, 1)$. We now use the statistical tables to write

$$\mathbb{P}(-2.46 \leq Z \leq 1.46) = \Phi(1.46) - (1 - \Phi(2.46) = 0.9279 - (1 - 0.99305) = 0.9201.$$

So

$$\beta = \mathbb{P}(\text{ fail to reject } H_0 \,; \mu = 105) = 0.9201.$$

The probability of a type II error is quite high here, this is because the value of $\mu = 105$ is close to the value of μ under the null hypothesis and the sample size is small. Therefore it is difficult to distinguish these two values.

Example 5.3. In a sequence of $n = 20$ independent trials we want to test the hypothesis

$$H_0 : p = 0.9 \text{ vs } H_1 : p = 0.6$$

where p is the probability of 'success'.
The test is based on the number of successes, X, in n trials and we reject H_0 when $X \leq 14$.
Find the probabilities of type I and type II errors.
Solution: The pivotal quantity here is X which is distributed according to $\mathrm{Bin}(20, p)$, and the null hypothesis is rejected of $X \leq 14$. Therefore a type I error occurs if $X \leq 14$ and $p = 0.9$ so

$$\alpha = \mathbb{P}(\text{ Type I error}) = \mathbb{P}(X \leq 14; p = 0.9).$$

In order to use the statistical tables we set $Y = n - X \sim \mathrm{Bin}(n, 1 - p)$ then

$$\mathbb{P}(X \leq 14; p = 0.9) = \mathbb{P}(Y \geq 6; p = 0.9) = 1 - \mathbb{P}(Y \leq 5; p = 0.9) = 1 - 0.9887 = 0.0113.$$

To calculate the probability of a type II we need to determine the probability of accepting H_0 if H_1 is true. That is, $X \sim \mathrm{Bin}(20, 0.6,)$ and

$$\beta = \mathbb{P}(\text{ Type II error}) = \mathbb{P}(X > 14; p = 0.6).$$

As before we use $Y = n - X \sim \mathrm{Bin}(n, 1 - p)$ to calculate this probability using the statistical tables

$$\beta = \mathbb{P}(X > 14; p = 0.6) = \mathbb{P}(Y < 6; p = 0.6) = 0.1256.$$

5.1.2 Probability Value (p-value) of a Test

In Example 5.1, the decision (conclusion) would be that we accept (or do not reject) H_0 at the 5% significance level. This does not describe in full detail the strength of the evidence in favour (or against) H_0.

A more informative approach is based on the following:

The p-value corresponding to an observed value of a test statistic, is the **probability to observe a test statistic as or more extreme as the value actually observed** (under the assumption that H_0 is true).

Also referred to as the tail probability or the observed significance level of the test, it gives the **lowest level of significance at which H_0 could have been rejected**.

For the purpose of this course we will consider the following guide for a test which has p-value p we can say:

- For $p < 0.001$: there is very strong evidence against H_0;
- For $0.001 < p < 0.01$: there is strong evidence against H_0;
- For $0.01 < p < 0.05$: there is some evidence against H_0;
- For $0.05 < p < 0.1$: there is weak evidence against H_0;
- For $p > 0.1$: there is no evidence against H_0.

Note that in practice what corresponds "strong evidence" will depend on the application and the standards of that field.

5.2 Hypothesis Testing in Practice

Once appropriate data are given, we can perform a statistical test for some population parameter following the steps below:

1. Formulate H_0 and H_1 and specify the significance level α.
2. Select an appropriate *test statistic*. This must involve the data and the parameter of interest and its distribution must be known.
3. Calculate the value of the test statistic under H_0 *using the data.*
4. **Either** determine a critical region of size α according to the chosen test statistic **or** Calculate the p-value that corresponds to the value of the test statistic. Remember:
   - The critical region depends on the sampling distribution of the test statistic, the significance level α and the nature of the alternative hypothesis H_1 (one or two-tailed test).
   - The p-value is the probability to observe a value of the test statistic as or more extreme as the value obtained from the sample data.
5. Reject the null hypothesis if **either** the value of the test statistic falls into the critical region **or** the p-value is less than or equal to the significance level α. Otherwise do not reject H_0 (or reserve judgement).

5.3 Hypothesis Testing For Means

5.3.1 Single sample situation

Test H_0: $\mu = \mu_0$
versus H_1: $\mu \neq \mu_0$ (or $\mu > \mu_0$, etc.)
at a specified significance level α.
With normal samples or large sample size n, use the 'z test' statistic

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim \mathrm{N}(0,1)$$

If σ is unknown and n is large use the sample standard deviation, S, instead.
Otherwise, for small, normal samples we perform a t-test, based on the test statistic

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t_{n-1}$$

Example 5.4. According to its specifications, the mean breaking strength of a certain kind of ribbon is 185 pounds. Five pieces are randomly selected and have breaking strengths (in pounds)

```
171.6  191.8  178.3  184.9  189.1
```

(a) Test the hypothesis that the mean is 185 pounds against the alternative that it is less than 185 pounds, at the 5% level of significance.

(b) Calculate the p-value for this test and give its interpretation.

(c) Now consider a test of H_0: $\mu = 185$ v H_1: $\mu \neq 185$ and repeat the exercise.

Solution:

(a) We will follow the steps given in Section 5.2:

1. Formulate H_0 and H_1: We are given

$$H_0 : \mu = 185 \text{ vs. } H_1 : \mu < 185$$

and $\alpha = 0.05$ is the significance level.

2. An appropriate test statistic in this case is

$$T(\underline{X}; \mu) = \frac{\bar{X} - \mu}{S/\sqrt{n}},$$

this involves the data and the parameter of interest. In this case the variance is unknown so we will assume that $T \sim t_{n-1}$.

3. Using the data we have

$$\bar{x} = 183.14$$
$$s = 8.219$$
$$n = 5.$$

So under the null hypothesis

$$T(\underline{x}; 185) = \frac{183.14 - 185}{8.219/\sqrt{5}} = -0.506.$$

4. To calculate the critical region of size α we need to find $t_{\alpha,4}$ such that

$$\mathbb{P}(T_4 \leq -t_{\alpha,4}) = \alpha, \text{ where } T_4 \sim t_4.$$

For $\alpha = 0.05$ we find from the statistical tables $t_{\alpha,4} = 2.132$. Therefore we reject the null hypothesis if $T(\underline{x}; 185) \leq -2.132$. As this is not the case there is not sufficient evidence to reject the null hypothesis at a 5% significance level.

(b) The p-value for this test is the smallest significance level at which we would reject the null hypothesis, this is

$$\mathbb{P}(T_4 \leq T(\underline{x}; 185)), \text{ where } T_4 \sim t_4.$$

Note here that $T(\underline{X}; 185)$ is a random variable but $T(\underline{x}; 185)$ is the observed value of the test statistic which is number and not a random quantity. In this case $T(\underline{x}; 185) = -0.506$ and

$$\mathbb{P}(T_4 \leq -0.506) = 0.320.$$

As the p-value is large there is very little evidence to reject the null hypothesis.

(c) Since the alternative hypothesis has changed we need to change the critical region, now we use a two sided test. In this case we find $t_{\alpha/2,4}$ such that

$$\mathbb{P}(-t_{\alpha/2,4} < T_4 \leq t_{\alpha/2,4}) = 1 - \alpha, \text{ where } T_4 \sim t_4.$$

For $\alpha = 0.05$ we have $t_{\alpha/2,4} = 2.776$, so we reject if $T(\underline{x}; 185) \leq -2.776$ or $T(\underline{x}; 185) \geq 2.776$. The value of the test statistic is unchanged and remains $T(\underline{x}; 185) = -0.506$. Therefore we still do not reject the null hypothesis.

Alternatively we could have used the p-value. Since we are using a two-sided test the p-value is now given by

$$p = \mathbb{P}(|T_4| \geq |T(\underline{x}; 185)|), \text{ where } T_4 \sim t_4.$$

Using the symmetry of the t-distribution

$$p = \mathbb{P}(|T_4| \geq |T(\underline{x}; 185)|) = 2\mathbb{P}(T_4 \leq -0.506) = 2(0.320) = 0.640.$$

Since the p-value is larger than the significance level we reject the null hypothesis.

5.3.2 Two samples: Differences between means

Test: H_0: $\mu_1 = \mu_2$ versus H_1: $\mu_1 \neq \mu_2$ (or $\mu_1 > \mu_2$, etc.) at a specified significance level α.
Assuming that the samples are normal and independent we can use the following test statistic

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \quad \sim \mathrm{N}(0, 1)$$

with s_1^2, s_2^2 substituting σ_1^2, σ_2^2 if unknown, providing both n_1 and n_2 are large enough.
With *small* n_1, n_2,we need to assume a common unknown variance, σ^2 for both populations, estimated by

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

and use the statistic

$$T = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad \sim t_{n_1+n_2-2}$$

We again need to assume that both populations are normal.

Example 5.5. The following two independent samples have been collected from normal populations with unknown parameters.

$$\underline{x} = (28, 26, 18, 16)$$
$$\underline{y} = (30, 32, 26, 24)$$

Test the hypothesis that the means of the two populations are equal.

Solution: Let μ_x denote the population mean of X (the corresponding to the data $\underline{x}$) and let μ_y denote the population mean of Y (the corresponding to the data $\underline{y}$). We use the hypothesis:

$$H_0 : \mu_x = \mu_y \text{ vs. } H_1 : \mu_x \neq \mu_y.$$

Our test statistic is:

$$T(\underline{X}, \underline{Y}; \mu_x, \mu_y) = \frac{\bar{X} - \bar{Y} - (\mu_x - \mu_y)}{S_p\sqrt{\frac{1}{n_x} + \frac{1}{n_y}}}.$$

Under the null hypothesis $T(\underline{X}, \underline{Y}; \mu_x, \mu_y) \sim t_{n_x+n_y-2}$.

Using the data we have

$$\bar{x} = 22, \bar{y} = 28, n_x = 4, n_y = 4, s_x^2 = 34.67, s_y^2 = 13.33, s_p^2 = 24.$$

So our observed value of the test statistic under the null hypothesis, $\mu_x = \mu_y$, is

$$T(\underline{x}, \underline{y}; \mu_x, \mu_x) = \frac{22 - 28}{\sqrt{24}\sqrt{\frac{1}{4} + \frac{1}{4}}} = -1.73.$$

We can now calculate the p-value

$$\mathbb{P}(|T_6| \geq |T(\underline{x}, \underline{y}; \mu_x, \mu_y)|) = 2\mathbb{P}(T_6 \geq 1.73; \mu_x, \mu_y) = 2(0.067) = 0.134$$

where $T_6 \sim t_6$ and 0.067 can be found using the statistical tables. There is insufficient evidence to reject the null hypothesis.

5.3.3 Two dependent samples

When the samples are not independent and paired, i.e. $(x_1, y_1), (x_2, y_2), ..., (x_n, y_n)$, we can work with the differences $d_i = x_i - y_i$ and the mean $\mu = \mu_x - \mu_y$.

Test: H_0: $\mu = 0$ versus H_1: $\mu \neq 0$ (or $\mu > 0$, etc.)

using the statistics presented before for the single sample situation, e.g.

$$T = \frac{\bar{D} - \mu}{S/\sqrt{n}} \sim t_{n-1} \qquad \text{for small } n.$$

Recall $D = X - Y$ is the population of differences, $\bar{D}$ is the sample mean of the differences and S^2 is the sample variance of the differences. Here, we need to assume that the differences come from a normal population.

5.4 Hypothesis Testing For Variances

5.4.1 Single sample situation

Test H_0: $\sigma^2 = \sigma_0^2$

versus H_1: $\sigma^2 \neq \sigma_0^2$ (or $\sigma^2 > {\sigma_0}^2$, etc.)

at a specified significance level α.

We use the test statistic

$$\chi^2 = \frac{(n-1)S^2}{{\sigma_0}^2} \sim \chi^2_{n-1}$$

5.4.2 Two samples (equality of two population variances)

Test: H_0: $\sigma_1^2 = \sigma_2^2$ (or $\frac{\sigma_1^2}{\sigma_2^2} = 1$) versus H_1: $\sigma_1^2 \neq \sigma_2^2$ (or $\sigma_1^2 > \sigma_2^2$, etc.)
The random samples are assumed **normal** and **independent**.
We use the test statistic

$$F = \frac{S_1^2}{S_2^2} \quad \sim F_{n_1-1,n_2-1}$$

5.5 Hypothesis Testing For Proportions

Hypothesis concerning the proportion P of a population having a specific characteristic.
Test: H_0: $P = p_0$ versus H_1: $P \neq p_0$ (or $P > p_0$, etc.)
With large sample size n, use the test statistic

$$Z = \frac{X - np_0}{\sqrt{np_0(1-p_0)}} \sim \mathrm{N}(0,1)$$

Where X is the 'number of successes in n trials' in the sample. Continuity correction may also be used.
For small n we can use the cumulative distribution function of the binomial distribution to obtain the exact p-value.

Example 5.6 (Belgian 1 Euro Coin)**.** The Guardian reported on January 4th 2002 that two Polish mathematicians claim that the Belgian one euro coin is 'struck asymmetrically'. They spin the Euro coin 250 times and found it landed heads up 140 times.
Is their claim statistically justified?
Solution: We use the hypotheses:

$$H_0 : p = 0.5 \text{ vs. } H_1 : p \neq 0.5$$

We could use the number, X, of times the coin lands heads up in 250 spins as our test statistic. We have that $X \sim \mathrm{Bin}(n,p)$ but since the number of trials is large we can use the central limit theorem to approximate the distribution of $X \dot{\sim} \mathrm{N}(250p, 250p(1-p))$. So under the null hypothesis $p = 0.5$ we have $X \dot{\sim} \mathrm{N}(125, 62.5)$. Therefore we will use the test statistic

$$T(X;p) = \frac{X - 250p}{\sqrt{250p(1-p)}}.$$

We will apply a continuity correction (as illustrated in Figure 13) and use as our observed test statistic $T(139.5; 0.5) = 1.834$.
From our data we have that the observed value of the test statistic is $T(140; 0.5) = 1.834$ so the p-value for this test is

$$\mathbb{P}(|Z| \geq 1.834) \approx \mathbb{P}(|Z| \geq 1.834) = 2\mathbb{P}(Z \geq 1.834) = 0.066.$$

Here $Z \sim \mathrm{N}(0,1)$. This shows there is weak evidence against H_0, in other words, there is weak evidence that the coin is biased.

5.6 Proportions of two samples

Given two independent observations $X_1 \sim \mathrm{Bin}(n_1, p_1)$ and $X_2 \sim \mathrm{Bin}(n_2, p_2)$ with n_1 and n_2 known but p_1 and p_2 unknown. We wish to test if the proportions p_1 and p_2 are the equal.

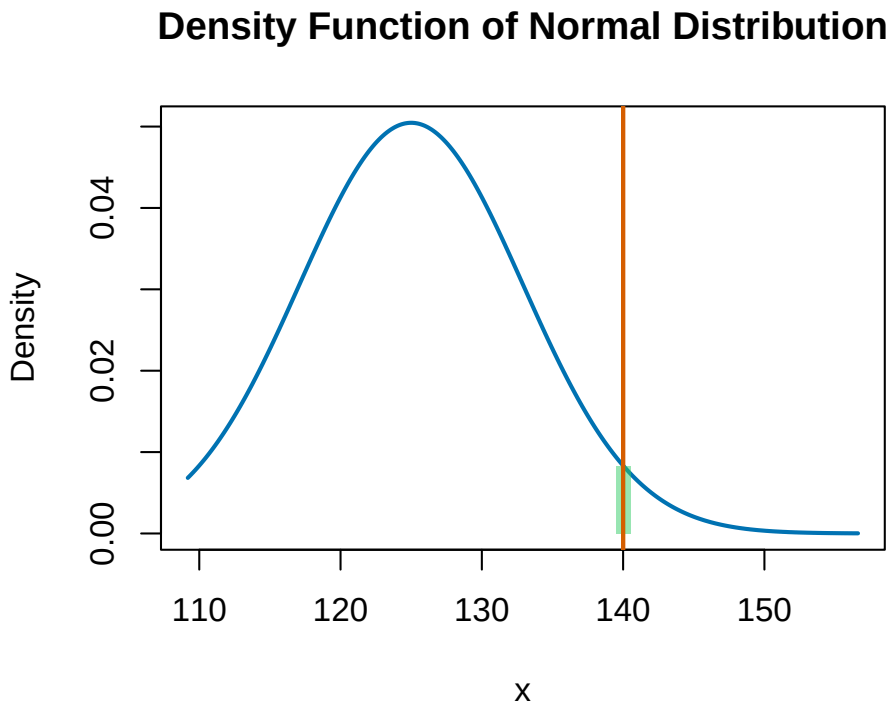


Figure 13: A plot of the probability density for the normal approximation $\tilde{X} \sim \mathrm{N}\left(np, np(1-p)\right)$ of X assuming the null hypothesis is true. In red is the observed value of X and in green are values of the normal approximation $\tilde{X}$ which are consistent with the observed value of X.

We use the hypotheses

$$H_0 : p_1 = p_2 \text{ versus } H_1 : p_1 \neq p_2 \text{ (or } p_1 > p_2 \text{ etc.)}$$

We will assume that both n_1 and n_2 are large then use the test statistic

$$Z = \frac{\frac{X_1}{n_1} - \frac{X_2}{n_2}}{\sqrt{\hat{p}(\underline{X})(1-\hat{p}(\underline{X}))\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}}, \quad \hat{p}(\underline{X}) = \frac{X_1 + X_2}{n_1 + n_2}.$$

Note if the null hypothesis is true then $p_1 = p_2$ and a sensible estimator for either is $\hat{p}(\underline{X})$. Moreover, under the assumption of the null hypothesis we have

$$Z \dot{\sim} \mathrm{N}\left(0, 1\right).$$

Example 5.7. In a psychological experiment, a collection of single people were sent £500 but the address on the letter was that of their neighbour.

- Out of 69 male recipients, 17 of them kept the £500.
- Out of 131 female recipients, 11 of them kept the £500.

Is there any difference in the behaviour of the two genders?

Solution: Let X_1 denote the number of male recipients who keep the money, and X_2 denote the number of female recipients who keep the money.

We test

$$H_0 : p_1 = p_2 \text{ versus } H_1 : p_1 \neq p_2$$

using the test statistic

$$Z(X_1, X_2) = \frac{\frac{X_1}{n_1} - \frac{X_2}{n_2}}{\sqrt{\hat{p}(\underline{X})(1 - \hat{p}(\underline{X}))\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}, \quad \hat{p}(\underline{X}) = \frac{X_1 + X_2}{n_1 + n_2}.$$

Our data is

$$n_1 = 69, x_1 = 17, n_2 = 131, x_2 = 11.$$

So our estimate of a common proportion, $\hat{p}$, is

$$\hat{p}(\underline{x}) = \frac{x_1 + x_2}{n_1 + n_2} = \frac{17 + 11}{69 + 131} = 0.14$$

and hence our observed value of the test statistic is

$$Z(x_1, x_2) = \frac{\frac{17}{69} - \frac{11}{131}}{\sqrt{0.14(1 - 0.14)\left(\frac{1}{69} + \frac{1}{131}\right)}} = 3.15$$

We can use R to find the p-value for this

$$\mathbb{P}(Z \geq 3.15 \text{ or } Z \leq -3.15) = 2\mathbb{P}(Z \geq 3.15) = 0.00165.$$

Therefore we have very strong evidence to reject the null hypothesis.

6 Relationships between two variables

Consider having data on two variables X and Y for a sample of n subjects (or objects). Correlation and regression involves investigating a possible relationship between X and Y.

Examples of such situations: 'before' and 'after' measurements; exam marks in different courses; dexterity and anxiety when performing a task; advertising and sales.

Correlation: this concerns measuring how strong a relationship is.

Regression: this concerns establishing a 'formula' for a relationship so that it can be used for prediction.

Consider paired data $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$. The first thing to do in order to determine any relationship between X and Y is to draw a scatterplot of the data. Immediately this lets us see if there is any relationship between x and y and what form that relationship might take.

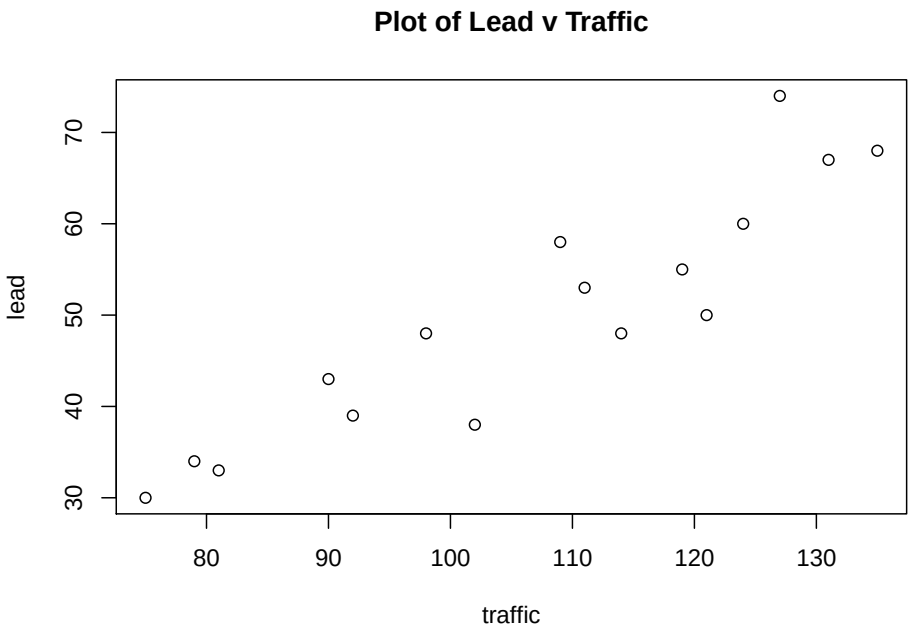
Notation :

$$s_{xx} = \sum_{i=1}^{n}(x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - \frac{1}{n}(\sum_{i=1}^{n} x_i)^2$$
$$s_{yy} = \sum_{i=1}^{n}(y_i - \bar{y})^2 = \sum_{i=1}^{n} y_i^2 - \frac{1}{n}(\sum_{i=1}^{n} y_i)^2$$
$$s_{xy} = \sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^{n} x_i y_i - \frac{1}{n}\sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i$$

Throughout this chapter we illustrate ideas using an example involving traffic pollution

Road	Lead (y_i)	Traffic(x_i)
1	34	79
2	58	109
3	43	90
4	60	124
5	53	111
6	48	114
7	74	127
8	38	102
9	50	121
10	55	119
11	67	131
12	68	135
13	30	75
14	39	92
15	33	81
16	48	98

Example 6.1. In an investigation of traffic pollution, information was collected from 16 busy roads. The data below are the amount of lead (in parts per million) in the ground level vegetation and the volume of traffic (hundreds of vehicles recorded per day):
We first draw a scatterplot:



The scatterplot suggests a positive linear relationship between traffic and lead amount. Now we formalise this by calculating the correlation coefficient r.
The **population** correlation coefficient is defined as

$$\rho = \frac{\mathbb{C}\text{ov}(X, Y)}{\sigma_x \sigma_y}$$

and the **sample** correlation coefficient is given by

$$r = r(\underline{x}, \underline{y}) = \frac{s_{xy}}{\sqrt{s_{xx}s_{yy}}}$$

Technically r is called the sample product-moment correlation coefficient or the Pearson correlation coefficient (there are others e.g. based on ranks).

Interpretation of r: Recall that r must lie in the range $-1 < r < +1$;

- $r = +1$ means that the points lie exactly on a line with positive slope;
- $r = -1$ means that the points lie exactly on a line with negative slope;
- r being large means that the points lie close to a straight line; smaller values of r mean that the points are more spread out;
- r near zero means that there is no linear relationship for these points.

Notice that r does **not** indicate 'cause and effect'.
For our example:

$$\sum x = 1708, \sum x^2 = 187890, \sum y = 798, \sum y^2 = 42454, \sum xy = 88714$$
$$s_{xy} = 88714 - \frac{1}{16}(1708)(798) = 3527.5$$
$$s_{xx} = 187890 - \frac{1}{16}(1708)^2 = 5561$$
$$s_{yy} = 42454 - \frac{1}{16}(798)^2 = 2653.75$$

So

$$r = \frac{s_{xy}}{\sqrt{s_{xx}s_{yy}}} = \frac{3527.5}{\sqrt{5561 \times 2653.75}} = 0.918.$$

This is large and positive as expected from the scatterplot.

Normal model and inference for ρ: Under the assumption that (x, y) has the **bivariate normal distribution**, we have two tests.
We can test

$$H_0 : \rho = 0 \quad \text{versus} \quad H_1 : \rho \neq \rho_0 \text{ (or } \rho > 0 \text{ etc)}$$

using the test statistic

$$T = \frac{r(\underline{X}, \underline{Y})\sqrt{n-2}}{\sqrt{1 - r(\underline{X}, \underline{Y})^2}} \sim t_{n-2}.$$

More generally (i.e. not just for $\rho = 0$), we can test

$$H_0 : \rho = \rho_0 \quad \text{versus} \quad H_1 : \rho \neq \rho_0 \text{ (or } \rho > \rho_0 \text{ etc)}$$

using the approximation

$$W = \frac{1}{2}\log\left(\frac{1 + r(\underline{X}, \underline{Y})}{1 - r(\underline{X}, \underline{Y})}\right) \dot{\sim} \mathrm{N}\left(\frac{1}{2}\log\left(\frac{1+\rho}{1-\rho}\right), \frac{1}{n-3}\right).$$

The same approximate sampling distribution may be used for confidence intervals for ρ.

Example 6.2. The sample correlation coefficient between two samples is found to be $r = 0.936$ ($n = 10$). Test the hypothesis that there is no correlation between the two corresponding populations at the 1% significance level.

We will solve this problem using both of the test statistics above.

Solution 1: We test

$$H_0 : \rho = 0 \text{ versus } H_1 : \rho \neq 0$$

at the 1% significance level.

From the approximation above our test statistic will be

$$T(\underline{X}, \underline{Y}; \rho) = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}.$$

Under the null hypothesis

$$T(\underline{X}, \underline{Y}; 0) \sim t_8.$$

Our observed value of the test statistic is

$$T(\underline{x}, \underline{y}; 0) = \frac{0.936\sqrt{10-2}}{\sqrt{1-0.936^2}} = 7.52.$$

We can either calculate the rejection region ($T \leq -3.355$ or $T \geq 3.355$) or determine the p-value. In this case 7.52 is larger than the values covered in the statistical tables however using R we have the p-value

$$\mathbb{P}(T \leq -7.52 \text{ or } T \geq 7.52) \approx 2\mathbb{P}(T_8 \geq 7.52) = 6.8 \times 10^{-5}.$$

Here $T_8 \sim t_8$. Therefore, since the p-value is very small we have very strong evidence against H_0.

Solution 2: We test

$$H_0 : \rho = 0 \text{ versus } H_1 : \rho \neq 0$$

at the 1% significance level.

From the approximation above our test statistic will be

$$T(\underline{X}, \underline{Y}; \rho) = \frac{\frac{1}{2}\log\left(\frac{1+r}{1-r}\right) - \frac{1}{2}\log\left(\frac{1+\rho}{1-\rho}\right)}{1/\sqrt{n-3}}.$$

Under the null hypothesis

$$T(\underline{X}, \underline{Y}; 0) \sim \mathrm{N}(0, 1).$$

Our observed value of the test statistic is

$$T(\underline{x}, \underline{y}; \rho) = \frac{\frac{1}{2}\log\left(\frac{1+0.936}{1-0.936}\right)}{1/\sqrt{7}} \approx 4.51.$$

We can either calculate the rejection region ($T \leq -2.576$ or $T \geq 2.576$) or determine the p-value. In this case 4.51 is larger than the values covered in the statistical tables however using R we have the p-value

$$\mathbb{P}(T \leq -4.51 \text{ or } T \geq 4.51) \approx 2\mathbb{P}(Z \geq 4.51) = 6.5 \times 10^{-6}.$$

Here $Z \sim \mathrm{N}(0, 1)$. Therefore, since the p-value is very small we have very strong evidence against H_0.

Confidence Interval for ρ Since the test statistic W given above has a known distribution for any value of ρ we can use it to calculate confidence intervals for ρ.

Using that W is approximately normally distributed we have

$$\mathbb{P}\left(-z_{\alpha/2} \leq \sqrt{n-3}\left(\frac{1}{2}\log\left(\frac{1+r}{1-r}\right) - \frac{1}{2}\log\left(\frac{1+\rho}{1-\rho}\right)\right) \leq z_{\alpha/2}\right) \approx 1-\alpha.$$

Here $z_{\alpha/2}$ are the quantiles of the standard normal distribution, i.e. such that $\mathbb{P}(Z \geq z_{\alpha/2}) = \alpha/2$ for $Z \sim \mathrm{N}(0,1)$.

To simplify the calculation we use that

$$\tanh^{-1}(x) = \frac{1}{2}\log\left(\frac{1+x}{1-x}\right)$$

to write

$$\mathbb{P}\left(-z_{\alpha/2} \leq \sqrt{n-3}\left(\tanh^{-1}(r) - \tanh^{-1}(\rho)\right) \leq z_{\alpha/2}\right) \approx 1-\alpha.$$

This we can rearrange to write

$$\mathbb{P}\left(\tanh^{-1}(r) - \frac{z_{\alpha/2}}{\sqrt{n-3}} \leq \tanh^{-1}(\rho) \leq \tanh^{-1}(r) + \frac{z_{\alpha/2}}{\sqrt{n-3}}\right) \approx 1-\alpha.$$

Since tanh is strictly increasing this implies

$$\mathbb{P}\left(\tanh\left(\tanh^{-1}(r) - \frac{z_{\alpha/2}}{\sqrt{n-3}}\right) \leq \rho \leq \tanh\left(\tanh^{-1}(r) + \frac{z_{\alpha/2}}{\sqrt{n-3}}\right)\right) \approx 1-\alpha.$$

This gives the $100(1-\alpha)\%$ confidence interval for ρ as

$$\left(\tanh\left(\tanh^{-1}(r) - \frac{z_{\alpha/2}}{\sqrt{n-3}}\right), \tanh\left(\tanh^{-1}(r) + \frac{z_{\alpha/2}}{\sqrt{n-3}}\right)\right).$$

6.1 Linear Regression

We may believe that one variable Y depends on the other variable x and that we may be able to predict the value of Y for a given value of x. Here Y is called the dependent (or response) variable and x is called the independent (or explanatory) variable.

Returning to the lead pollution example we might expect the volume of traffic to influence the amount of lead in vegetation. So Y would be taken as 'amount of lead' and x as 'volume of traffic'. Regression concerns fitting a function to represent this relationship. We consider only linear regression in which we fit a straight line to our (x, y) data. Note that the values of x are regarded as constant - they are controlled by the experimenter - so x is **not** a random variable.

Model for Regression Let

$$Y_i = \alpha + \beta x_i + \varepsilon_i \quad \text{for } i = 1, \ldots, n.$$

Here the ε_i's are independent error terms such that $\mathbb{E}[\varepsilon_i] = 0$ and $\mathbb{V}\mathrm{ar}(\varepsilon_i) = \sigma^2$.

This model has three parameters:

$$\begin{aligned} \beta &: \text{ the slope coefficient} \\ \alpha &: \text{ the constant or intercept} \\ \sigma^2 &: \text{ the error variance} \end{aligned}$$

The error variance allows for variation in the data about a straight line.

We fit the line $Y_i = \alpha + \beta x_i + \varepsilon_i$ by the method of least squares i.e. we seek the values of α and β to minimise the 'error sum of squares'. This is seen graphically in the scatter plot in Figure 14.

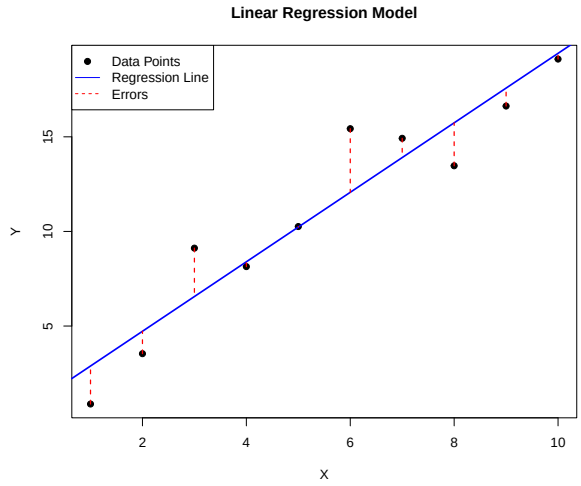


Figure 14: The plot illustrates a simple linear regression model. Each black dot represents a data point, while the blue line indicates the regression line that best fits the data. Red vertical dashed lines represent the residuals or errors, showing the difference between the observed data points and the predicted values from the model.

Fitting the Model When there is indication that the relationship between Y and x is linear (e.g from a plot), estimate the regression parameters α and β using the method of least squares (LS).

Note that the regression model can also be written as:

$$\begin{aligned}\mathbb{E}(Y_i) &= \alpha + \beta x_i, i = 1, 2, \ldots, n \\ \mathbb{V}\text{ar}(Y_i) &= \sigma^2 \\ \mathbb{C}\text{ov}(Y_i, Y_j) &= \mathbb{C}\text{ov}(\varepsilon_i, \varepsilon_j) = 0, i \neq j \text{ (independent observations).}\end{aligned}$$

The fitted regression line is given by $\hat{y} = \hat{\alpha} + \hat{\beta}x$ and gives the estimated value of Y for fixed x. The aim of least square estimation is to minimise the differences between observed and fitted values. We therefore have to find values of α and β that minimise

$$S(\alpha, \beta) = \sum_{i=1}^{n} (y_i - (\hat{\alpha} + \hat{\beta}x_i))^2.$$

Solve

$$\frac{\partial S}{\partial \alpha} = 0 \Rightarrow \sum_{i=1}^{n} y_i = \hat{\alpha}n + \hat{\beta}\sum_{i=1}^{n} x_i$$

This can be rearranged to give $\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$ (so the fitted line passes through $(\bar{x}, \bar{y})$).

Also solve

$$\frac{\partial S}{\partial \beta} = 0 \Rightarrow \sum_{i=1}^{n} x_i y_i = \hat{\alpha}\sum_{i=1}^{n} x_i + \hat{\beta}\sum_{i=1}^{n} x_i^2$$

Solving simultaneously gives

$$\hat{\beta} = \frac{\sum_{i=1}^{n} x_i y_i - \frac{1}{n}\sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} x_i^2 - \frac{1}{n}(\sum_{i=1}^{n} x_i)^2} = \frac{s_{xy}}{s_{xx}} \qquad \hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$$

Properties of LS estimator for β: Recall we consider $\underline{x}$ to be known and fixed, while Y is random, with this in mind we write

$$\hat{\beta}(\underline{x}, \underline{Y}) = \frac{s_{xY}}{s_{xx}}, \quad \hat{\alpha}(\underline{x}, \underline{Y}) = \bar{Y} - \hat{\beta}(\underline{x}, \underline{Y})\bar{x}.$$

This gives us the following estimator for the fitted value of Y_i as

$$\hat{Y}_i = \hat{\alpha}(\underline{x}, \underline{Y}) + \hat{\beta}(\underline{x}, \underline{Y})x_i.$$

It can also be shown that :

1. The estimator $\hat{\beta}$ is unbiased, i.e. $\mathbb{E}(\hat{\beta}(\underline{x}, \underline{Y})) = \beta$
2. The variance of $\hat{\beta}$ is given by $\mathbb{V}\text{ar}(\hat{\beta}(\underline{x}, \underline{Y})) = \frac{\sigma^2}{s_{xx}}$.
3. The estimator $\widehat{\sigma^2}$ of σ^2 given by

$$\widehat{\sigma^2}(\underline{x}, \underline{Y}) = \frac{1}{n-2}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2 = \frac{1}{n-2}\sum_{i=1}^{n}(Y_i - \hat{\alpha}(\underline{x}, \underline{Y}) - \hat{\beta}(\underline{x}, \underline{Y})x_i)^2$$

   is unbiased.

Proof. Before we establish these properties we observe that

$$s_{xY} = \sum_{i=1}^{n}(x_i - \bar{x})(Y_i - \bar{Y}) = \sum_{i=1}^{n}(x_i - \bar{x})Y_i - \bar{Y}\sum_{i=1}^{n}(x_i - \bar{x}).$$

Note that $\sum_{i=1}^{n}(x_i - \bar{x}) = n\bar{x} - n\bar{x} = 0$ so we have

$$s_{xY} = \sum_{i=1}^{n}(x_i - \bar{x})Y_i.$$

Therefore we can write $\hat{\beta}$ as

$$\hat{\beta}(\underline{x}, \underline{Y}) = \frac{1}{s_{xx}}\sum_{i=1}^{n}(x_i - \bar{x})Y_i \tag{7}$$

1. Taking the expectation of (7) (using that x_i are known) we have

$$\mathbb{E}[\hat{\beta}(\underline{x}, \underline{Y})] = \frac{1}{s_{xx}}\sum_{i=1}^{n}(x_i - \bar{x})\mathbb{E}[Y_i] = \frac{1}{s_{xx}}\sum_{i=1}^{n}(x_i - \bar{x})(\alpha + \beta x_i) = \frac{\beta}{s_{xx}}\sum_{i=1}^{n}(x_i - \bar{x})x_i.$$

   Here we have used that $\sum x_i - \bar{x} = 0$ to show that $\alpha\sum x_i - \bar{x}$ vanishes. Observe that

$$\sum_{i=1}^{n}(x_i - \bar{x})x_i = \sum_{i=1}^{n}x_i^2 - \frac{1}{n}\left(\sum_{i=1}^{n}x_i\right)^2 = s_{xx}$$

   so $\mathbb{E}[\hat{\beta}(\underline{x}, \underline{Y})] = \beta$, i.e. $\hat{B}$ is an unbiased estimator of β.

2. Using properties of the variance and that $Y_1, \ldots, Y_n$ are independent

$$\mathbb{V}\text{ar}(\hat{\beta}(\underline{x}, \underline{Y})) = \frac{1}{s_{xx}^2}\sum_{i=1}^{n}(x_i - \bar{x})^2\mathbb{V}\text{ar}(Y_i) = \frac{1}{s_{xx}^2}\sum_{i=1}^{n}(x_i - \bar{x})^2\sigma^2 = \sigma^2.$$

3. Since $\hat{\alpha}(\underline{x}, \underline{Y}) = \bar{Y} - \hat{\beta}(\underline{x}, \underline{Y})\bar{x}$ we can write

$$\widehat{\sigma^2}(\underline{x}, \underline{Y}) = \frac{1}{n-2}\sum_{i=1}^{n}(Y_i - \bar{Y} - \hat{\beta}(\underline{x}, \underline{Y})(x_i - \bar{x}))^2$$

Expanding the square we can write this as

$$\widehat{\sigma^2}(\underline{x}, \underline{Y}) = \frac{1}{n-2}(s_{YY} - 2\hat{\beta}(\underline{x}, \underline{Y})s_{xY} + \hat{\beta}(\underline{x}, \underline{Y})^2 s_{xx}).$$

Using that $s_{xY} = s_{xx}\hat{\beta}(\underline{x}, \underline{Y})$ we can simplify this to

$$\widehat{\sigma^2}(\underline{x}, \underline{Y}) = \frac{1}{n-2}(s_{YY} - s_{xx}\hat{\beta}(\underline{x}, \underline{Y})^2).$$

By the definition of Y_i we have

$$\begin{aligned}
s_{YY} &= \sum_{i=1}^{n}(Y_i - \bar{Y})^2 \\
&= \sum_{i=1}^{n}((\alpha + \beta x_i + \varepsilon_i) - (\alpha + \beta\bar{x} + \bar{\varepsilon}))^2 \\
&= \sum_{i=1}^{n}(\beta(x_i - \bar{x}) + \varepsilon_i - \bar{\varepsilon}))^2 \\
&= \beta^2 s_{xx} + 2\sum_{i=1}^{n}\beta(x_i - \bar{x})(\varepsilon_i - \bar{\varepsilon}) + \sum_{i=1}^{n}(\varepsilon_i - \bar{\varepsilon})^2.
\end{aligned}$$

Here

$$\bar{\varepsilon} = \frac{1}{n}\sum_{i=1}^{n}\varepsilon_i$$

is the sample mean of $\varepsilon_1, \ldots, \varepsilon_n$. Observe that

$$\mathbb{E}[\varepsilon_i - \bar{\varepsilon}] = 0$$

and

$$\begin{aligned}
\mathbb{E}[(\varepsilon_i - \bar{\varepsilon})^2] &= \mathbb{E}\left[\left(\frac{1}{n}\sum_{j=1}^{n}(\varepsilon_i - \varepsilon_j)\right)^2\right] \\
&= \frac{1}{n^2}\sum_{j\neq i}\sum_{k\neq i}\mathbb{E}\left[(\varepsilon_i - \varepsilon_j)(\varepsilon_i - \varepsilon_k)\right] \\
&= \frac{1}{n^2}\sum_{j\neq i}\sum_{k\neq i}\mathbb{E}\left[\varepsilon_i^2 - \varepsilon_j\varepsilon_i - \varepsilon_i\varepsilon_k + \varepsilon_j\varepsilon_k\right] \\
&= \frac{1}{n^2}\sum_{j\neq i}\sum_{k\neq i}(\sigma^2 + \mathbb{E}\left[\varepsilon_j\varepsilon_k\right]) \\
&= \frac{1}{n^2}\sum_{j\neq i}(\sigma^2(n-1) + \mathbb{E}\left[\varepsilon_j^2\right])
\end{aligned}$$

In the final inequality we used that $\mathbb{E}[\varepsilon_j\varepsilon_k] = \sigma^2$ if $j = k$ and is 0 otherwise. This now simplifies to

$$\mathbb{E}[(\varepsilon_i - \bar{\varepsilon})^2] = \frac{n-1}{n}\sigma^2.$$

Using these expressions we have

$$\begin{aligned}\mathbb{E}[s_{YY}] &= \beta^2 s_{xx} + 2\sum_{i=1}^{n}\beta(x_i-\bar{x})\mathbb{E}(\varepsilon_i-\bar{\varepsilon}) + \sum_{i=1}^{n}\mathbb{E}(\varepsilon_i-\bar{\varepsilon})^2 \\ &= \beta^2 s_{xx} + (n-1)\sigma^2.\end{aligned}$$

This allows us to write the expectation of $\widehat{\sigma^2}(\underline{x},\underline{Y})$ as

$$\begin{aligned}\mathbb{E}[\widehat{\sigma^2}(\underline{x},\underline{Y})] &= \frac{1}{n-2}(\mathbb{E}[s_{YY}] - s_{xx}\mathbb{E}[\hat{\beta}(\underline{x},\underline{Y})^2]) \\ &= \frac{1}{n-2}(\beta^2 s_{xx} + (n-1)\sigma^2 - s_{xx}\mathbb{E}[\hat{\beta}(\underline{x},\underline{Y})^2]) \\ &= \frac{1}{n-2}((n-1)\sigma^2 - s_{xx}\mathbb{V}\mathrm{ar}(\hat{\beta}(\underline{x},\underline{Y}))) \\ &= \frac{1}{n-2}((n-1)\sigma^2 - \sigma^2) = \sigma^2.\end{aligned}$$

□

Example 6.3. Find the equation of the least squares regression line for the example on lead pollution on vegetation.

Solution: Recall from Example 6.1 that we have

$$\begin{aligned}&\bar{x} = 106.75, \quad \bar{y} = 49.875 \\ &s_{xy} = 3527.5, \quad s_{xx} = 5561, \quad s_{yy} = 2653.75.\end{aligned}$$

Then we find

$$\begin{aligned}\hat{\beta} &= \frac{s_{xy}}{s_{xx}} = \frac{3527.5}{5561} = 0.634, \\ \hat{\alpha} &= \bar{y} - \hat{\beta}\bar{x} = 49.875 - (0.634)(106.75) = -17.8.\end{aligned}$$

Therefore, our least squares estimator is

$$\hat{y} = -17.8 + 0.634x.$$

This line is plotted in Figure 15.

Residuals

The difference between the response y_i and the fitted value $\hat{y}_i$

$e_i = y_i - \hat{y}_i$

$= y_i - \hat{\alpha} - \hat{\beta}x_i, \quad i = 1, 2, ..., n$ is called the **residual at** x_i**.**

The residuals estimate the true errors

$\varepsilon_i = y_i - \alpha - \beta x_i, \quad i = 1, 2, ..., n$

Properties

1. $\sum_{i=1}^{n} e_i = 0$
2. $\sum_{i=1}^{n} e_i x_i = 0$
3. $$\sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} y_i^2 - \hat{\alpha}\sum_{i=1}^{n} y_i - \hat{\beta}\sum_{i=1}^{n} x_i y_i = s_{yy} - \frac{s_{xy}^2}{s_{xx}}$$

(Use this form in $\hat{\sigma}^2 = \frac{1}{n-2}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \frac{1}{n-2}\sum_{i=1}^{n} e_i^2$)

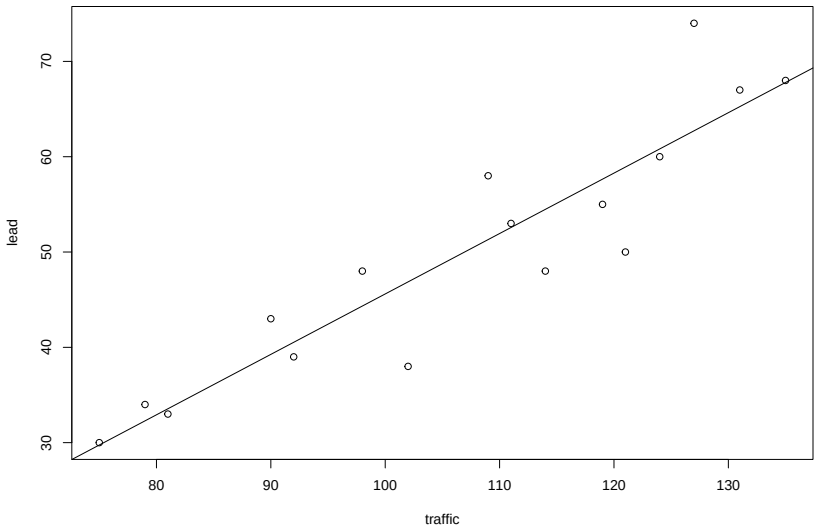


Figure 15: The lead pollution data from Example 6.1 with a linear regression line included.

Example 6.4. Find the value of $\hat{\sigma}^2$ for the example on lead pollution on vegetation.
We first calculate $s_{yy} - \frac{s_{xy}^2}{s_{xx}}$
This gives the value $2653.75 - \frac{3527.5^2}{5561} = 416.157$
$\hat{\sigma}^2 = \frac{1}{n-2}\sum_{i=1}^{n} e_i^2 = \frac{1}{n-2}(s_{yy} - \frac{s_{xy}^2}{s_{xx}}) = \frac{1}{14} \times 416.157 = 29.726$

Variability explained by the model For each data point y_i we can estimate the distance to the mean $\bar{y}$ as

$$y_i - \bar{y} = (y_i - \hat{y}_i) + (\hat{y}_i - \bar{y}).$$

That is, we decompose the error between y_i and $\bar{y}$ into the error between y_i and the estimated value $\hat{y}_i$ (red residual in Figure 16) plus the error between $\hat{y}_i$ and the mean value $\bar{y}$ (orange explained variability in Figure 16).
Squaring and summing the expression for the error gives

$$\sum_i (y_i - \bar{y})^2 = \sum_i (y_i - \hat{y}_i)^2 + \sum_i (\hat{y}_i - \bar{y})^2 + \sum_i (y_i - \hat{y}_i)(\hat{y}_i - \bar{y})$$

Using that $e_i = y_i - \hat{y}_i$ is the residual and $\hat{y}_i - \bar{y} = \hat{\beta}(x_i - \bar{x})$ due to the definition of $\hat{\alpha}$ we can write

$$\sum_i (y_i - \hat{y}_i)(\hat{y}_i - \bar{y}) = \sum_i \hat{\beta} e_i (x_i - \bar{x}) = \hat{\beta} \sum_i e_i x_i - \hat{\beta}\bar{x} \sum_i e_i.$$

This gives zero by our properties of the residual and hence we have

$$\sum_i (y_i - \bar{y})^2 = \sum_i (y_i - \hat{y}_i)^2 + \sum_i (\hat{y}_i - \bar{y})^2.$$

We will label these terms:

$\text{SS}_\text{T} = \sum_i (y_i - \bar{y})^2$ "total sum of squares" - total variability in the responses

$\text{SS}_\text{RES} = \sum_i (y_i - \hat{y}_i)^2$ "residual sum of squares" - variability remaining and "unexplained" by the model

$\text{SS}_\text{REG} = \sum_i (\hat{y}_i - \bar{y})^2$ "regression sum of squares" - variability "explained" by the model

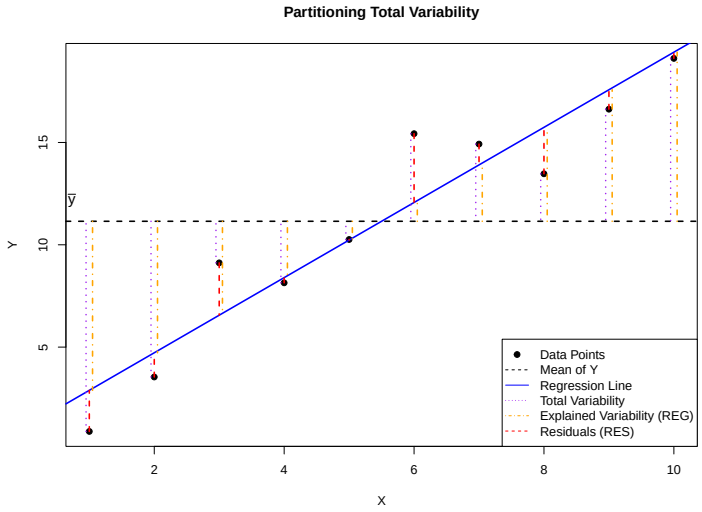


Figure 16: This plot illustrates the partitioning of total variability in the response variable y, with lines slightly offset for clarity. The purple dashed lines (left of each point) represent the total variability, the orange dashed lines (right of each point) represent the explained variability (REG), and the red dashed lines (centered) represent the residuals (RES). The black line indicates the mean of y, and the blue line represents the regression line.

For the purpose of calculations we note that

$$\text{SS}_\text{T} = s_{yy}$$
$$\text{SS}_\text{REG} = \sum_{i=1}^{n} [(\hat{\alpha} + \hat{\beta} x_i) - (\hat{\alpha} + \hat{\beta}\bar{x})]^2 = \hat{\beta}^2 s_{xx}.$$

Therefore

$$\text{SS}_\text{RES} = \text{SS}_\text{T} - \text{SS}_\text{REG} = s_{yy} - \frac{s_{xy}^2}{s_{xx}}.$$

Remark 6.1. An alternative way to calculate SS_RES is to do so directly, i.e. calculate the fitted values $\hat{y}_i$, then the residuals $y_i - \hat{y}_i$ then square and sum them.

Coefficient of Determination This is the proportion of the total variability of the responses that is 'explained' by the regression model and is given by:

$$R^2 = \frac{\text{SS}_\text{REG}}{\text{SS}_\text{T}} = \frac{s_{xy}^2}{s_{xx} s_{yy}}$$

Notice that in the simple linear regression model $R^2 = r^2$ (where r is the sample correlation coefficient).

The full normal model We now add the assumptions that the error terms ε_i are independent and normally distributed. So the response variables Y_i are independent with $Y_i \sim \mathrm{N}\left(\alpha + \beta x_i, \sigma^2\right)$. In the context of regression analysis, this is called the "full normal model".

We are now in a position to write down a likelihood function for our data $\underline{y}$ and so to find Maximum Likelihood Estimators of the parameters.

Maximising the normal likelihood function is equivalent to minimising the sum of squares S (as done earlier) so we get $\hat{\alpha}$ and $\hat{\beta}$ as before (i.e. same as the least squares estimators). But the maximum likelihood estimator of σ^2 turns out to be a simple multiple of the earlier estimator, $\frac{n-2}{n}\widehat{\sigma^2}$.

Inferences on the slope β Using a matrix formulation, we can show that $\hat{\alpha}(\underline{x}, \underline{Y})$ and $\hat{\beta}(\underline{x}, \underline{Y})$ are independent of $\widehat{\sigma^2}(\underline{x}, \underline{Y})$. We also have

$$\frac{(n-2)\widehat{\sigma^2}(\underline{x}, \underline{Y})}{\sigma^2} \sim \chi^2_{n-2}.$$

Under the full normal model

$$\frac{\hat{\beta}(\underline{x}, \underline{Y}) - \beta}{\sqrt{\sigma^2/s_{xx}}} \sim \mathrm{N}\left(0, 1\right).$$

Therefore

$$\frac{\hat{\beta}(\underline{x}, \underline{Y}) - \beta}{\sqrt{\sigma^2/s_{xx}}} \sqrt{\frac{\sigma^2}{\widehat{\sigma^2}(\underline{x}, \underline{Y})}} \sim t_{n-2}.$$

That is,

$$\frac{\hat{\beta}(\underline{x}, \underline{Y}) - \beta}{\sqrt{\widehat{\sigma^2}(\underline{x}, \underline{Y})/s_{xx}}} \sim t_{n-2}.$$

The term $\sqrt{\widehat{\sigma^2}(\underline{x}, \underline{Y})/s_{xx}}$ is called the estimated standard error of $\hat{\beta}$ from which we can get confidence intervals for β and/or carry out tests on the value of β.

Example 6.5. Find the 95% confidence interval for β in the example on lead pollution on vegetation. Also test the null hypothesis that the slope coefficient is equal to 0 against the alternative that it is not equal to 0.

Solution: Recall we have previously found (in Examples 6.1, 6.3, 6.4) :

$$s_{xx} = 5561, \quad \hat{\beta} = 0.634, \quad \widehat{\sigma^2} = 29.726.$$

To find the confidence interval we use the pivotal quantity

$$h(\underline{x}, \underline{Y}; \beta) = \frac{\hat{\beta}(\underline{x}, \underline{Y}) - \beta}{\sqrt{\frac{\widehat{\sigma^2}(\underline{x}, \underline{Y})}{s_{xx}}}}.$$

Note that this is indeed a pivotal quantity because

1. It is a function of both the data (through $\hat{\beta}(\underline{x}, \underline{Y})$, and $\widehat{\sigma^2}(\underline{x}, \underline{Y})$) and the parameter of interest β.
2. It has known (or approximately known) distribution, in this case t_{14}.
3. It is monotonic in the parameter of interest, in this case it is linear in β.

Now we can find the quantile $t_{14,0.025} = 2.145$ from the tables and this gives

$$\mathbb{P}\left(-2.145 \leq \frac{\hat{\beta}(\underline{x}, \underline{Y}) - \beta}{\sqrt{\frac{\widehat{\sigma^2}(\underline{x}, \underline{Y})}{s_{xx}}}} \leq 2.145\right) = 0.95$$

Rearranging

$$\mathbb{P}\left(\hat{\beta}(\underline{x}, \underline{Y}) - 2.145\sqrt{\frac{\widehat{\sigma^2}(\underline{x}, \underline{Y})}{s_{xx}}} \leq \beta \leq \hat{\beta}(\underline{x}, \underline{Y}) + 2.145\sqrt{\frac{\widehat{\sigma^2}(\underline{x}, \underline{Y})}{s_{xx}}}\right) = 0.95$$

Therefore we have a confidence interval

$$\left(\hat{\beta}(\underline{x}, \underline{Y}) - 2.145\sqrt{\frac{\widehat{\sigma^2}(\underline{x}, \underline{Y})}{s_{xx}}}, \hat{\beta}(\underline{x}, \underline{Y}) + 2.145\sqrt{\frac{\widehat{\sigma^2}(\underline{x}, \underline{Y})}{s_{xx}}}\right)$$

Using the data our observed confidence interval is

$$\begin{aligned}
&\left(\hat{\beta}(\underline{x}, \underline{y}) - 2.145\sqrt{\frac{\widehat{\sigma^2}(\underline{x}, \underline{y})}{s_{xx}}}, \hat{\beta}(\underline{x}, \underline{y}) + 2.145\sqrt{\frac{\widehat{\sigma^2}(\underline{x}, \underline{y})}{s_{xx}}}\right) \\
&= \left(0.634 - 2.145\sqrt{\frac{29.726}{5561}}, 0.634 + 2.145\sqrt{\frac{29.726}{5561}}\right) \\
&= (0.477, 0.791).
\end{aligned}$$

Next we test the hypotheses:

$$H_0 : \beta = 0 \text{ vs } H_1 : \beta \neq 0.$$

We use the test statistic

$$T(\underline{x}, \underline{Y}; \beta) = \frac{\hat{\beta}(\underline{x}, \underline{Y}) - \beta}{\sqrt{\frac{\widehat{\sigma^2}(\underline{x}, \underline{Y})}{s_{xx}}}}.$$

Note that under the assumption of the null hypothesis $T(\underline{x}, \underline{Y}; \beta)$ is distributed according to a t_{14} distribution.
Our observed value of the test statistic is

$$T(\underline{x}, \underline{y}; 0) = \frac{\hat{\beta}(\underline{x}, \underline{y})}{\sqrt{\frac{\widehat{\sigma^2}(\underline{x}, \underline{y})}{s_{xx}}}} = \frac{0.634}{\sqrt{\frac{29.726}{5561}}} = 8.67.$$

Then our p-value is

$$\mathbb{P}(|T(\underline{x}, \underline{Y}; \beta)| \geq 8.67) = 2\mathbb{P}(T_{14} \geq 8.67) = 5.3 \times 10^{-7}.$$

Here T_{14} is a random variable with t_{14} distribution.

Properties of $\hat{\alpha}$ and inference about α

Remember: $\hat{\alpha}(\underline{x}, \underline{y}) = \bar{y} - \hat{\beta}(\underline{x}, \underline{y})\bar{x}$ (LS and MLE).
We can prove that:

1. $\mathbb{E}(\hat{\alpha}(\underline{x}, \underline{Y})) = \alpha$ (unbiased estimator)

2. $\mathbb{V}\text{ar}(\hat{\alpha}(\underline{x}, \underline{Y})) = \sigma^2(\frac{1}{n} + \frac{\bar{x}^2}{s_{xx}})$

Under full normal model

We have $\hat{\alpha}(\underline{x}, \underline{Y}) \sim \text{N}\left(\alpha, \sigma^2(\frac{1}{n} + \frac{\bar{x}^2}{s_{xx}})\right)$ and $\frac{\hat{\alpha}(\underline{x},\underline{Y})-\alpha}{\hat{\sigma}(\underline{x},\underline{Y})\sqrt{\frac{1}{n}+\frac{\bar{x}^2}{s_{xx}}}} \sim t_{n-2}$ which can be used for hypothesis tests and confidence intervals.

Assumptions and Validity

Technically the model for regression $Y_i = \alpha + \beta x_i + \varepsilon_i$, for $i = 1, 2, ..., n$, requires the x values to be controlled constants i.e. we observe values of the variable Y for fixed chosen values of X. However in practice in the majority of regression applications this is not the case (as in Example 6.1) and we still use the methodology.

For validity we focus on the assumptions: the ε_i values are independent and $\varepsilon_i \sim N(0, \sigma^2)$. There are three features here:

- Independence;
- Normality;
- Constant variance.

To check these we use 'residual plots':

- For normality use a dotplot / histogram;
- For independence and constant variance use a 'residuals v. fitted values' plot. Here we want a random or 'pattern-free' plot.

Examples

In the plots of Figure 17 we show 4 possible behaviours. In panel (a) we see the behaviour we would like to see, there is no discernable pattern and the behaviour appears random. In panel (b) the spread around the residual varies with x, showing that there is not a constant variance in te residuals. In panel (c) there are a number of large outliers which makes us question the normality assumption. In panel (d) the residuals show a clear pattern suggesting that the assumption of a linear relationship is not valid.

Prediction

An obvious use of a fitted regression line is to predict the value of Y for a new value of X. Recall the fitted or predicted values are obtained using $\hat{y}_i = \hat{\alpha} + \hat{\beta} x_i$.

Clearly for a new value of X, say x_0, the actual predicted value is $\hat{\alpha} + \hat{\beta} x_0$, but what about its accuracy?

It depends on what we are predicting:

(A) The expected (or mean) Y at x_0

or

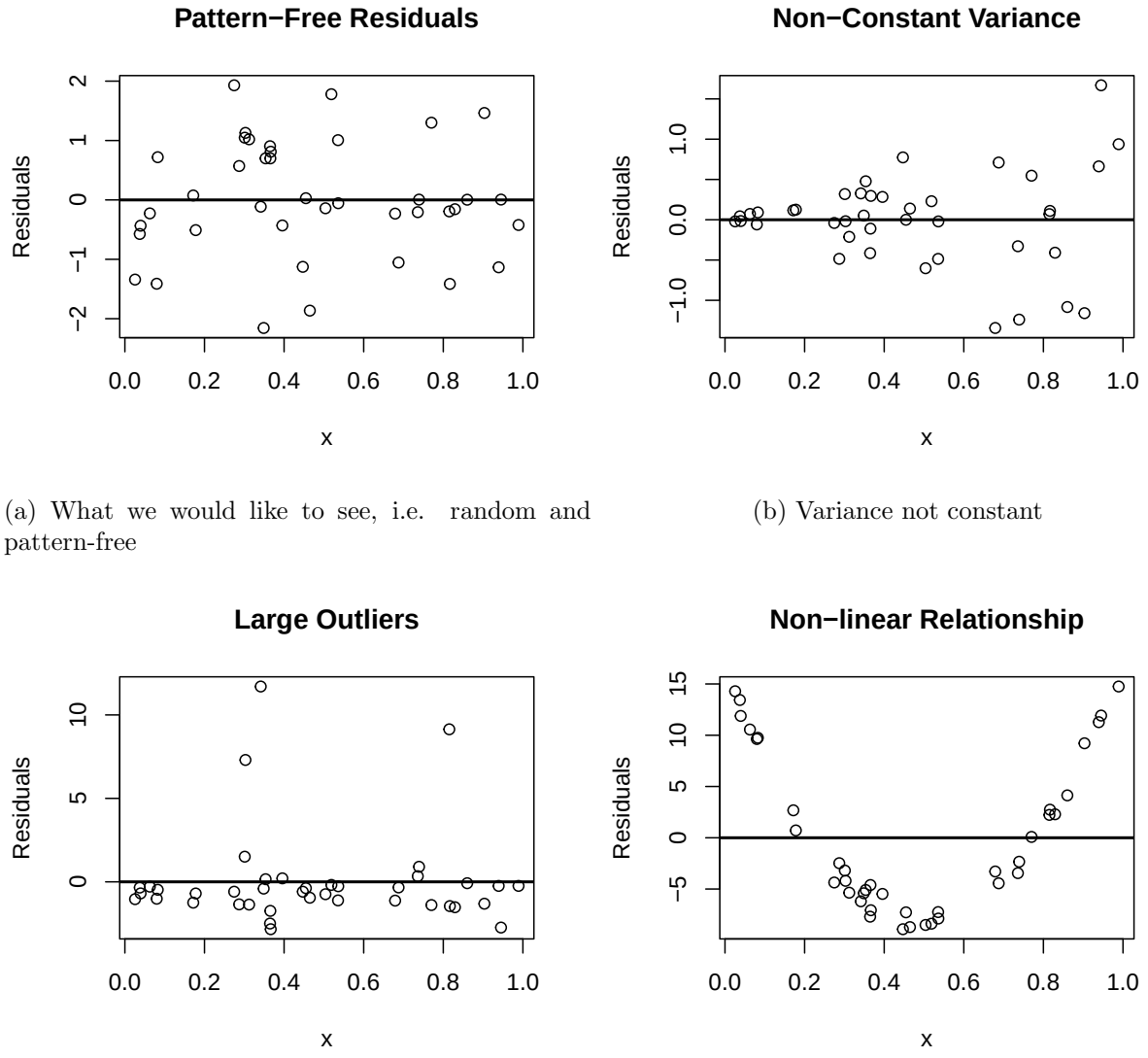
(B) An individual value of Y at x_0

Both have the same estimate, but the latter will be subject to more variation and will have a greater standard error.

In (A) we are estimating $\alpha + \beta x_0$

In (B) we are estimating $\alpha + \beta x_0 + \varepsilon$, i.e. an additional 'error term'

In (A) the standard error of the estimate is $\hat{\sigma}\sqrt{\frac{1}{n} + \frac{(x_0-\bar{x})^2}{s_{xx}}}$



(a) What we would like to see, i.e. random and pattern-free

(b) Variance not constant

(c) Outliers

(d) Not linear but a curved relationship

Figure 17: Example plots showing possible behaviours of the residuals. Panel (a) shows what we would like to see while (b), (c) and (d) are various ways in which the assumptions we use might be violated.

In (B) the standard error of the estimate is $\hat{\sigma}\sqrt{\frac{1}{n} + \frac{(x_0-\bar{x})^2}{s_{xx}} + 1}$

Further note the size of the $\frac{(x_0-\bar{x})^2}{s_{xx}}$ term - it is zero at $x_0 = \bar{x}$ (the middle of the x values) but increases as x_0 moves away from $\bar{x}$.

Therefore, 95% confidence limits for the expected (mean) Y at $X = x_0$ are given by:

$$(\hat{\alpha}(\underline{x}, \underline{Y}) + \hat{\beta}(\underline{x}, \underline{Y})x_0) \pm t_{n-2,0.025}\hat{\sigma}(\underline{x}, \underline{Y})\sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{s_{xx}}}$$

and the 95% confidence limits for an individual Y at $X = x_0$ are given by:

$$(\hat{\alpha}(\underline{x}, \underline{Y}) + \hat{\beta}(\underline{x}, \underline{Y})x_0) \pm t_{n-2,0.025}\hat{\sigma}(\underline{x}, \underline{Y})\sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{s_{xx}} + 1}$$